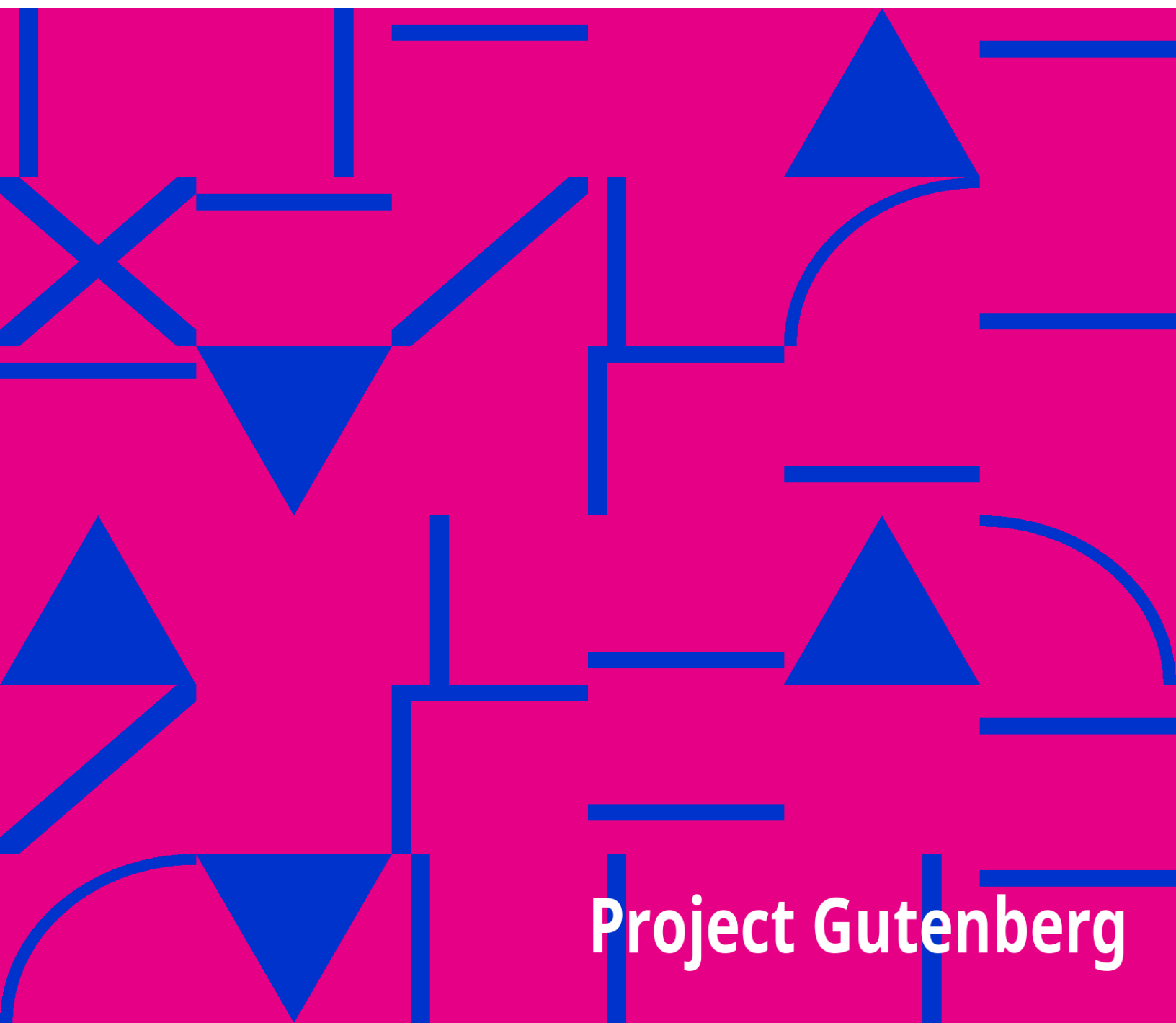


# The Adventures of a Grain of Dust

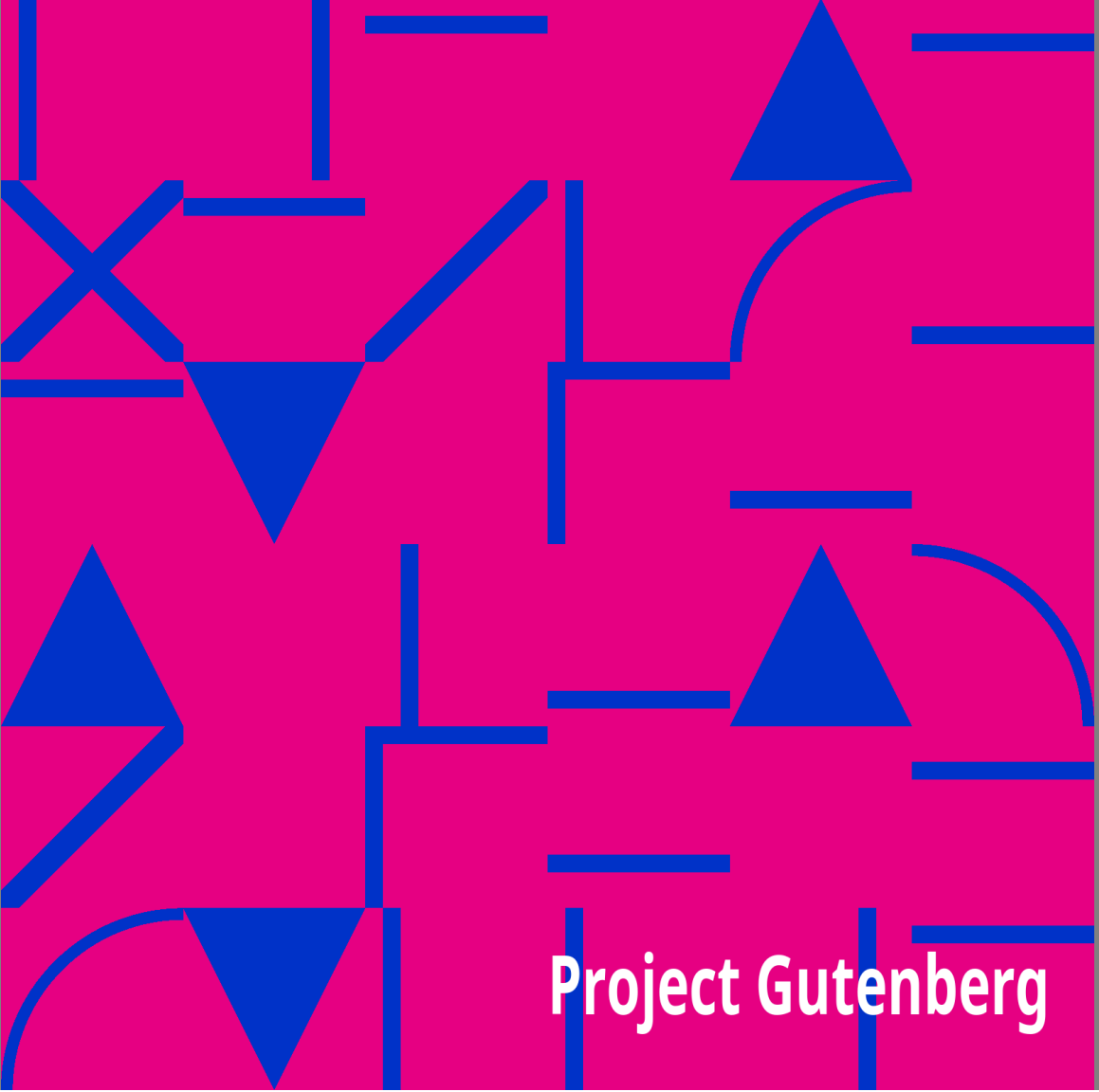
Hallam Hawksworth



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# The Adventures of a Grain of Dust

Hallam Hawksworth



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ADVENTURES OF A GRAIN OF DUST \*\*\*

STRANGE ADVENTURES IN NATURE'S WONDERLANDS

# THE ADVENTURES OF A GRAIN OF DUST

BY  
HALLAM HAWKSWORTH

AUTHOR OF "THE STRANGE ADVENTURES OF A PEBBLE"

CHARLES SCRIBNER'S SONS  
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## JUST A WORD

I don't want you to think that I'm boasting, but I *do* believe I'm one of the greatest travellers that ever was; and if anybody, living or dead, has ever gone through with more than I have I'd like to hear about it.

Not that I've personally been in all the places or taken part in all the things I tell in this book—I don't mean to say that—but I do ask you to remember how long it is possible for a grain of dust to last, and how many other far-travelled and much-adventured dust grains it must meet and mix with in the course of its life.

The heart of the most enduring grains of dust is a little particle of sand, the very hardest part of the original rock fragment out of which it was made. That's what makes even the finest mud seem gritty when it dries on your feet. And the longer these sand grains last the harder they get, as you may say; for it is the hardest part that remains, of course, as the grain wears down. Moreover, the smaller it gets the less it wears. If it happens to be spending its time on the seashore, for example, the very same kind of waves that buffet it about so, waves that, farther down the beach hurl huge blocks of stone against the cliffs and crack them to pieces, not only do not wear away the sand grains, to speak of, but actually save them from wear. The water between the grains protects them; like little cushions. And the sand in the finer dust grains carried by the wind is protected by the material that gathers on its surface.

Why, if a pebble of the size of a hickory-nut may be ages and ages old—almost in the very form in which you see it,<sup>[1]</sup> think what the age of this long-enduring part of a grain of dust must be.

Then remember what the ever-changing material on the surface of these immortal grains is made of; the dust particles of plants and animals, of buried Cæsars and still older ancients, such as those early settlers of Chapter II.

Finally, if what we call flesh and blood can think and talk, why not a grain of dust? In fact, what is flesh and blood but dust come back to life? Says the poet—and the poets know:

"The very dust that blows along the street  
Once whispered to its love that life is sweet."

You see it's as likely a thing as could happen—this whole story.

THE GRAIN OF DUST.

(Per H. H.)

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# CONTENTS

| CHAPTER |                                                                           | PAGE |
|---------|---------------------------------------------------------------------------|------|
| I.      | <a href="#"><i><u>The Little Old Man of the Rock</u></i></a>              | 1    |
| II.     | <a href="#"><i><u>Some Early Settlers and Their Bones</u></i></a>         | 19   |
| III.    | <a href="#"><i><u>The Winds and the World's Work</u></i></a>              | 37   |
| IV.     | <a href="#"><i><u>The Bottom-Lands</u></i></a>                            | 55   |
| V.      | <a href="#"><i><u>What the Earth Owes to the Earthworm</u></i></a>        | 75   |
| VI.     | <a href="#"><i><u>The Little Farmers with Six Feet</u></i></a>            | 92   |
| VII.    | <a href="#"><i><u>Farmers with Four Feet</u></i></a>                      | 114  |
| VIII.   | <a href="#"><i><u>Water Farmers Who Help Make Land</u></i></a>            | 137  |
| IX.     | <a href="#"><i><u>Farmers Who Wear Feathers</u></i></a>                   | 162  |
| X.      | <a href="#"><i><u>The Busy Fingers of the Roots</u></i></a>               | 186  |
| XI.     | <a href="#"><i><u>The Autumn Stores and the Long Winter Night</u></i></a> | 204  |
| XII.    | <a href="#"><i><u>The Brotherhood of the Dust</u></i></a>                 | 225  |
|         | <a href="#"><i><u>Index</u></i></a>                                       | 247  |

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It will be understood, as stated in the preface, that, like "The Strange Adventures of a Pebble," this is an autobiography. In other words, it is

the grain of dust itself that tells the story of the life of the soil of which it is a part.



# THE ADVENTURES OF A GRAIN OF DUST

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# CHAPTER I

(JANUARY)

In truth you'll find it hard to say  
How it could ever have been young  
It looks so old and grey.

*Wordsworth.*

## THE LITTLE OLD MAN OF THE ROCK

Some say it was Leif Ericson, some say it was Columbus, but *I* say it was The Little Old Man of the Rock.

And I go further. I say he not only discovered America but Europe, Asia, and Africa, and the islands of the sea. I'll tell you why.

### I. HOW LITTLE MR. LICHEN DISCOVERED THE WORLD

As everybody knows, we must all eat to live, and how could either Columbus or anybody else—except Mr. Lichen—have done much discovering in a world where there was nothing to eat? When the continents first rose out of the sea<sup>[2]</sup> there wasn't anything to eat but rock. Rock, to be sure, makes very good eating if you have the stomach for it, as Mr. Lichen has. It contains sulphur, phosphorus, silica, potash, soda, iron, and other things that plants are fond of, but ordinary plants can't get these things out of the rock—let alone human beings and other animals; and that's why Mr. Lichen had the first seat at the table and always does.

On bare granite boulders in the fields, on the rocky ruins at the foot of mountains, and even on the mountain tops themselves, on projecting rocks far above the snow line, you find the lichens. On rock of every kind they

settle down and get to work. They never complain of the climate—hot or cold, moist or dry. When the land goes dry they simply knock off, and then when a little moisture is to be had they're busy again. A little goes a long way with members of the family who live in regions where water is scarce. Indeed, most of them get along with hardly any moisture at all. The very hardiest of them are so small that a whole colony looks like a mere stain upon the rock.

While lichens are generally gray—they seem to have been *born* old, these queer little men of the rock—you can find some that are black, others bright yellow or cream-colored. Others are pure white or of various rusty and leaden shades. Some are of the color of little mice. To make out any shapes in these tiny forms, you must look very close; and if you have a hand lens you will be surprised to find that this fairy-land of the lichens isn't so drab as it seems to the naked eye. For there are flower gardens—the tiny spore cups. Some of them are vivid crimson and, standing out on a background of pure white, they're very lovely. Some of the science people believe the colors attract the minute insects that the lens shows wandering around in these fairy flower gardens. But just what the insects can be there for nobody knows, since the lichens are scattered, not by insects, but by the wind.

As a rule lichens grow only in open, exposed places, although some are like the violets—they enjoy the shade. Some varieties grow on trees, some on the ground, others on the bleached bones of animals in fields and wastes and on the bones of whales cast up by the sea.

Of course the whole country was awfully wild when the continents first came out of the sea, but that just suited Mr. Lichen, for there is one thing he can't stand, and that is city life, with its smoke and bad air.

"Why, one can't get one's breath!" he says.

#### **WHY THE LICHENS DISLIKE CITY LIFE**

So, while you will not meet Mr. Lichen in cities—at least, until after the people are all gone; that is to say, on ruins of cities of the past—you will find him beautifying the ancient walls of abbeys, old seats of learning like Oxford, and the tombstones of the cities of the dead.

Mr. Lichen always travels light. On the surface of the lichens are what seem to be little grains of dust, and these serve the purpose of seeds. A puff of wind will carry away thousands of them, and so start new colonies in lands remote.

You see, the fact that he requires so little baggage must have been a great advantage to Mr. Lichen in those early days, when he had to discover not only America but all the rest of the world map, spread out so wide and far. You can just imagine how the grains of lichen dust, the seed of the race, must have gone whirling across the world with the winds.

But if a breath of wind would carry them away so easily, how could they *stay* on a rock, these tiny lichen travellers? Especially as they have no roots? They have curious rootlike fibres which absorb food by dissolving the rock, and this dissolved rock, hardening, holds them on. The fibres of lichens that grow on granite actually sink into it by dissolving the mica and forcing their way between the other kinds of particles in the rock that they can't eat. Thus they help break it up.

As we all know, little people are great eaters in proportion to their size, but it is said the lichens are the heartiest eaters in the world. They eat more mineral matter than any other plant, and all plants are eaters of minerals.

Yet, you'd wonder what they do with the food they eat—most of them grow so slowly. A student of lichens watched one of them on the tiled roof of his house in France—one of the kind of lichens that look like plates of gold—and in forty years he couldn't see that it had grown a single bit, although he measured it carefully.

#### **HOW MR. LICHEN EATS UP STONES**

But how could such feeble creatures, as they seem to be, ever eat anything so hard as rock? Well, they couldn't if it wasn't for one thing—they understand chemistry. At least they carry with them, or know how to make, an acid, and it's this acid which enables them to dissolve the rock so that they can absorb it. The acid is in their fibres—what answer for roots. And the dissolved rock not only gives them their daily bread, but, as I said a moment ago, holds them on. This use of acid is their way of eating;

chewing their food very fine, and mixing it with saliva, as all of us young people are taught to do.

The first and smallest of the lichen family spread and decay into a thin film of soil. This decay makes more acid, just as decaying leaves do to-day—they learned it, no doubt, from the lichens—and this acid of decay also eats into the rock and makes more soil. (You see nature, from the start, has been helping those that help themselves, just as the old proverb has it.) Then, after the first tiny lichens—mere grains of dust that have just begun to feel the stir of life—come somewhat larger lichens which can only live where there is a little soil to begin with. These in turn die, which means a still deeper layer of soil, still more acid of decay, and so on up to larger lichens and later more ambitious plants. Then, on the soil made by these successive generations of lichens, higher types of plants—plants with true roots—get a foothold.

Besides making soil themselves, the lichens help accumulate soil by holding grains of rock broken up by their fibres and loosened by the action of the heat and cold of day and night and change of season. These little grains become entangled in the larger lichens and are kept, many of them, from being washed away by the heavy rains. So held, they are in time crumbled into soil by the action of the acids and by mixture with the products of plant decay. To this day, go where you will, over the whole face of the earth, and you'll find the lichens there ahead of you, dressed in their sober suits, some gray as ashes, others brown, but some are as yellow as gold; for even these old people like a little color once in a while. As travellers they beat all.

"Their geographical range is more extended than that of any other class of plants."

That's how the learned lichenologists put it. For these lichens, these humble little brothers of our dust, that many of us never looked at twice on the stones of the field, or the gray stumps and dead limbs in the wood, are so interesting when you've really met them—been properly introduced—that a whole science has grown up around them called "lichenology." And exciting! You ought to hear the hot discussions that lichenologists get into. You read, for instance, that such and such a theory "was received with a



storm of opposition" (as most new theories are, by the way, particularly if they are sound).

But the tumults and the strifes of science, of politics, or of wars don't disturb little old Mr. Lichen himself. There on his rock he'll sit, overlooking the scenery and watching life and the seasons come and go for 100, 200, 500 years, and more. For while they grow so slowly the lichens make up for it by living to an extreme age.

#### **THE LICHENS AND THE ROMAN EMPIRE**

Why, do you know that during the lifetime of certain lichens that are still hale and hearty, not only a long line of Cæsars might rise, flourish, die, and, with their clay, stop holes to keep the wind away, as Mr. Shakespere put it, but the vast Roman Empire could and did come into being, move across the stage with its banners and trumpets and glittering pomp and go back to the dust again.

Some lichens, growing on the highest mountain ranges of the world, are known to be more than 2,000 years old!



## THE SEQUOIAS; THE SUNLIGHT AND THE SHADE

Wonderful sunlight effect, isn't it? We are here in Sequoia National Park and those big trees are sequoias, members of the pine-tree family.

## II. THE MARCH OF THE TREES

Of course I don't mean to say it takes any 2,000 years for the average lichen to die and turn to dust. These long-lived lichens are the Methuselahs of their race. Most kinds die much younger, as time goes among the lichens, and in a comparatively few years, a century say, after their first settlement on the rock, the lichens have become soil. All this time the heating of the rock by day and the cooling off at night, the work of frost and the gases of the rain and the air<sup>[3]</sup> have also helped to make more soil and by and by there is enough for lichens of a larger growth; and mosses begin to get a foothold. These, in turn, die and, in decaying, make acids, as did the little lichens before them, and this acid joins hands with all the other forces to work up the rock into soil. Presently there is enough soil to let certain adventurers of the Weed family drop in. The picking is very thin, to be sure, but some of these Weed people have learned to put up with almost anything. Don't suppose, however, that all weeds are alike in this respect. Oh, dear, no! They come into new plant communities just as the trees do, not haphazard, but according to a certain more or less settled order. Some of them, the adventurer type, will, it is true, settle down and seem contented enough on land so poor that to quote the witty Lady Townshend "you will only find here and there a single blade of grass and two rabbits fighting for that"; while other weeds will have nothing to do with soil that, in their opinion, is not good enough for people of their family connections.



#### EARLY SETTLERS IN THE DESERT

Besides earning their own living under hard conditions, these sturdy pioneers of the desert are preparing the way for plants of a higher kind, as the next two pictures will tell you.

It has long been known that the character of soil may be told, to a considerable degree, by the kind of weeds that grow on it. An old English writer pointed this out in his quaint way some 200 years ago:

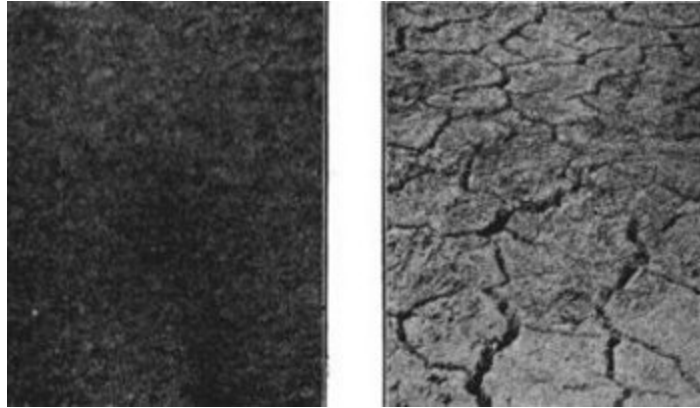
"Ground which, though it bear not any extraordinary abundance of grass yet will load itself with strong and lusty weeds, as Hemlocks, Docks, Nettles and such like, is undoubtedly a most rich and fruitful ground for any grain whatsoever."

But, he goes on to say:

"When you see the ground covered with Heath, Broom, Bracken, Gorse and such like, they be most apparent signs of infinite great barrenness. And, of these infertile places, you shall understand, that it is the clay ground which for the most part brings forth the Moss, the Broom, the Gorse and such like."

Wherever soil is coarse and bouldery the weeds also are of a sturdy breed. In his long, delightful days among the mountains Muir<sup>[4]</sup> tells us what a brave show the thistles made in this new world of soil; how royal they

looked in their purple bloom, standing up head and shoulders above the other plants, like Saul among the people.



#### WHAT THE DESERT PIONEERS DO FOR FUTURE GENERATIONS

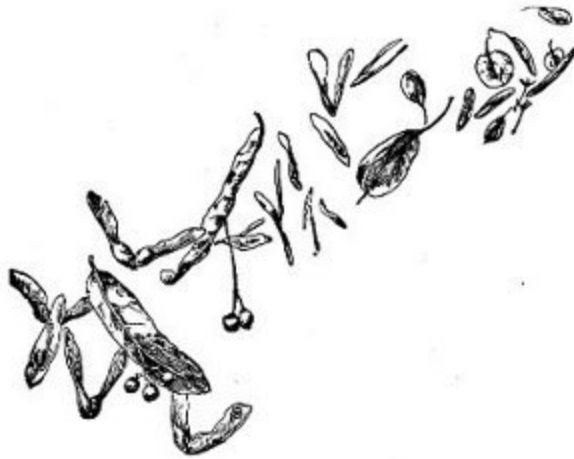
Only the sturdiest kinds of shrubs and weeds, such as you see in the desert, can earn their keep in sandy soil, always thirsty, like that on the right. But the desert vegetation, dying and decaying—it is then called "humus"—not only knits the soil together but absorbs moisture and ammonia from the air and so helps grow good crops.

#### HOW PLANT PEOPLE PAY THEIR TAXES

In all these plant republics each citizen must pay something into the common treasury for its board and keep. This fund not only meets "national expenses" during the lifetime of the ones who pay these taxes, but it helps prepare the land for the great citizens of the future—the trees. In another hundred years—making two hundred in all, after the arrival of the very first lichens—low shrubs and bushes often find spots in these new communities where the soil is thick enough for their needs.

It is very curious how members of the plant world, growing side by side, seek their food at different depths, and send out their roots accordingly. It reminds one of the rigid class distinctions below stairs in a nobleman's household where the chef has his meals in his own private apartment, the

kitchen maids in their quarters, the chauffeurs, footman, under butler, and pantry boys in the servants' hall.



THE LEADERS OF THE GRAND MARCH

But most striking, it has always seemed to me, is the settled order in which trees march into the land. Why shouldn't the oaks come before the maples? Or the maples before the beeches? Or the beeches before the pines? Why is it that, with the exception of a straggler here and there, the first trees to climb the stony mountainsides are the pines? Then close behind come such trees as the poplars, and along the streams below, the willows. Still farther down the valley are the beeches; farther still the maples, and last of all the oaks.

So it is they advance in a certain regular way, each in its own place in the ranks. At first it seems as strange as the coming of Birnam wood to Dunsinane that gave poor Macbeth such a turn that time. But, after all, the explanation is quite simple and no doubt you have guessed it already.

The reason such trees as the pines, poplars, and willows come first is that the seeds are so light they are easily carried by the winds and so reach new soil ahead of other trees with winged seeds like the beeches and the maples; for, although these seeds also travel on the wind, they are much larger than

the winged seeds of the pine and they travel much more slowly and for shorter distances.

Moreover, at the end of their first journey, having once fallen to the ground, they are apt to stay. Then there is no further advance, so far as these particular seeds are concerned, until trees have sprung from them and they, in turn, bear seeds. In the case of very light seeds, like those of the pines, the wind not only carries them far beyond the comparatively slow and heavy march of the beech and the maple, but if they fall on rock with little or no soil the next wind picks them up and carries them farther, so that they may strike some other spot where there is soil and perhaps a little network of grass and weeds to secure them until they can take root and so hold their own. It is not only a great advantage to the pine seeds to be so small, so far as getting ahead of other trees is concerned, but it is an advantage in another way. Because they are so small they require comparatively little soil to start with, are more easily covered up, and so they soon begin to sprout. The very winds that carry them up among the mountain rocks are quite likely to cover them with enough dust to start on, and I myself have helped raise many a giant of the mountain forests in this way. It is really wonderful how little soil a pine-tree can get along with; if, say, its fortunes are cast on some mass of mountain rock. Somehow it manages to get a living among the cracks and at the same time to hold its own in the bitter struggle with the winds.

"The pine trees," says Muir, "march up the sun-warmed moraines in long hopeful files, taking the ground and establishing themselves as soon as it is ready for them."



*From the painting by Rousseau in the Metropolitan Museum of Art.*

### THE EDGE OF THE WOODS

Last of all come tramping along the sturdy old oaks.

Last of all come tramping along the sturdy old oaks and the nut-bearing trees. Their seeds are so heavy they get little help from the winds, and then only in the most violent storms. They must advance very slowly indeed, with occasional help from absent-minded squirrels who carry away and bury nuts and acorns and then forget where they put them.



### HOW SQUIRRELS HELP OAKS TO MARCH

Sometimes they bury acorns and forget just where. When frightened they often drop them and run away.

### ROUGH CITIZENS AMONG THE PIONEERS

The beginnings of a forest are stunted because the soil is thin. Moreover, the company in which the trees find themselves is very miscellaneous, like the population of all pioneer communities—weeds, grasses, briers, shrubs. High up on a mountainside you can find all these types of vegetation. Pines growing clear to the snow line; farther down the mountain, in crannies, sumach and elder bushes with field daisies and goldenrod scattered among them; while on the barren rocks are the lichens and the mosses.

Not only do the citizens of the plant world follow a certain fixed order in coming into new regions, but also in giving place to one another. All plants of a higher order can live only on the remains of those of a lower, and it is most interesting to note the process by which each lower form comes, does its work, passes on, and is replaced by a superior type. The shrubs, which can only grow after the weeds and grasses have made enough soil for them, at length shade out these smaller pioneers. Haven't you often noticed, when picnicing in deep woods, that the grasses and flowers are to be found only in the sunny spaces, where there are no trees?

But these thickets themselves, after a while, disappear, and pines take their places. I am speaking now of the growth of forests, where the soil-making has so far advanced that forests are possible. The thickets, with their good soil and the shade which keeps it damp, are just the places for the pine seeds brought in by the wind to get a foothold and sprout up. When they grow into big trees they gather with their high branches so much of the sunshine for themselves that little of it gets through to the shrubs below, so these shrubs disappear, surviving only in the sunny open spaces or along the borders of the wood.

But now notice what happens to the pines. When the trees become larger, the young pines that spring up beneath their shade can't get enough sunshine, so, as the big trees grow old and die, there are fewer and fewer young pines to take their places. Now comes the turn of the spruces. For spruces require more and better soil than the pines and they don't mind a reasonable amount of shade. So, as the woods grow thicker and shadier, the pines gradually disappear and the spruces take their places.

At first, in the reign of the spruces, some of the old residents begin to come back. A spruce forest, not being so dense in the beginning as a pine forest,



lets in a good deal of sunlight, and you'll find scattered through its aisles and byways gentians, bluebells, daisies, goldenrod.

In course of time, however, the leaves and branches of the spruces become so thick that hardly a sunbeam can get through and you have a forest where noontime looks like twilight; a forest of deep shade and silence with its thick carpet of brown needles, and where all the shrubs and grasses and flowers have disappeared, except in the open spaces. It was in such a forest and in one of these sunny glades, no doubt, that the knight the little girl tells of in Tennyson:

"... while he past the dim lit woods  
Himself beheld three spirits mad with joy  
Come dashing down on a tall wayside flower  
That shook beneath them as the thistle shakes  
When three gray linnets wrangle for the  
seed."

#### HOW NATURE RESTORES ABANDONED FARMS

So it is that new lands pass from barren rock to forest, and deep rich soil, and so it is that worn-out soils, the result of reckless farming are finally restored. Hardly any soil is too poor for some kind of a weed. These weeds springing up, die and make soil that better kinds of weeds can use. Later come a few woody plants. In the course of fifteen or twenty years the soil is deep enough to support trees; and in fifty years there is a young forest. At the end of a century fine timber can be cut, the land cleared, and the old place may be as good as new.

But it's a long time to wait! It's a much better plan to take care of the land in the first place.

#### HIDE AND SEEK IN THE LIBRARY

One of the strangest things about Mr. Lichen, as you will see by looking up the subject in any botany or encyclopædia, is that he is really *two* people—two different plants that have grown into

partnership; and that one of the partners supplies water for the firm while the other furnishes the food.

The part of "him" that supplies the food is green, or blue-green, and that is why it is able to do this. This idea that Mr. Lichen is really two people was one of those that was "received with a storm of opposition," but certain lichenologists actually took two different kinds of plants, put them together and *made* a lichen themselves, as you will see when you look the matter up.

As to just who among these two kinds of plants shall go into partnership—that usually depends on chance and the winds; although in the case of some lichens, the parents determine upon these partnerships, just as they often do in human relations.

If you want to continue this interesting study and become Learned Lichenologists, you will be interested to know that there are a lot of things to be learned, including not only no end of delightful names, such as *Endocarpon*, *Collema*, *Pertusaria*, not to speak of *Xanthoria parietina*, and loads of others, but there are still things unknown that *you* may be able some day to find out. For instance, while they know that the two kinds of vegetation that together make a lichen, feed and water each other, it's not known exactly *how* they do it; although the "Britannica" article has a picture showing the two partners in the very act of going into partnership. The article in the "Americana" shows some striking forms of lichens, and how nature from these very dawnings of life begins to dream of beauty. You will be surprised at the forms shown in the "Americana," they are either so graceful, symmetrical, or picturesque. One of them looks like a very elaborate helmet decoration, or plume of a knight.

This article also tells what an incredible number of species of lichens there are—enough to make quite a good-sized town, if they were all real people.

It also tells why the orange and yellow lichens take to the shady side of the rock; and something about how the lichens get those remarkable decorations and sculpturings, and what the weather has to do with it.

There you will also get a probable explanation of the fact that the manna which the Israelites found on the ground in the morning appeared so suddenly.

In the article in the "International" you will find another picture of how the two partners—the fungus and the alga—make the lichen, and you will learn that Mr. Lichen's name, like Mr. Lichen himself, is centuries old; being the very name given him by the Greeks, and afterward by the Romans.

In the "Country Life Reader" there is an article on the soil that has a very close relationship to the subject of the lichens and their work. It tells, among other things, about the value of humus—decayed leaves, grass, etc.—to the soil. It was the lichens, you know, who *started* the humus-making business.

The article in the reader on "Planting Time," by L. H. Bailey, expresses the wonder we must all feel when we stop to think about it, at the magic work of the soil in changing a little speck of a seed into a plant.

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## CHAPTER II

(FEBRUARY)

Behold a strange monster our wonder engages!  
If dolphin or lizard your wit may defy.  
Some thirty feet long, on the shore of Lyme-Regis  
With a saw for a jaw and a big staring eye.  
A fish or a lizard? An Ichthyosaurus,  
With a big goggle-eye and a very small brain,  
And paddles like mill-wheels in chattering chorus  
Smiting tremendous the dread-sounding  
main.

—*Professor Blackie.*

### SOME EARLY SETTLERS AND THEIR BONES

But a farm where nothing but plants grow isn't much of a farm. Every good farmer knows that nowadays, and so he stocks his place with horses and cows and chickens and things. Mother Nature understood this principle from the beginning, and the plants and animals on her farm have always got on well together.

For one thing the plant and the animal each help the other to get its breath. That is to say, plants, when they take in the air, keep most of the carbon there is in it and give back most of the oxygen, which is just what the animal world wants; while the animals, when they breathe, keep most of the oxygen and give back most of the carbon—just the thing that plants grow on.

But the service of the animals to the plants is very important after they have stopped breathing altogether; since their flesh and bones, like the dead bodies of the plants, go back to enrich their common dust. The bones and

bodies and shells of members of the animal kingdom, however, are far richer food for soils than is dead vegetation. The shell creatures of the sea to which we owe our wonderfully fertile limestone soils are—many of them—so small that you can only make them out with a microscope; while certain other contributors to our food-supply were so big that one of them, walking down a country road, would almost fill the road from fence to fence.

## **I. MR. DINOSAUR AND HIS NEIGHBORS**

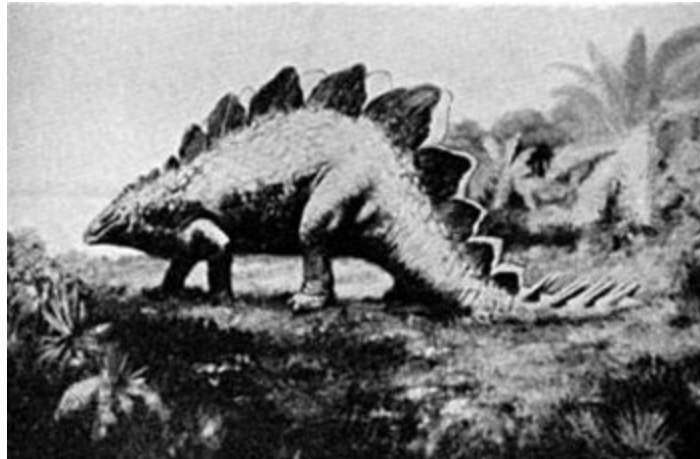
### **A STRANGE FACE IN THE MEADOW**

Now let's take a look at some of these big fellows. How would you like to have such a creature as the one at the right of this page come ambling up to meet you at the meadow gate of an evening when you went to milk the cows? Yet more than likely either this gentle animal, or some of his kin, browsed over the very field where now the cattle pasture, for he, too, was a grass-eater, and with an appetite most hearty. If you kept him in a barn his stall would have to be eighty feet long, and it would be necessary to fill his rack with a ton of fodder every third day. But, assuming there was a market for him in the shape of steaks and roasts, you would be well repaid; for, in prime condition, he weighed twenty tons.



## IN THE LAND OF HIS FATHERS

These monsters who ate grass, and other monsters who ate them, and still other monsters who lived in the sea, appeared comparatively late in the life of the world.



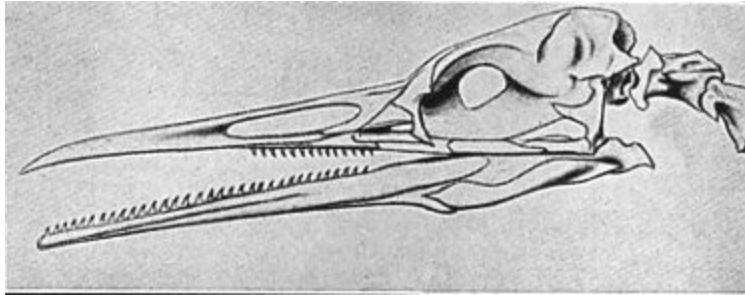
### NO WONDER HE NEVER WORRIED!

Quite aside from the fact that he had so little brain to worry with, it seems highly improbable that the Stegosaurus ever felt any apprehension about attacks from the rear, in the frequent military operations which distinguished the times in which he lived. In addition to the horny plates down his back he had those horny spines which were swung by a tail some ten feet long.

## TONS AND TONS OF ANCIENT BONES

It is only about 15,000,000 years ago, for example, that the biggest of them all, the Dinosaurs, lived, while the earth itself is now supposed to be some 100,000,000 years old. Their numbers were enormous, and it is probable there is not an acre of ground from the Atlantic to the Pacific, and from Alaska to the tip end of South America that has not been fertilized by their bones. In fact, of certain species I have found the bones scattered all the way from Oregon to Patagonia; so this must have been their pasture.

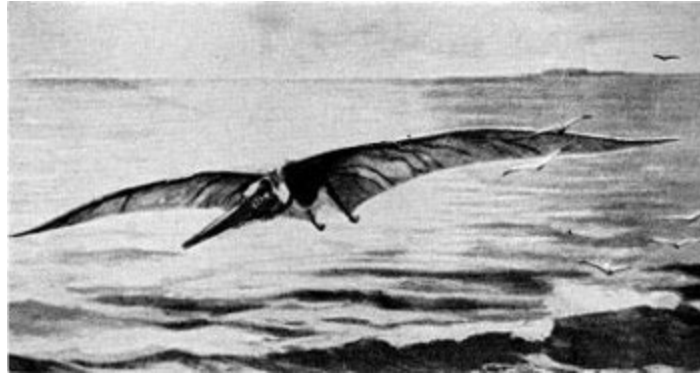
They were not only all over the land, but in the lakes and in the great sea that once extended right through North America from the Gulf of Mexico to the Arctic Ocean. And they were along the shores of the sea and in the swamps. The bones of the ancestors of the whale were found in such quantities in some of the Southern States that they were used to build fences until it was found they were much more valuable to enrich the fields themselves.



THE HEAD OF HESPERORNIS

"Then there was a great toothed, diving creature with wings. They've named him the Hesperornis, which means 'western bird,' because the fossils of the best-known species were found in the chalk-beds of Kansas."

In the great American inland sea of those days swam one kind of fierce fish-lizard that took such big bites he had to have a hinge in his jaw. Because of this hinge he could open his mouth wider without putting anything out of place, don't you see? He was called the Mesosaur. But he never bit the Archelon, who was in his crowd, because he couldn't. The Archelon was the king of turtles, and, like all the turtle family, wore heavy armor. He was over twelve feet long. And sharks—no end of them! A shark at his best is bad enough, but the sharks of those days were almost too terrible to think about. Such jaws! And teeth like railroad spikes! Then there was a great toothed diving creature with wings. They've named him the "Hesperornis," which means "western bird." He was given the name because the fossils of the best-known species were found in the chalk-beds of Kansas.



#### GREATEST OF ANCIENT FLYING MACHINES

Mr. Pterodactyl, on his way to dinner, looked like this. He was the largest of all flying-machines before the days of the Wright brothers. He would have measured—if there had been anybody to measure him—twenty feet across the wings! Like the Hesperornis, he always dined on fish.

Over the waters flew another bird-like, fish-like, bat-like thing called the Pterodactyl. Look at his picture and you will see how he got his nickname. It means "finger-toe." He was the largest of all flying-machines until the days of the Wright brothers. It was over twenty feet across his wings, from tip to tip; and, like the Hesperornis, he always had fish for dinner.





#### A BIG "LITTLE FINGER" AND WHAT IT WAS FOR

Mr. Pterodactyl means "finger toe." What is our little finger was the longest of his five digits. It helped support and operate that big bat-like wing extending from his arms to his toes.

#### THE EARLIEST RULERS OF THE SEA

The first monsters, like the first of almost everything else, including the land itself, were in the sea. <sup>[5]</sup> For a time giant fish, armor-plated like a man-of-war, and with awful appetites, just about ran everything. Then came the reign of the sharks. Some of them had jaws that opened to the height of a door—six feet or over. Next in succession, as rulers of the sea, were the fish-lizards, of whom that hinge-jawed Mesosaur was one. Of another of these fish-lizards a famous teacher of Edinburgh University, Professor Blackie, wrote that funny verse at the head of this chapter. The bones of this particular specimen were found sticking out of a cliff at Lyme-Regis, a popular watering-place in the English Channel, by a pretty English girl who was strolling along the beach.



#### A FAMILY PARTY

The imagination of the artist enables us to picture this family party—Mrs. Ichthyosaurus and her children out for a stroll in prehistoric waters.

The Ichthyosaurus, as Professor Blackie says in his verse, was some thirty feet long, with a comparatively large head—like an alligator's—set close to his body. Another fish-lizard, well and unfavorably known by his neighbors of the sea, was the Plesiosaurus. Instead of fins he had big paddles resembling those of the seal. He was a kind of side-wheeler, like the Mississippi River steamboats, and he could go like everything! His neck was long and he darted after the smaller creatures he lived on.

#### REIGN OF THE LIZARD FAMILY

But these queer fish seem to have just been getting ready to land; for, by being lizards, they after a while managed it. A lizard, you know, belongs to the reptile family, and out of these sea reptiles there grew, in course of time, reptiles which lived, not in the sea but in the swamps along the sea. These reptiles were the Dinosaurs, and they are related to the Minosaurs and the

Ichthyosaurus, and the rest of the Saur, as you can see by the family name; for "saur" means lizard. Dinosaur means "terrible lizard." Don't you think he looks it?

Although some of these Dinosaurs were no larger than chickens, others were by far the largest creatures that ever were, on sea or land. Many of the biggest lived on grass, just like an old cow, while the flesh-eating Dinosaurs lived on them. Some of these Dinosaurs went on all fours, while others ran about on their hind legs, and when they stood still, propped themselves up on their big, thick tails as do kangaroos. The Camptosaurus, one of whose favorite resorts was the land that is now Wyoming, was thirty feet long. Another called the Brontosaurus, was sixty feet long. The Atlantosaurus, one of the pioneers of Colorado, measured eighty feet from the end of his nose to the end of his tail, and all of them were built in proportion. The Stegosaurus, also an early settler in Wyoming, had huge bony plates, like ploughshares, sticking out all along his back from the nape of his neck to the end of his tail. He seems to have gone about looking quite ugly and humpbacked, as our old cat does when she has words with the dog.

After the swamps dried up and the lizards could no longer make a living, came the reign of the mammals; including the Mastodons and the Mammoths, marching in countless herds, trumpeting through the forests.

#### **HOW SOME MONSTERS PLOUGHED THE FIELD**

But besides what they did in the way of fertilizing the land with their flesh and bones some of the mammals did a good deal of ploughing. Among these early ploughmen were the Mastodons and the Mammoths, and another elephant-like creature with two tusks, that he wore, not after the fashion among elephants to-day, but curving down from his chin, somewhat like Uncle Sam's goatee. He used these tusks, it is supposed, not only for self-defense, but for grubbing up roots which he ate. If so, they must have been about as good ploughs as those crooked sticks that were used by the early farmers among men, and that are still in use among primitive peoples.

#### **THE ELEPHANT FAMILY AS PLOUGHMEN**

What makes it more likely that the creature with the down-curving tusks stirred the soil with them is that his cousins, the elephants of to-day, are themselves great ploughmen. Elephants feed, not only on grass and the tender shoots of trees, but on bulbs buried in the soil, which they hunt out by their fine sense of smell. In digging these bulbs they turn up whole acres of ground. Elephants also do a great deal of ploughing by uprooting trees so as to make it more convenient to get at their tender tops. Sir Samuel Baker, the explorer, says the work done by a herd of elephants in a mimosa forest in this way is very great and that trees over four feet in circumference are uprooted. In the case of the biggest trees several elephants work together, some pulling the tree with their trunks, while others dig under the roots with their tusks. To be sure, the mimosa-trees have no tap roots, but tearing them out of the ground is no small job, nevertheless. It takes strength and it takes engineering.

Another early ploughman was a bird, the Moa. The Moa had no wings, but his muscular legs were simply enormous, and so were his feet. New Zealand seems to have been the headquarters of the Moas. There used to be loads of them as shown by the huge deposits of their bones. They are supposed to have been killed in countless numbers during the Ice Ages in the Southern Hemisphere; for there were Ice Ages in the Southern as well as the Northern Hemisphere. In one great morass in New Zealand abounding in warm springs, bones of the Moas were found in such countless numbers, layer upon layer, that it is thought the big birds gathered at these springs to keep warm during those great freezes.

#### **THE MILLSTONES OF THE MOAS**

Besides the work they did with feet and bills you may imagine how much nice fresh stone the Moas must have ground up in their crops during the millions of years they existed. It was a regular mill—the gizzard of a Moa—full of pebbles as big as hickory nuts. Scattered about the springs where their bones are found are little heaps of these pebbles, each the contents of a gizzard. Like miniature tumuli, they mark the spots where the bodies of the Moas returned to dust.

Perhaps some of those flesh-eating Dinosaurs did a little ploughing once in a while, too; for one theory is that those ridiculous little arms were used for

scratching out a nest for the eggs, just as the crocodiles and the alligators and the turtles dig nests for their eggs to-day. For all these animals, as did the Dinosaurs, belong to the reptile family, and show the family trait of digging out nests for their eggs.

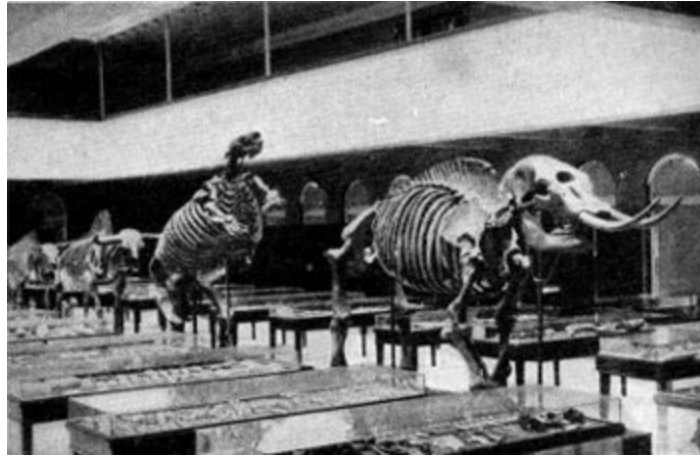


A PUZZLE PAGE FROM THE GREAT STONE BOOK

Talk about your cut-out puzzles! Here is a specimen of the kind of puzzle Nature and the course of things in the darkest ages of world history have cut out for the paleontologists. It is a find of ancient bones in the asphalt deposits near Los Angeles.

Although the Dinosaurs roamed the swamps and lowlands of all the ancient world, their favorite resort was the territory now occupied by our Western States—judging from the quantities of bones they left—while that old Mediterranean Sea of ours was full of their kin, the sea-lizards. Professor Marsh, of Yale, who was among the first explorers of the graves of these monarchs of the past, says that one day, while riding through a valley in the Rocky Mountains, he saw the bones of no less than seven sea-lizards staring at him from the cliffs. Yet, only here and there by the wearing through of the rocks by flowing streams has nature opened up these vast mausoleums, the mountains and the cliffs. What enormous quantities of bones, then, must still be buried there, what tons and tons must have given their lime and

phosphate to the soil. So you see this story of old bones, even from a farming standpoint, is no light matter.



#### HOW THE WISE MEN ANSWER THE PUZZLES

By their marvellous skill and their knowledge of the mechanics of monster anatomy the paleontologists fit one bone fragment to another, supply the missing parts in artificial material, and behold! the monsters take their places in the long procession of the ages. There has been nothing equal to it since the vision of the prophet in the Valley of Dry Bones. (Ezekiel 37:1-10.)

## II. HOW THE MONSTERS DIED AND RETURNED TO DUST

"But you said these monsters lived in the sea and in swamps. Then how, in the name of common sense, did their bones get up into the mountains?"

#### WHEN THE INLAND SEA WENT DRY

Well, it's like this: As I said a while back, in the days of the monster fish and the monster lizards, there was a great sea reaching clear from the Gulf of Mexico to the Arctic Ocean, and with swamps along the borders extending far into lands that afterward became the Rocky Mountains. When the land began to rise, due to the shrinking of the earth—a thing that has been going on ever since the earth was born—the sea and the swamps went

dry, and far to the west the land wrinkled up into the Rocky Mountains. In these layers of rock that made the mountains were the bones of the monsters that had died when the rocks were still mud, in the swamps and along the borders of the inland sea.

Not only did the land under the western portion of the sea slowly rise until the waters were completely closed in on the west, and the sea thus made that much narrower, but the rise of the land on the south cut off connection with the great salt ocean which surrounds the continents to-day. So the salt-water fish, for lack of salt water, died, and with them the monsters like the Ichthyosaurus that lived on the salt-water fish that lived in this salt sea.

But it wasn't alone that the seas grew narrower and more shallow because of the elevation of the lands. The mountains rising in the west, cut off the rain-laden winds which blew from the Pacific in those days just as they do now. Thus the seas dried up so much the faster. But first, before the sea went entirely dry, its place was taken by the lakes and swamps into which it shrivelled up. Low, swampy land is just what reptiles like, so this was their Golden Age, just as the previous time of the wide, deep sea was the Golden Age of the big fish and the fish-lizards.

Then, as the land still rose and the climate grew dryer, the reptiles passed away, and in came the mammal family, to which the cows and the horses and the cats and the kittens, and all the rest of us, belong.



THE TIGER WITH THE SABRE TEETH

Tigers like this lived ages ago in both the Old World and the New. They had canine teeth, curved like a sabre, in the upper jaw.)

### TOO MUCH BRAWN, TOO LITTLE BRAIN

Of course, even where they didn't die with their boots on, so to speak, as so many of them did in those lawless days, there came a time for each monster, in the order of nature, when he drew his last breath. But what seems so strange is that all these monsters—the biggest and strongest of them—entirely disappeared and left no descendants!<sup>[6]</sup> The whole of the mystery has not been unravelled yet, even by the wise men of science, but still they have learned a good deal. For one thing, they know that most of the reptiles and the fish-lizards disappeared because so much of the land where they lived went dry. They had to get a new boarding-place, and there wasn't any to get! Another thing was that these big fellows, although they *were* so big, and got along finely while everything was just so, had so little brain they couldn't change their habits to meet new conditions, as our closer and cleverer cousins, the mammals, did. Why, do you know that one of these monsters, who was twenty-five feet long if he was an inch, and twelve feet high, had a brain no bigger than a man's fist? All the monsters of those days were like that—tons of bone and muscle, but a very small supply of brains.

So when things went against them, they just had to give up, and, like a queer dream, they faded away. But their history makes one of the most interesting chapters in the whole wonderful story of the dust.

Of all the live stock that have fed on the great world-farm and helped enrich it with their bones, these animals were surely the strangest that ever were seen!

### HIDE AND SEEK IN THE LIBRARY

"But since these monsters passed away many millions of years ago, and all that is usually found is a piece of them here and there, how do the men of science know so much about them—how they looked, and how they ate, and how they treated one another?"



That's a good question. It *does* seem strange. Why, to hear them talk, you'd suppose these men, learned in ancient bones, had actually *met* the monsters! And, speaking of meeting them, I must tell you a little story. It's a good story and it will answer your question.

Baron Cuvier, one of the most famous of the paleontologists, awoke from a deep sleep to see standing by his bed a strange, hairy creature with horns and hoofs. And it said:

"Cuvier! Cuvier! I have come to eat you!" But the baron, taking in the form of the monster at a glance, only laughed.

"Horns and hoofs? You can't. You're a grain-eater!"

See the point? The baron argued that because the monster had horns and hoofs he must be a grain-eater; for all creatures with both horns and hoofs are grain-eaters. This particular creature, to be sure, was an eater of both meat and grain—being one of Cuvier's students who was trying to play a trick on him. But the principle holds good. The scientists, *knowing* one thing, *infer* another. Because animals with both horns and hoofs eat no meat Cuvier knew his visitor couldn't eat *him*, even if he'd been real and not just made up.

For another instance, take our queer old friend that Professor Blackie wrote the funny rhyme about—the Ichthyosaurus "with a saw for a jaw and a big staring eye." The scientists figure, just from looking into the hollow socket where the eye used to be, that he could see at night like a cat—and right through muddy water, too; that he spent most of his time in shallows near the shore; that it didn't make any difference to him whether a fish was near or far, provided it wasn't too far, of course, for he could see it and catch it, just the same. They also said—these learned men, after peering into the dark hollow where that remarkable eye used to be—that Mr. Ichthyosaurus spent a great deal of time diving and a great deal of time with his homely face just above the surface of the water.

Why they could reason all this from a hollow eye socket and some bony, flexible plates around the outer edge of it, you will see by referring to such books as "Animals of the Past," by F. A. Lucas,

director of the American Museum of Natural History; "Creatures of Other Days" and "Extinct Monsters," by Hutchinson; "Extinct Animals," by Lankester; "Mighty Animals," by Mix; the chapter "When the World was Young," in Lang's "Red Book of Animal Stories," and "Restoring Prehistoric Monsters" in "Uncle Sam, Wonder Worker," by Du Puy.

Here are some more conclusions they draw from certain facts. See how near you can come to reasoning them out for yourself before looking them up in the books that tell.

Why it is supposed the Dinosaurs swam like Crocodiles. (Look at the picture of Mr. I., and pay *particular* attention to his tail.)

Why it is they say that the sea-lizards with long necks must have had small heads.

Why it is argued that because the Mesosaurus had a hinge in his jaw he must have had a big, loose, baggy throat.

"Keeping Up the Soil," in "The Country Life Reader," deals with the subject of the use of fertilizers on the farm—how easy it is to waste them, how easy it is to save them, and how important it is that they should be saved; while the article on "Acid Soils" tells how the lime in the bones of the monsters has helped keep the soil from getting "sour stomach," and also how they unlocked the potash and phosphorus in the soil so that the plants could get at them.

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### FERTILE FIELDS THAT RODE ON THE WIND

The winds that now help grow the corn and wheat on these broad fields by carrying the pollen from one plant to another, also brought the soil on which they grew. These are the loess plains of Nebraska. There are 42,000 acres of them.

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## CHAPTER III

(MARCH)

... the busy winds  
That kept no intervals of rest.

*Wordsworth.*

Except wind stands as never it stood  
'Tis an ill wind turns none to good.

—*Tusser.*

### THE WINDS AND THE WORLD'S WORK

That saying "idle as the winds" must have started in the days when they didn't know; for if ever there was a busy people, it's the Winds.

Not only do they help plant the trees of the forest, sow the fields with grass and flowers, and water them with rain, but they make and carry soil all over the world. And, like everything else in Nature, they have a sense of beauty and the picturesque. Rock, for example, weathered away into dust by the help of the winds, as it is, takes on all sorts of picturesque shapes. And, of course, the winds love music; everybody knows that. Before we get through with this chapter we're going to end a happy day outdoors with a grand musical festival in the forest, with light refreshments—spice-laden winds from the sea. There'll be nobody there but the trees and the winds and John Muir and us; all nice people.

#### I. SUCH CLOUDS OF DUST!

March leads the procession of the dusty months because the warming up of the land, as the sun advances from the south, brings the colder and heavier winds down from the north. These winds seem to have a wrestling match with the southern winds and with each other, and among them they kick up a tremendous dust, because there's so much of it lying around loose; for the snows have gone, and the rainy season hasn't begun, and the fields are bare.

#### ABOUT THE DUST WE GET IN OUR EYES

Most people think these March winds a great nuisance because some of us dust grains are apt to get into their eyes; but dust in the eye is only the right thing in the wrong place. Just think of the amount of dust going about in March that *doesn't* get into your eye; and how nice and fine it is, and how mixed with all the magic stuff of different kinds of soil, thus brought together from everywhere.

An English writer on farming says he thinks the fact that English farms have done their work so well for so many centuries is due, in no small degree, to the March winds that have brought us world-travelled dust grains from other parts of the globe.

And the wind is a good friend to the good farmer, but no friend to the poor one; for it carries away dust all nicely ground from the fields of the farmer who doesn't protect his soil and carries it to farmers who have wood lots and good pastures and winter wheat, and leaves it there; for woods and pastures and sown fields hold the soil they have, as well as the fresh, new soil the winds bring to them.

Most of the fine prairie soils in our Western States owe not a little of their richness to wind-borne dust. In western Missouri, southwestern Iowa, and southeastern Nebraska are deep deposits of yellowish-brown soil, the gift of the winds. And, my, what apples it raises! It is in this soil that many of the best apple orchards of these States are located. And now, of course, the apple-growers see to it that this soil stays at home.

But there's another kind of dust that deserves special mention, and that's the kind of dust that comes from volcanoes. Volcanoes make a very valuable kind of soil material, often called "volcanic ash." It isn't ashes, really. It's the very fine dust made by the explosion of the steam in the rocks thrown

out by the volcano. The pores of the rocks, deep-buried in the earth, are filled with water, and when these rocks get into a volcanic explosion, this water turns to steam, and the steam not only blows out through the crater of the volcano, but the rocks themselves are blown to dust. This dust the winds catch and distribute far and wide. Sometimes the dust of a volcanic explosion is carried around the world. In the eruption of Krakatoa, in 1883, its dust was carried around the earth, not once but many times. The progress of this dust was recorded by the brilliant sunsets it caused. It is probable that every place on the earth has dust brought by the wind from every other place. So you see if you happen to be a grain of dust yourself, and keep your eyes and ears open, you can learn a lot, as I did, just from the other little dust people you meet.

#### THE WINDS AND VOLCANOES

But that isn't all of this business—this partnership—between the volcanoes and the winds. Did anybody ever tell you how the volcanoes help the winds to help the plants to get their breath? It's curious. And more than that, it's so important—this part of the work—that if it weren't carried on in just the way it is, we'd all of us—all the living world, plants and animals—soon mingle our dust with that of the early settlers we read about in the last chapter. In other words, all the *plant* world would die for lack of fresh air and all the *animal* world would die for lack of fresh vegetables. So they say!

According to that fine system—the breath exchange between the people of the plant and animal kingdoms—the plants breathe in the carbon gas that the animals breathe out; you remember about that. But the amount of carbon gas in the air is never very large, and if there were no other supply to draw on except the breath of animals and the release of this same gas when the plants themselves decay, we'd very soon run out.

Now this needed additional supply comes from the volcanoes. Every time a volcano goes off—and they're always going off somewhere along the world's great firing-line—it throws out great quantities of this gas, and this also the winds distribute widely and mix through the atmosphere.

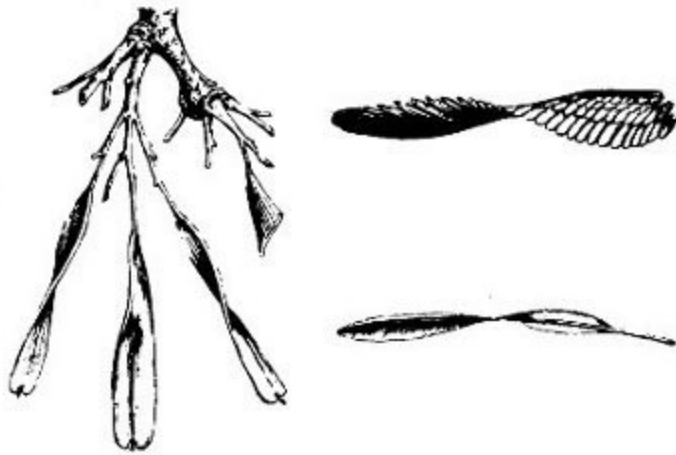
And another thing: This carbon in the air helps crumble up the rocks already made, and it enters into the manufacture of the limestone in the rock mills of the sea. This limestone will make just as rich soil for the farmers of the future as the limestones of other ages have made for the famous Blue-Grass region of Kentucky, for example.

All of which only goes to show how first unpleasant impressions about people and things are often wrong. A "dusty March day," you see, isn't just a dusty March day. It's quite an affair!

## **II. THE DUST MILLS OF THE WIND**

But wind is not alone a carrier for other dust-makers; it has dust mills of its own. The greatest of these mills are away off among the mountains and in desert lands, but after making it in these distant factories the winds carry much of this fresh new soil material to lands of orchard and pasture and growing grain.

Not long ago two of the professors at the University of Wisconsin found a good illustration of what an immense amount of soil is distributed in this way, and what long distances it travels. Among the weather freaks of a March day was a fall of colored snow that, it was found, covered an area of 100,000 square miles, probably more. The color on the snow was made by dust blown clear from the dry plains of the Southwestern States, a thousand miles away. The whole of this dust amounted to at least a million tons; and may even have amounted to hundreds of millions of tons, so the professors think.



#### TYPES OF NATURE'S SCREW PROPELLERS

You can see for yourself (from the picture on the left) that long before man ever thought of driving his ships through the water with screw propellers or pulling his flying machines through the air by the whirligigs on the end of their noses, some flying seeds, such as those of the ash here, had screw propellers of their own. And do you know that Nature also employs the propeller principle, not only in the operation of the wings of birds but in the wing feathers themselves? The two pictures on the right show the action of the wing and the wing feathers when a bird is in flight.

#### LITTLE MILLSTONES IN BIG BUSINESS

For grinding rocks to get out ore, or for making cement in cement mills, men use big machines, somewhat on the style of a coffee-mill. These machines are called "crushers." The winds, in their enormous business of soil-grinding, however, stick to the idea you see so much in Nature, that of using *little* things to do *big* tasks; as in digging canyons and river beds, and spreading out vast alluvial plains by using raindrops made up into rivers; in working the wonders of the Ice Ages with snowflakes; and building the bones and bodies of those big early settlers, and of all animal life, and the giant trees of the forest out of little cells. For, what do you suppose the winds take for millstones in grinding down the mountains into dust? Little grains of sand!



And with the help of the sun and Jack Frost it makes these fairy millstones for itself. The outside of a big rock grows bigger under the warm sun, in the daytime, and then when the sun goes down and the rock cools off it shrinks, and this spreading and shrinking movement keeps cracking up and chipping off pieces of rock of various sizes. Up on the mountain tops, among the peaks, the change of temperature between night and day is very great, and even in midsummer you can always hear a rattling of stones at sunrise. The heat of the rising sun warms and expands the rock, and so loosens the pieces that Jack Frost has pried off with his ice wedges during the night.

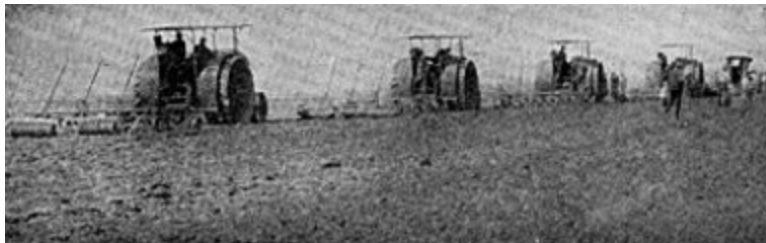
Then also during periods of alternate freezing and thawing in Spring and Fall, the rock is slivered up. These changes in the weather as between one day and another are due to the winds. In January and February, for example, thaws and freezes are common. When the winds blow from the south, the snow melts, water runs into cracks in the rock and fills their pores; then a shift of the winds to the north, a freeze, and the water in the crevices and the pores turns to ice, expands, and breaks off more rock.

And what muscles Jack has! Freezing water exerts a pressure of 138 tons to the square foot; so there's no holding out against him once he gets his ice wedges in a good crack. He sends huge blocks tumbling down the mountainside. The larger blocks, striking against one another, break off smaller fragments. The smallest fragments the wind seizes. Others are washed down by the rains. The largest, carried away by mountain torrents, bump together as they thunder along, and so break off more fragments and grind them so small that the wind can pick them up along the banks when the torrents shrink, or in their beds when these sudden streams go dry.

#### **RUNNING WATER AND THE WINDS**

In changing rock into soil, running water and the winds each have an advantage over the other. Water weighs a great deal more than air—over 800 times as much—and so grinds faster with its tools of pebbles and sand. The winds, on the other hand, get over a great deal more territory, and they, like the lichens, understand chemistry. Two of the gases they always carry right with them—carbon dioxide and oxygen—help decay the rocks.

As I said, the winds do most work in dry and desert regions, but when you remember that over a fifth of the globe is just that—dry as a bone most of the time—you see this is a great field. It has been so from the beginning, for it is thought probable that there was always about the same proportion of desert lands. Night and day the winds have been busy through all these ages. Dust is carried up by ascending air currents. Then the same force that keeps the earth in its orbit—gravity—pulls down on a grain of dust. But its fall is checked by the friction of the air. You see there's a lot of mechanics involved in moving a grain of dust; and Nature goes about it as if it were the most serious business in the world; handles every grain as if the future of the universe depended on it. In the case of sand or coarse dust, unless the winds are very strong, gravity soon gets the best of it, and down the dust grain comes to the ground again; then up with another current, then down again—carried far by stiff breezes, only a short distance by puffs—a kind of hop, skip, and jump. But fine dust getting a good lift into the upper currents at the start may stay in the air for weeks.



*Courtesy of The Dunham Company.*

#### TO KEEP MOISTURE AND SOIL AT HOME

In the broad fields of the West, where "dry-farming" is practised, they have these huge machines. They are called "Cultipackers." They are cultivators with big, broad-brimmed wheels that pack the surface of the soil after the blades of the cultivator have stirred it. This not only prevents the moisture in the soil from evaporating as fast as it would otherwise do, but keeps the winds from carrying away the soil itself.

In very wild wind-storms it has been figured out that there may be as much as 126,000 tons of dust per cubic mile; several good farms in the air at once,

over every square mile of the earth below!

### **III. THE STORM PLOUGHS OF THE WIND**

#### **TWO KINDS OF WOODEN PLOUGHS**

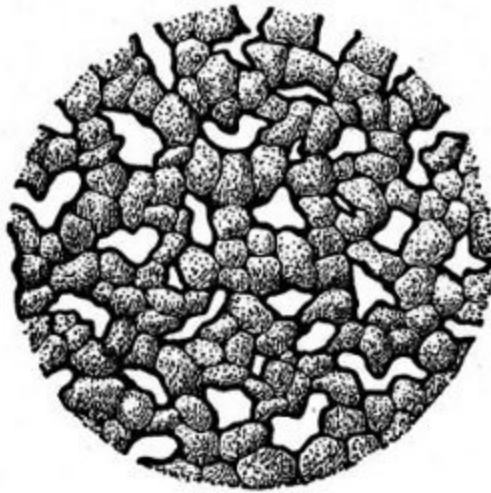
They use wooden ploughs, these winds, just as primitive man did, and as primitive peoples do now; but not quite in the same way, and the ploughing they do is much better. For man's wooden plough is a crooked stick made from the branches of a tree while the winds use the whole tree—roots and all, and both on mountainsides and on level lands the amount of ploughing they do is immense.

Almost all forests are liable to occasional hurricanes which lay the trees over thousands of acres in one immense swath. A large number of these trees, owing to their strong trunks, do not break off but uproot, lifting great sheets of earth. Soon, by the action of its own weight and the elements, this soil falls back. The depth to which this natural ploughing is done depends, of course, on the character of the tree, but as it is the older and larger trees that are most likely to be overturned, since they spread more surface to the wind, the ploughing is much deeper than men do with ordinary ploughs.

The result is that new unused soil is constantly being brought to the surface; and not only this, but air is introduced into the soil far below the point reached by ordinary ploughing. The soil needs air just as we do; for the air hurries the decay of the soil and its preparation for the uses of the plant. The immediate purpose of ploughing is to loosen the soil so that the roots of the plants can get their food and air more easily. It also helps to keep the fields fertile by exposing the lower soil to more rapid decay.

But here's the trouble: While the ordinary plough introduces air into the soil for a few inches from the surface, the subsoil, which is very important to the prosperity of the plant, is practically left out of it, so far as getting needed fresh air is concerned. The long roots of the trees that, among other things opened for it channels to the air, are gone. The burrowing animals that used to loosen up the earth, man has driven away. More than that, the foot of the plough which has to press heavily on the subsoil in order to turn

the furrow, smears and compacts the earth into a hard layer, which shuts out the air, and also—to a certain extent—the water from the lower levels.



#### HOW THE SOIL GETS ITS BREATH

Plants must have air to breathe, both above and below the soil, and the microscope is showing us here how a sandy loam allows the air to reach the roots.

In mountain regions these "storm ploughs," as we may call them, not only help to renew and prepare the soil in the valleys, but are a part of the machinery of delivery of new soil from mountain to valley. When trees on the mountainside are overturned, they not only bring up the soil, which the mountain rains quickly carry to the valleys, but the roots having penetrated—as they always do—into the crevices of the rocks, bring up stones already partly decayed by the acids of the roots. These stones, as the roots die, decay and so release their hold, and also go tumbling down toward the valley.

Consider how much of this storm-ploughing must be done in the forests of the world in a single year, and that this has been going on ever since trees grew big on the face of the earth. In a storm in the woods of California, Muir heard trees falling at the rate of one every two or three minutes. And, as I said, it is precisely the trees that can do the most ploughing—the older

and larger trees—that are most apt to go down before the wind. Younger trees will bend while older and stiffer trees hold on to the last. Before a mountain gale, pines, six feet in diameter, will bend like grass. But when the roots, long and strong as they are, can no longer resist the prying of the mighty lever—the trunk with its limbs and branches—swaying in the winds, down go the old giants with crashes that shake the hills. After a violent gale the ground is covered thick with fallen trunks<sup>[7]</sup> that lie crossed like storm-lodged wheat.

There are two trees, however, Muir says, that are never blown down so long as they continue in good health. These are the juniper and dwarf pine of the summit peaks.

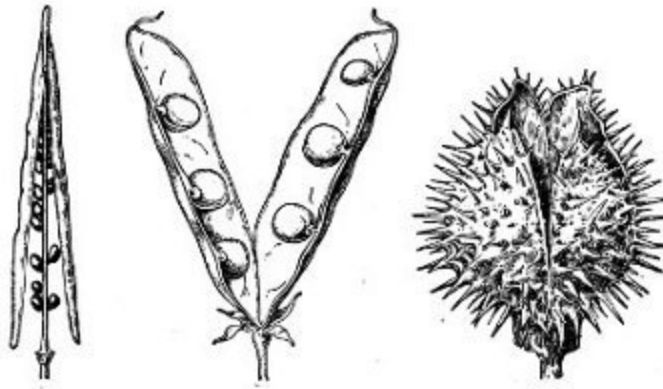
"Their stout, crooked roots grip the storm-beaten ledges like eagle's claws, while their lithe, cord-like branches bend round completely, offering but slight holds for winds, however violent."

#### **AT THE STORM FESTIVAL WITH MR. MUIR**

Trees were among Muir's best friends, and he spent a large part of his life chumming with them. What do you think that man did once? He was always doing such things. He climbed a tree in a terrific gale so that he could see right into the heart of the storm and watch everything that was going on. Just hear him tell about it:

"After cautiously casting about I made choice of the tallest of a group of Douglas spruces that were growing close together like a tuft of grass, no one of which seemed likely to fall unless the rest fell with it. Being accustomed to climb trees in making botanical studies, I experienced no difficulty in reaching the top of this one, and never before did I enjoy so noble an exhilaration of motion."

And such odors! These winds had come all the way from the sea, over beds of flowers in the mountain meadows of the Sierras; then across the plains and up the foot-hills and into the piny woods "with all the varied incense gathered by the way."



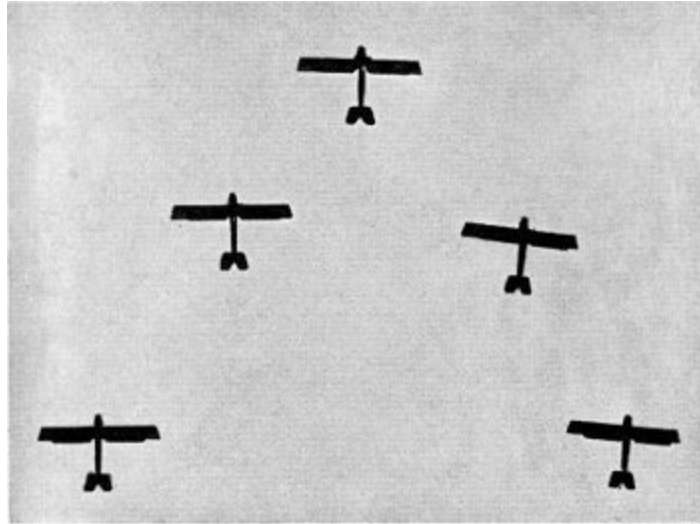
### THREE KINDS OF SEED THAT THE WIND SHAKES FREE

Here are three kinds of seed adapted for dispersal by the shaking action of the wind.

Though comparatively young, these trees—the one Mr. Muir climbed into and its neighbors—were about 100 feet high, and "their lithe, brushy tops were rocking and swirling in wild ecstasy." In its greatest sweeps the top of Muir's tree described an arc of from twenty to thirty degrees, but he felt sure it wouldn't break, and so he proceeded to take in the great storm show.

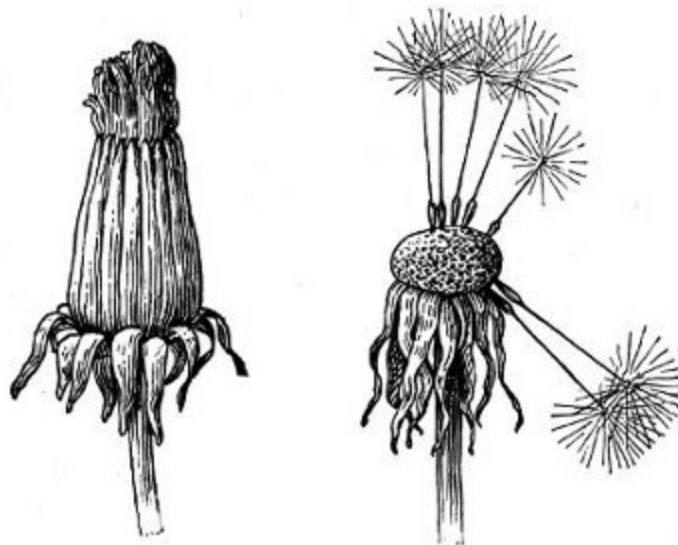
"Now my eye roved over the piny hills and dales as over fields of waving grain, and felt the light running in ripples across the valleys from ridge to ridge, as the shining foliage was stirred by the waves of air. Oftentimes these waves of reflected light would break up suddenly into a kind of beaten foam and finally disappear on some hillside, like sea waves on a shelving shore."

This was his impression of the forest as a whole, a dark green sea of tossing waves. But if we study trees as long and lovingly as Muir did, we can pick out the different members of the family a mile away—even several miles away—by their gestures, their style of grave and graceful dancing in the wind.



### TYPES OF FLYING MACHINE

Here is the type of flying machine that carries men. On the opposite page is the kind that carries the dandelion seeds.



### THE DANDELION-SEED FLYING MACHINE

The dandelion on the left shows how the seeds are kept in the "hangar" at night and on rainy days, shut up tight to prevent

them from getting wet with rain or dew and so made unfit for flying.

Muir especially mentions the sugar-pines as interpreting that storm to him. They seemed to be roused by the wildest bursts of the wind music to a "passionate exhilaration," as if saying "*Oh*, what a glorious day this is!"

This was the picture part of it—the glorious moving-picture show. Now listen to some of the music:

"The sounds of the storm corresponded gloriously with the wild exuberance of light and motion. The profound bass of the naked branches and boles booming like waterfalls, the quick, tense vibrations of the pine-needles, now rising to a shrill, whistling hiss, now falling to a silky murmur. The rustling of laurel groves in the dells, and the keen metallic click of leaf on leaf—all this was heard in easy analysis when the attention was calmly bent.

"Even when the grand anthem had swelled to its highest pitch I could distinctly hear the varying tones of individual trees—spruce, fir, pine, and oak—and even the infinitely gentle rustle of the withered grasses at my feet."

When the winds began to fall and the sky to clear, Muir climbed down and made his way back home.

"The storm tones died away, and turning toward the east I beheld the countless hosts of the forests hushed and tranquil, towering above one another on the slopes of the hills like a devout audience. The setting sun filled them with amber light, and seemed to say while they listened:

"My peace I give unto you."

## HIDE AND SEEK IN THE LIBRARY

Did you know that the ash and maple seeds actually have screw propellers, like a ship, so that they can ride on the wind? Pettigrew's



great work, "Design in Nature," makes this very plain, both in word and picture.

In what way does the wind help to *produce* the seed of grasses as well as carry and plant them? (Any encyclopædia or botany will tell you how plants are fertilized.)

How could a tempest that blew down a tree help its seeds to get a start? Wallace, in his "World of Life," says that on a full-grown oak or beech there may be 100,000 seeds that are thus given a better chance of life.

Speaking of "wind ploughs," what is the object of ploughing anyway? The article on preparing the seed bed in "The Country Life Reader" tells about what ploughing means to the soil and also:

Why good soil takes up more room than poor.

Why it is a good thing to plough deep, but a bad thing, if you don't do it just right.

And farther on there is a most inspiring poem about the history of the plough from the days of early Egypt to the present. It begins like this:

"From Egypt behind my oxen,  
With their stately step and slow,  
Northward and east and west I went,  
To the desert and the snow;  
Down through the centuries, one by one,  
Turning the clod to the shower,  
Till there's never a land beneath the sun  
But has blossomed behind my power."

The deserts have helped to make western China fertile. How did they do it? (Look at your geography map and remember that the prevailing winds of the world are westerly.)

You'll find many interesting things about the winds and the soil in Keffer's "Nature Studies on the Farm" and Shaler's "[Outlines of Earth's History](#)." Shaler's "Man and the Earth" says a single gale may blow away more soil from an unprotected field than could be made in a

geological age, and an hour's rain may carry off more than would pass away in a thousand years if the land were in its natural state. He also tells what to do to prevent the best part of ploughed fields from being carried off by the wind.

Have you any idea how far seed may be carried by a hurricane? Wallace, in his "Darwinism" deals with this question, and it's very important in the story of the earth. Beal's admirably written and illustrated little book on "[Seed Dispersal](#)." tells a world of interesting things about the wind as a sower. For instance:

How pigweed seeds are built so that wind can help them toboggan on snow or float on water;

How wind and water work together in the distribution of seeds;

About seeds that ride in an ice-boat;

About the monoplane of the basswood;

About the "flail" of the buttonwood, and how the wind helps it to whip out the seeds; and how the seeds then open their parachutes.

Dandelions go through quite a remarkable process in preparing for flight. I wonder if you have ever noticed it. Before the seeds get ripe Mother Dandelion blankets them at night and puts a rain-cloak on them on rainy days, and just won't let them get out, as shown on page 51. And do you know how she opens the flowers for the bees on sunshiny days?

There is no island, no matter how remote, that isn't supplied with insects. How do you suppose they get there? You may be sure the wind has something to do with it or I wouldn't mention the subject at the end of this chapter. (Wallace: "Darwinism.")

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### THE WEST WINDS AND THE RAINS

On the western slopes of this mountain the trees, with the help of the winds and the rain, climb to the very summit, while the other side of the mountain remains only a barren rock. The moisture-laden winds from the west glide up the slope, the air expands as it rises, the expansion cools it and down comes the rain! But the eastern slope gets little or none of it.

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## CHAPTER IV

(APRIL)

The higher Nilus swells  
The more it promises; as it ebbs, the seedsman  
Upon the slime and ooze scatters his grain,  
And shortly comes the harvest.

—*Shakespeare: "Antony and Cleopatra."*

### THE BOTTOM-LANDS

All that wind was bound to blow up rain. I said so at the time. And, sure enough, here it is; right where we want it, at the beginning of April, a month famous for its rains.

The work of the rains is going to make one of the most interesting chapters in the long story of the dust. At least I hope so. But don't think I intend to tell it all. Why, it would make a whole book in itself. But you can believe every single thing I do tell, no matter how it makes you open your eyes; for, if I've helped it rain once I've helped it rain a million times!

#### I. THE MARCH DUST AND THE APRIL RAINS

##### HOW RAIN GOES UP BEFORE IT COMES DOWN

It's this way: You remember how you can "see your breath," as we say, on a cold morning? Well, that's because the moisture in your breath is condensed by the cold. Now as the waters of the earth—the seas, lakes, rivers, ponds, and so on—are warmed by the sun, the air above them is filled with moisture, for the heating of the air causes it to expand and draw in moisture from the water like a sponge. Expansion makes it lighter also, and it rises. Rising, it turns cooler, and the moisture condenses and comes down as rain.

Mountains usually have clouds around them because moist air striking the mountainside is driven up the slope, cooling as it rises. So rain and snow fall often in mountain regions, and that's why so many rivers rise in mountains. The moist air is also condensed when it meets other and cooler air currents. But right here is where the work of the dust comes in. For to make rain you've got to have clouds, and clouds are due to this moisture collecting around the little particles of dust of which the air is full. When these little motes of matter become cooler than the air that touches them the moisture in the air condenses into a film of water around them. Fairy worlds with fairy oceans floating in the sky!

Each of these baby worlds is falling toward the big world below. But very slowly; only a few feet a day, so that even if nothing happened it might be months—yes, years—before it would come to the ground, even in still air. But when air is very thick with moisture the water films on these dust particles grow rapidly, and thus increasing in weight, they fall faster and faster, and finally strike the earth as raindrops.

But here's another thing that helps. On the way down two or more raindrops, falling in with each other, will go into partnership—melt into one—and then they hurry down so much the faster. That's why the sky grows darker and darker just before a rain, and why the lower part of a rain-cloud is the darkest: the little raindrops are forming into bigger raindrops as they fall.

#### **THE LITTLE ARTISTS THAT SHAPE THE CLOUDS**

But the shapes of clouds are supposed to be due to another thing, the mysterious force we call electricity, and that other mysterious force we call gravity. Just as the worlds attract each other by gravity so these raindrops—or dust grains growing into raindrops—are drawn toward one another. Here's where Electricity steps in. These rain particles are full of electricity and when two of these electrified particles meet in the air—unless they strike one another in falling, in which case, as I said a moment ago, they blend into one—they get very close together and yet keep dancing around one another without touching! It is this dancing about that makes all those strange and beautiful and ever-changing forms in the vast picture-gallery of the sky.

Of course the wind currents help to change these shapes, but I'm talking about the original designs.

## **II. THE RAINDROPS AND THE RIVER MILLS**

So much for the dust that helps make raindrops; now for the raindrops that help make dust. This the raindrops do in several ways. Falling on big rocks or decaying pebbles, for example, they pound loose with their patter, patter, patter, any little bits of soil and grains of sand that have been made by the other soil makers—the sun, the wind, the lichens, the chemists of the air, and so on. This soil and these sand particles, if there is already any depth of earth there, they carry down into the ground. Some of this soil, with various stops and mixings with other soils on the way, finally reaches the sea, where it helps to make the rich limestone soils for the Kentuckies of millenniums yet to be, by supplying food for sea creatures and lime for their shells. For these shells become limestone when the shell-fish are through with them. Mother Nature, in addition to feeding her big, hungry families of to-day in the plant and animal world, is always laying by something for the future. But before it gets back to the sea, by far the greatest part of the ground-up soil the rivers carry is spread out in the lowlands in those "alluvial plains" your geography tells about and that make a large proportion of the fertile farms of the world. If the raindrops fall on comparatively barren rock—in the mountains, say—they carry some of this fresh soil to the mountain valleys below, and some of it they may spread in bottom-lands a thousand miles away, where the new soil helps feed the plants. The sand grains in it not only help the soil to get its breath by making little air spaces, but these sand grains themselves slowly decay and so make more soil.



#### WHAT IRRIGATION DOES FOR DESERTS

It is such land as this, in the arid regions of the West, that irrigation converts from a desert to a garden of abundance. The soil is rich in all the substances that plant life needs.

But it isn't alone that they carry away the soil already made and bury the sand grains. Some of the raindrops soak into cracks in stones and dissolve the material that binds the rock particles together, and so get them ready to give way under the fairy hammers of the next shower that comes along.

After Nature finally gets an original waste of barren rock all nicely set with grass and flowers and trees and things, the raindrops help to make soil in still another way. Soaking through the decaying leaves, they pick up acids which are just the thing for eating into rock and crumbling it into soil. To be sure, the water soaking into the soil and coming out of springs carries some plant food away with it; but it takes it to lands farther down the river valleys, and more than makes up for what it carries away by the new soil made by its acids from the rocks, as it soaks into their pores and runs among the cracks.

#### HOW RAINDROPS MANAGE TO GRIND UP THE ROCKS

Moreover, raindrops actually grind up rocks. In order to do this a lot of raindrops have to get together, to be sure, and become rivers; but after all

it's the raindrops that do it. There'd never be any rivers if it weren't for the rains and, of course, the snows.

Well, anyhow, the rivers, besides running other people's mills, have mills of their own; and millstones. Most of these stones originally came from mountains and were brought into the milling business by mountain streams, with the help of Jack Frost. For the frost not only pries stones from the mountains and so sends them tumbling down the slopes, but it keeps edging them along and edging them along, farther down, after they have fallen. You'd hardly think that, would you? Yet it's simple enough. The water in the pores of the rock expands when it freezes and that makes the whole rock expand, for the time being. Then when the frozen water in the rock pores thaws out, the rock contracts, and this spreading out and pulling together, small as it is, causes the rock to keep hitching along down the incline; oh, say a fraction of an inch a year. But still, in the course of the ages, these inches foot up, and after a while this tortoise-like gait lands the stone—lands tens of thousands of such stones—in the beds of the mountain torrents that run along at the bottom of these inclines. There they get ground together and so grind out more soil material, particularly when the floods are on, with the melting of the snows in spring and the falling of the heavy and frequent rains.





#### AN OLD RIVER MILL

It used to do a lot of business—this old river mill. Its grist was ground-up rock that helped make fine farming land in the bottoms along the river's course. Such mills, called "pot holes," are found in the rocky floors of rapid streams, where the eddying current or the water of a waterfall wears depressions in the bed. Into these depressions stones are washed, and then by the whirl of the flowing water kept going round and round, grinding themselves away and grinding out the sides and bottom of the mill.

Another curious thing is how the river mills help themselves to new millstones when they need them. If a river hasn't enough for its work, it has a way of drawing on its banks for more. Whenever the stones in its bed get scarce, so that it can make comparatively little new soil—having so few stones to grind together—it proceeds to dig its own bed deeper, since this bed is no longer protected by a rock pavement in the bottom. This, of course, deepens its channel, and so adds to the steepness of the slope of its banks. Then, owing to this increase in the incline of the slope, more rocks tumble in, and the "milling business" picks up again.

#### THE GOVERNOR IN THE RIVER MILL

But there may be too much of a good thing; the rocks may come in faster than the river mill can take care of them. Then the river bottom becomes so completely paved over that the channel stops wearing down at all, to speak of, and the river remains at the same level until the rains and the wind and other workers have worn the banks down and lessened the incline. Then, with fewer and fewer fresh stones tumbling in, the river gets a chance to catch up with its work.

It is this ground-up rock stuff of the mountain river mills, made by the grinding of the running streams all the way down, that has helped form the rich bottom-lands of the Mississippi Valley. For uncounted ages, the water of the Mississippi and its tributaries have been at work, and by the time you get down into southern Louisiana you come to the delta where this rich soil has been piled up for more than 1,000 feet above the bottom of the old Mediterranean Sea, that used to reach north and south across the country.

You remember the lines, don't you:

"Little drops of water, little grains of sand  
Make the mighty ocean and the pleasant land."

Well, this is how they do it; all this that I've been telling you.



*Courtesy of the Scientific American.*

#### THOUSANDS OF FARMS POURED INTO THE GULF

The Father of Waters is a good farmer in some respects but needs training in others. The Mississippi's floods, like those of Father Nile, enrich the bottom lands, but the river is apt to break all bounds and do a lot of damage. Moreover, every year it carries away thousands of acres of good soil and pours it into the Gulf. How to teach the Mississippi to work in harness, as the Nile has been taught to do in recent years, is one of the problems which will require all of Uncle Sam's ingenuity and skill to solve. A good deal of the yearly waste could be prevented, however, by the various means employed by good farmers.

### III. HOW THE RIVERS ACT AS BANKERS FOR THE FARMERS AND THE SEA

We speak of river banks and the kind of banks that handle those promissory notes our arithmetics tell about as if they were entirely different; and so they are, I suppose, if one just looks at the surface of the thing. But if we dig into the subject a little we shall see that they are much alike in the fact that one of the principal businesses of both kinds of banks is to make loans at interest. Men's banks loan money, to be sure, while the river banks loan pebbles, but if it were not for these pebble loans there would be a mighty sight less money for the banks to loan, or the farmer to borrow; and the way both banks do business ought to be a good lesson to certain farmers I know, who seem to think they can always be cashing checks on their banks—the farm lands—by hauling away the crops without ever putting anything back.



#### WHERE THE RIVERS ACT AS BANKERS

Here is a fine piece of bottom land, one of those "banks" where the rivers keep "checking accounts" for the farmers and the sea; using pebbles for currency, as explained in this chapter.

#### HOW THE RIVERS PLACE PEBBLES ON DEPOSIT

The rivers make loans to the soil by depositing pebbles in the broad bottom-lands along their banks, and then draw interest by carrying along to other lands, from time to time, some of the fine rich soil these pebbles help make by their decay. And the river does this in regular banking style, "checking out" the pebbles from time to time, and then depositing other pebbles in

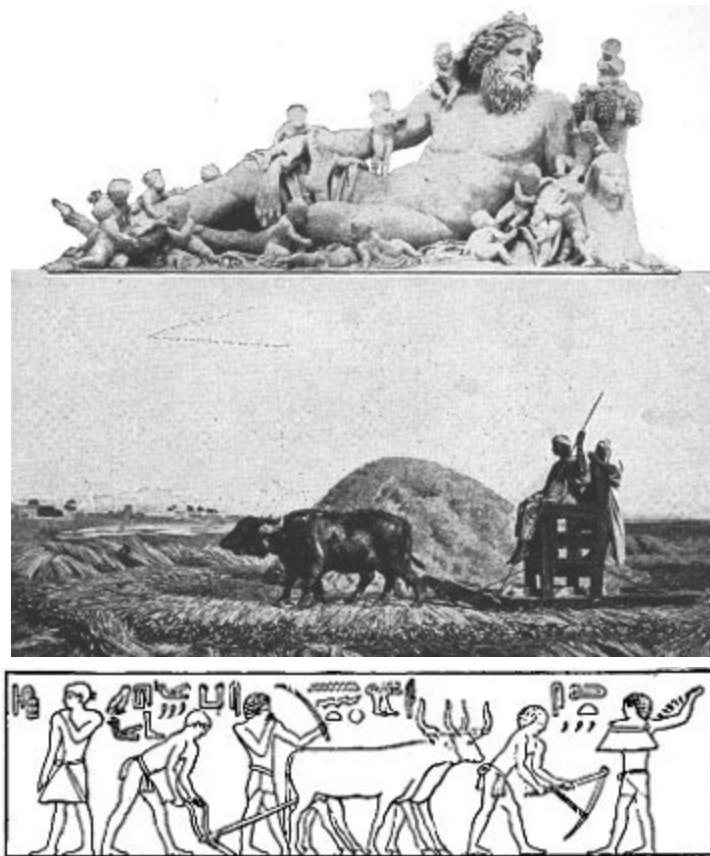
their places. Take the banks and bottom-lands of the Mississippi River, for example. It has been estimated that it requires about 40,000 years for a pebble to make the journey to the Gulf from the mountains of a tributary stream where it was first broken from the rock as a sharp fragment.

The first part of the journey in the mountains is over steep down grades, and so is comparatively fast, but as the river gets farther from the mountains, the slope of its bed becomes less and less, the onward movement is slower and slower, and more of the pebbles stop to rest. In times of flood they are carried far away from the regular channel and spread over the wide flood-plain of the river. Then, as the flood goes down, they are left buried there under a coating of mud. So buried, they decay and enrich the soil. Then the next flood that comes along sweeps the pebbles with it—checks them out of the bank—but at the same time carries away not only some of the soil richness which these pebbles helped to make but the soil material made by the decay of the vegetation these pebbles thus helped to grow, such as the roots and blades of wheat and corn and stubble and chaff left in the fields. That's the interest on the loan. Then, when the flood subsides, the pebbles are again deposited farther along in the river's course, but meanwhile the same flood has brought fresh deposits of pebbles from up-stream, and these are left in place of those taken away.

#### **RIVER BANKING AND HUMAN CIVILIZATION**

This banking business has been going on for ages and is a very important part of the history of civilization. Here and there along the sides of the older and larger river valleys are found the remains of ancient plains. These plains are now, many of them, quite a distance above the level of the stream. This means that they were at one time the bottom-lands of that same stream, but the stream, as it dug deeper and deeper into its bed, grew narrower, and so abandoned its old flood-plains. As savage man gradually settled down and took to farming, he found these bottom-lands, with their rich, mellow soil, just the thing for his crooked-sticks and stone hoes—the only kinds of ploughs and hoes there were in those days. With such crude farming tools he couldn't have managed to scratch a living on any other kind of soil. When the river floods came along, all these crooked-stick farmers had to do was to keep out of the way until the floods went down,

and there were their fields all fertilized for them, as good as new, and they could go on for thousands of years working the same fields without ever bothering their heads as to whether they needed any lime or potash or nitrogen, or anything; for they didn't. The river floods attended to all that.



#### FATHER NILE AND THE MAKING OF EGYPT

"Egypt," said Herodotus, "is the gift of the Nile"; and it is true so far as her fertile lands are concerned. The ancients attributed the annual floods to the god of the Nile, as shown in that statue of Father Nile in the Vatican. Below is a threshing scene in Egypt painted by Gerome. The last picture, from a carving in the tomb of an Egyptian noble, shows how they ploughed and sowed in the Pyramid age.

So, in course of time, civilizations such as those of Egypt and India and Persia grew up, and in further course of time these civilizations spread into

Europe, and finally to the New World.

#### **HOW RIVER BANKS GO BANKRUPT**

Now all this is very well, this leaving it to Nature to fertilize the fields, where everything is just right for it, as it is along the Nile, but in most lands it won't do it all. The trouble is that, in raising the grain foods, the ground must be kept free of grass and weeds, and well ploughed during the rainy season. But the same rains that water the fields wash more or less good soil into the streams; much more than Nature alone can put back. For instance, down in Italy where, if the old forests were still there, the rains wouldn't wash away more than a foot of soil in 5,000 years, this soil is being carried into the Po, and by the Po emptied into the sea so fast—a foot in less than 1,000 years—that if you visit Italy to-day, say, and then go back in ten years, you'll see bare rocks on many a hillside that is now clothed in green. On such rocks the soil is already thin, and in ten years more it is all gone; all washed away! This thing is going on all around the shores of the Mediterranean. You are constantly coming on sections of country that used to be covered with great forests and prosperous farming communities where the soil has vanished, and many stretches of barren, rocky land where hardly a weed can find a foothold.



#### **WHAT HAPPENS TO THE LAND WHEN THE TREES ARE GONE**

Could anything be more desolate? You can see from this example how vital to our national life is the forest conservation

work of our government. Trees, by the network of their roots, keep the soil from washing away, retain moisture by their shade, and absorb the water of the rains and the melting snows so that it reaches the rivers and the creeks gradually. But when the trees are gone the water, unchecked, rushes down the slopes in floods, washing away the precious soil and leaving them as barren as a desert.

"But, what are you going to do about it?" you say. "You can't change the slope of the hills, can you? And the farmer has *got* to plough his land—you just said so yourself."

Yes, he's got to plough his land, to be sure; but so has he got to have pasture for his live stock. If he hasn't any live stock, that just shows what kind of a farmer he is. Every farmer ought to have live stock. Corn always brings a great deal more when it goes to market "on four feet," as the saying is; and, besides, the live stock give back to the fields, in the shape of manure, a large part of what they eat. Now, if you have live stock you must have pasture, and all land with a slope of more than one foot in thirty should be used partly for pasture and partly to grow wood for the kitchen stove, and hickory-nuts and walnuts for winter firesides. Although the land slopes, the mat made by the grass roots will keep it from washing away.

"But suppose you lived where there wasn't any land to speak of that didn't tip up; in New England, say—what would you do then?"

Leave the upper part of the slopes in the woods. Then the water that carries off the soil will not run entirely away, as it does in ploughed fields, but will creep down slowly, and, charged with the decay of the woods, help fertilize the lower lands and change the rocks beneath them into soil—the acids from the decaying vegetable matter eating into them.

"But still," you say, "there are farm lands that must be ploughed even if they do wash away; they're all the land a man has, sometimes. What then?"

Plough deep. Then the soil soaks up more of the rain and lets the water pass away in clear springs. This not only saves soil but, as we have just said, helps to decompose the subsoil and the bed rock.

Then there's another thing that good farmers do in such cases. They plough ditches along the hillside leading by a gentle slope to the natural watercourses; so the water of the rains, instead of going down the hills with a rush, and going faster the farther it runs—like a boy on a toboggan—is caught and checked in these sloping ditches, and much of the soil it contains deposited before it reaches the streams.



#### HOW THE FRENCH PROTECT THEIR HILLSIDE FARMS

This is how the French peasant keeps the mountain torrents from carrying off his precious soil.

The best way of all, of course, is to build terraces, as they do in the thickly settled parts of Europe. But this is only profitable for the more valuable crops and not for ordinary grains.

**SUCH SPENDTHRIFTS OF GOD'S GOOD SOIL!**



My, but it's a shame the way we've wasted soil in this country. What spendthrifts! To start with—when the country was first settled—there seemed no end to the fine land, and every one could have a good farm for the asking. All he had to do was to make his wants known to Uncle Sam and then go out and help himself. What happened then? Why, what always happens? Easy come, easy go. These pioneer farmers worked their farms for all there was in them; didn't bother, many of them, even to haul the barn manure into the fields. Then when the old farm was exhausted they moved off to new lands and did the same thing over again.



A HOME IN THE DESERT

Doesn't look much like a home in the desert, does it? But it is—a lovely home in what the old geographies called "The Great American Desert." In the Sahara oases are few and far between, but modern irrigation engineering makes oases to order—thousands and thousands of acres of them!

They ploughed on steep hillsides; they allowed gulches to form, as they will quickly do on sloping ploughed land, if you don't watch out; they cut away

the timber. It's easy in a hill country like the eastern part of the United States to have all the good top-soil washed away in twenty years after the forests have been destroyed; the good soil that it probably took 2,000 years to make.

Doctor Shaler<sup>[8]</sup> estimated that in the States south of the Ohio and the James Rivers more than 8,000 square miles of originally fertile land had, by this shiftless and thoughtless way of doing things, been put into such a state that it wouldn't grow anything; and over 1,500 square miles of this, actually worn down to the subsoil, and even to the bed rock, so that it may never be profitable to farm again—at least not in our time—no matter what they do!

I knew a farmer with a small son to whom he intended to leave the farm when he grew up, who did things like that for twenty years. By the time the little boy was old enough to vote, there was no farm to leave; all the good part of it was gone.

Serious thing for that little boy, wasn't it?

## HIDE AND SEEK IN THE LIBRARY

What have burrowing animals to do with the drainage system of the land? (Keffer's "Nature Studies on the Farm.")

How do angleworms help drain the soil?

How do the forests help make good use of the rain that falls, not only for themselves but for the rest of us?

How do the rains help to warm the ground in the spring? The heat they carry into the soil is produced in two ways. The book mentioned above tells of one of these ways, and Russell's little book, "The Story of the Soil," tells of another.

Beale's "[Seed Dispersal](#)." tells how the raindrops (working together, of course) help plant maple, elm, sycamore, willow, and other trees that grow by the waterside, to scatter their seeds.

You'd be surprised what a series of adventures the seeds of a bladderwort have before they get planted on some new shore, after

having left the parent shrub. First, they float down-stream, as you know, but when autumn comes on, what do you suppose they do? They go to bed. Where? Right in the bottom of the stream. Then how do they ever get up and get planted on the shore? Well, you just look it up in that Beale book and see.

Do you know how the rains help to get the mineral food up into the plant?

And why swamps are such poor producers?

And how the sun acts as a pump for the plant world?

You will find answers to all these questions in Shaler's "[Outlines of Earth's History](#)." and in your books on botany and agriculture.

Russell's book on the soil tells how the ancient Gauls and Britons used to fertilize their land with marl, and how the tides help to fertilize England. It's just the reverse of the way Father Nile looks after Egypt, as you will see.

If you want to read an interesting description of the difficulties of farming on wet lands, you will find it in this meaty little book.

If you don't know how serious a thing it is to let gullies form in land, look it up in Shaler's "Man and the Earth" and you will see.

How do you suppose deserts that get so little rain themselves could *help make it rain* in other places? For example, the desert of Thibet is the chief cause of the monsoon rains that do so much for India. That part of your geography that explains the circulation of the air will help you figure this out; particularly with a map under your eye that shows the relative location of the desert and the Indian Ocean, over which the monsoon winds blow.

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#### AN EXAMPLE OF MAN'S DEBT TO THE EARTHWORM

Much of the earth's Maytime bloom and beauty is due to the labor of our humble little brother of the dust, the earthworm; a striking fact which was never recognized until the great Charles Darwin looked into the matter and wrote a book about him. This picture by Millet is called "Springtime" and hangs in the Louvre, in Paris.

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## CHAPTER V

(MAY)

It may be doubted whether there are many other animals which have played so important a part in the history of the world as these lowly organized creatures.

—Darwin: "[The Formation of Vegetable Mould.](#)"

### WHAT THE EARTH OWES TO THE EARTHWORM

Suppose father had a hired hand who would plough his fields, fertilize them at his own expense, build his own house, board himself, and for all this ask only the privilege of living on the place, studying Botany, Geology, and Geometry, and enjoying the scenery.

"Where can I get a man like that?" I imagine father saying.

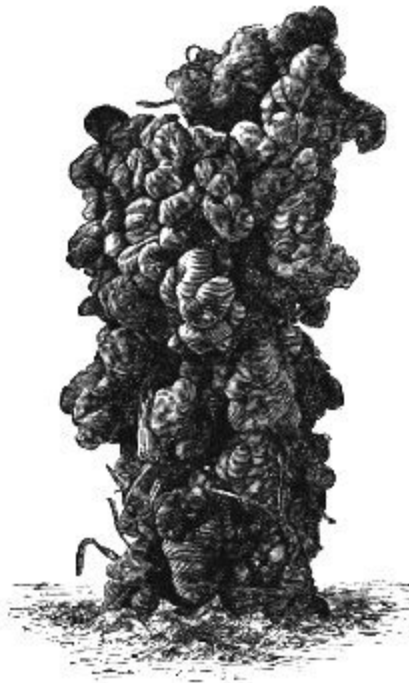
"You've got him now," you might reply. "He's already working for you—thousands of him, and has been working for you—millions of him—for thousands and millions of years."

We have all known him well from boyhood by several names—angleworm, fishworm, earthworm. He also, as you will find in the dictionary, has a nice long Latin title. And it is particularly fitting that his name should be so associated with antiquity, since he belongs to one of the oldest families in the world; a family far older than the Roman Empire itself, which his people long ago helped grind back into the dust from which it came.

And, speaking of Romans, every few years Mr. Earthworm does what Julius Cæsar did, captures the whole of England—all the best parts of it—and then, unlike Cæsar, gives it back to the English, made over again, better than it was before, as you will see.

## I. THE CITIES OF WORMS

If you happen to be a high school boy you, of course, know about a certain city of Worms and what great things took place there once upon a time, but there are many cities of worms on any good farm, and each has more inhabitants than the famous city of Worms of history—something like 25,000 to the acre; and, in garden soil, 50,000!



ANOTHER "CATHEDRAL OF WORMS"

In the story of the Reformation in your history you will read of a certain Cathedral of Worms and what took place there once upon a time. Here is a "cathedral of worms" as interesting to the student of nature as that famous edifice is to the historian and the architect. It is the tower-like casting of a big earthworm and was found in the Botanic Garden at Calcutta. The picture is "life-size."

Did you ever notice how big boulders in a field are frequently sunk into the ground as if dropped from a great height? It is the earthworms that help sink

them in the course of their soil-making. They like the moist shelter of the stones and burrow under them. Finally the weight of the stones crushes the burrows, and so the stones sink down.

#### PIONEER LIFE AMONG THE EARTHWORMS

Poor soil, as every boy knows, is a poor place to look for fishworms. But you have noticed that the mounds the worm throws up on such soil are larger than those on rich soil. The reason is that the soil, being less nutritious, the worm must eat more of it and, in so doing, pulverizes and fertilizes it. But a menu of earth alone not being to the earthworm's liking, undesirable regions have fewer of these farmers working underground; and this, for the same reason that these regions are sparsely settled on the surface—it is so hard to make a living.

So the earthworms may be said to have a decided taste in landscape. They don't care for desert scenery like Gerome's picture of the lion's big front yard,<sup>[9]</sup> but they are very fond of orchards where the soil is rich and leaves are plenty. The pathways artists are fond of putting in landscapes would also probably attract the eyes of earthworms—if they had any, for the worms prefer soil a little packed, as it is in pathways, because it makes more substantial burrows. And, singularly enough, the worms also like most the very thing that the artist emphasizes to lead the eye into his picture—the border lines that *define* the path. It is along the edges of a pathway that you find most worms.



*Painted by F. O. Sylvester.*



*Painted by Westman.*

### THE EARTHWORM'S TASTE IN SCENERY

Two features common to both these pictures—the trees and the pathways—appeal to earthworms as well as artists, for reasons you have learned in this chapter.

The earthworm, in addition to working over and fertilizing the soil already made, actually helps make soil out of rock. He does this in two ways: (1) With acids—for, like the Little Old Man of the Rock, he is a chemist; (2) by grinding up rock in a little mill he always carries with him.

### HOW THE EARTHWORM COOKS HIS MEALS

The earthworm's favorite diet is leaves and he has a way of cooking them. It is not quite like our way of cooking beet or dandelion leaves, but it answers the same purpose—it partially digests them. In glands, in his "mouth," he secretes a fluid which, like our saliva, contains an alkali. But the



earthworm's alkaline solution is much stronger, and when he covers a fresh green leaf with it—as he is usually obliged to do in Summer when there are so few stale vegetables, the kind he prefers, in his market—the leaf quickly turns brown and becomes as soft as a boiled cabbage.

Of course, there are always dead leaves in the woods, and these, which even the cow with her fine digestive outfit cannot handle, are a delight to the earthworm; for he also has a much larger supply of pancreatic juice than the higher animals, and this takes care of the leaves after he has swallowed them. He swallows bit by bit; just like a nice little boy who has been taught not to bolt his food.

The acids in the earthworm's "stomach," acting on the leaves, help make other acids which remain in the soil after it has passed through the earthworm's body and help dissolve those fine grains of sand which make your bare feet so gritty when mud dries on them. And, not only that, but this coating of soil lying upon the bed rock hastens its decay; for the earthworm's burrow runs down four to six feet, sometimes farther.

Besides the soil he thus grinds up and fertilizes so well with leaf-mould—what your text-book on agriculture calls "humus"—the earthworm does a lot of useful grinding in connection with the building of his house. He begins, as we do, by digging the cellar; but there he stops, for *his* house is *all* cellar! He makes it in two ways: (1) By pushing aside the earth as he advances; (2) by swallowing earth and passing it through his body, thus making the little mounds you see on the surface.

#### THE EARTHWORM SYSTEM AT PANAMA

A principle similar to his swallowing operations is frequently employed in engineering; as in making the Panama Canal, where dredging machinery dug out swamps and pumped the mud through a tube into other swamps to fill them up and help get rid of the mosquitoes.

In pushing the earth away the worm uses the principle of the wedge, stretching out his "nose"—as you have often seen him do when crawling—and poking it into the crevices in the ground; much as the wheat roots poke *their* little noses through the fertile soil the earthworm makes.

And, as in human engineering and the work of the ant, the earthworm doesn't throw the dirt around carelessly. He casts it out, first on one side and then on the other; using his tail to spread it about neatly.

#### **THE TILING IN THE EARTHWORM'S HOUSE**

The walls of the earthworm's house are plastered, too. At first they are made a little larger than his body. Then he coats them with earth, ground very fine, like the clay for making our cups and saucers, and for making the beautiful white tiling on the walls at the stations of a city subway. When this earthworm "porcelain" dries it forms a lining, hard and smooth, which keeps the earthworm's tender body from being scratched as he moves up and down his long hallway. It also enables him to travel faster because it is smooth, and it strengthens the walls.

The burrows which run far down into the ground, as all finally do toward Autumn, end in a little chamber. Into this tiny bedroom the worm retires during the hot, dry days of August and there he spends the Winter—usually with several companions, all sound asleep, packed together for warmth.

#### **AND RUGS ON THE FLOORS!**

Sometimes the Summer and Winter residences are quite ambitious, several burrows opening into one large chamber and each tunnel having two, sometimes three, chambers of its own—like a fashionable apartment with its main reception-room, and still more like the central sitting-rooms in Greek and Roman palaces. And the earthworm seems even to have some idea of mosaics, for it is the general practice to pave these chambers with little pebbles about the size of a mustard-seed. This is to help keep the worm's body from the cold ground. In addition to the mosaic floors the earthworms have rugs with lovely leaf patterns like the Oriental rugs that are so highly prized; and, as in the case of genuine Oriental rugs, no two patterns are alike. These rugs are leaves which the earthworm drags into his burrow, not for food but for house furnishing. When used for house furnishing they are placed in the entrance-hall; that is to say, they are used to coat the mouth of the burrow to prevent the worm's body from coming in contact with the ground. The mouth of the burrow, of course, is just where it is coldest at night in the Summer, the time of year when the earthworm

spends a great deal of his time in the front of his house. The surface of the earth, you know, cools very rapidly after sunset and the dew on the grass in the morning is so cold it makes your bare feet ache. The worm requires damp earth around him because he breathes through his skin and must keep it moist, but at the same time he is sensitive to cold.

And to drafts. Ugh!

#### **PEBBLE-FORT DEFENSES AGAINST THE FOE**

So he is very careful to keep the front door closed. This he does by stopping it up with leaves, leaf stems, and sticks. He also protects the door with little heaps of smooth round pebbles; but these pebbles are of a larger size than those he uses for paving the floor of his chamber. Besides helping to keep out drafts these pebbles serve another purpose. As our ancestors, the cave-builders, barred the door with boulders to keep out bears and other unwelcome callers, so the earthworms are protected by the pebbles, to a certain extent, from one of their enemies—the thousand-legged worm. Because of these little forts, the earthworms can remain with more safety near the doorway and enjoy the warmth of the morning sun. (So we might have reproduced Corot's "Morning" as a kind of landscape the earthworm enjoys!)

## **II. THE MIND OF THE EARTHWORM**

From all of which you can see the earthworm, for what small schooling he gets, is a very bright boy! If we were as bright, according to our opportunities, we would probably have answered long ago such puzzles as the question whether there is really anybody at home in Mars, how to keep stored eggs from tasting of the shell, and other great scientific problems of our day.

#### **WHERE MR. EARTHWORM KEEPS HIS BRAIN**

Just as we have little brains in the tips of our fingers, the earthworms have brains in the ends of their "noses." They have neither eyes nor ears, but, like that wonderful girl, Helen Keller, they make up for the lack of these senses,

to a remarkable degree, by the development of the sense of touch. They acquire quite a little knowledge of Botany, for example. They not only know that leaves are good to eat, but they know which is the "petiole" and which is the "base." They always drag leaves into their burrows by the smallest ends, because this makes it easier to get them through the door. And it is not by mere instinct that they do this. Supply worms with leaves of different form from those which grow in the region where they live, and they will experiment with them until they find just the best way in which to pull them into the burrows. After that they will always take hold of them so, without further experiment. That is the majority of them will do this; for earthworms are like other little people—all of them are not equally ambitious or studious.

And the earthworm also knows something about Geometry. Cut paper into little triangles of various shapes and pretend to the worms that they are leaves by scattering them near the mouths of the burrows. Then remove the leaves with which the burrows are stopped. The worms will pull in the slips to close the door and they will—most of them—take hold by the apex of the triangle because that is the narrowest point.

#### **THE EARTHWORM'S TASTE IN MUSIC**

So you see the earthworm is a very cultivated country gentleman with his knowledge of Botany and Geometry, and his taste for landscape. But this is not all. He also has opinions about music. There are certain notes that apparently get on his nerves. Put worms in good soil in a flower-pot, and some evening when they are lying outside their burrows set the pot on the piano and strike the note C in the bass clef. Instantly they will pull themselves into their burrows. They will do the same thing at the sound of G above the line in the treble clef. Although they cannot hear, they are sensitive to vibrations, and these are carried from the sounding-board of the piano into the pot. They are less sensitive when the pot itself is tapped. The music seems to go right through them.

#### **WHY THE EARLY BIRD GETS THE WORM**

Except in rainy weather worms ordinarily come out of their burrows only at night. By early morning they have withdrawn into their holes and lie with

their noses close to the surface to get the warmth of the morning sun. Then the early bird gets *them*! The reason a robin cocks his head in such a funny way—like a lord with a monocle—just before he captures a worm, is not because he is *listening*, as many people think; for the worm isn't saying a word and he isn't moving, and wouldn't make a bit of noise if he did move. The robin's eyes are on each side of his head and not in the middle of his face like ours, so he must turn his head in order to bring his eye in line with the hole where he sees the tip of Mr. Earthworm's nose.



### THREE EARLY BIRDS. FIND THE THIRD

Don't they look happy—these two tow-heads? They are evidently going fishing in the early morning. Another early bird—several of him—that we are saying a good deal about in these pages is to be found in the can. Still another, the one at the bottom of the page, is taking advantage of the earthworm's family habit of warming his "nose" in the early sun rays.



And many people also believe that earthworms come down with the rain. Even park policemen believe it. At least, one said to me, in Central Park:

"In dhry spells ye won't see wan. But let there come a little shower an' th' walks and the dhrives will be covered wid them; like the fairy stones that fall wid the rain in the ould counthry."

#### **DO EARTHWORMS COME DOWN WITH THE RAIN?**

The reason you see so many worms after a rain is that earthworms like moisture, and the rain seems to make them feel particularly good and breed a spirit of adventure. So out of their holes and away they go! A rain is their shower-bath; and you know how a shower-bath makes you feel. The mornings when the earthworms are apt to be thickest are those following a comparatively light rain in early Spring when the worms have recently awakened from their long Winter nap. With the beginning of the rainy season in the Fall, the worms also do a good deal of travelling into foreign lands, but in both Spring and Fall you will usually find more worms after a light shower than after a long, heavy downpour. If the worms were drowned out it would be the other way around, don't you see?

To be sure, you will often find dead worms in shallow pools by the roadside; particularly after Autumn rains. These are sick worms and the chill was too much for them. But it's remarkable how low a temperature a good husky angleworm can stand. A professor in the University of Chicago, near which I live, tells me he has often found the ground in the neighboring park covered with worms after November rains when his hands, and those of the students who were helping him gather them for study, were numb with the cold.

And how much work do you suppose these farmers do in grinding up and fertilizing the soil? In many parts of England the whole of the best land—the vegetable mould—passes through their bodies every few years, and they are doing similar work all over the world.

They not only fertilize the earth by mixing it with the leaves they eat and those that decay in their burrows, but their castings help to bury fallen leaves and twigs and dead insects, and they also bring up lower soil to the

surface, thus increasing its fertility. And by loosening the soil they let in more air. Remember that roots, like people, must have air.

### III. THE MILL OF THE EARTHWORM

For the grinding up of the earth and the leaves, the earthworm has, as I have already said, a little mill that he always carries with him. Do you know what a gold mill is? Well, a gold mill is a mill that grinds up rock and so grinds out the gold. The earthworm's mill, in a manner of speaking, also grinds out gold, for it grinds the little particles of stone in the soil, and this soil grows fields of golden grain.

The earthworm's mill is his gizzard. This gizzard is made and works very much like the gizzard of the chicken. And like the chicken the earthworm swallows little stones to help his digestion. So these stones, too, are ground into soil.

Like the chicken's gizzard the gizzard of the earthworm is lined with a thick, tough membrane, and it has muscles—such muscles! There are two sets of these muscles and they cross each other somewhat like the warp and woof of the cloth in your clothes. The muscles that run lengthwise are not so very strong, for all they have to do is to help the earthworm swallow, but the muscles that run around the gizzard are wonderfully strong. They are about ten times as thick as the other muscles. One of Mr. Earthworm's French biographers<sup>[10]</sup> calls these muscles "veritable armatures"; that is, freely translated, "veritable hoops of steel."

I said, in the second paragraph above this, that worms swallow grains of sand and stones to help their digestions, as chickens do. But the earthworm saves time, for he takes the stones with his meals; just as some Englishmen, fat old squires, when they get along in years, or for any other reason are a little weak in their digestive regions—keep pepsin on the table with the pepper and salt.

And—believe it or not—the earthworm actually makes his *own* millstones sometimes! The chalk in the chalky fluid of the glands that help him digest his meals frequently hardens into little grains in grinding the food. It's

almost as if the saliva in our mouths, in addition to acting directly on the food, also made a new set of teeth for us!

Suppose we had a stomach like the earthworm, wouldn't it be fun? We could digest the biggest dinners at Thanksgiving and Christmas and picnics and birthdays. We could even eat apples without waiting for them to get quite ripe. Haven't you done it to your sorrow? And no stomachache and no mince-pie nightmares!

#### WHY THE EARTHWORM NEVER HAS NIGHTMARES

By the way, the earthworm, although he has his troubles like the rest of us, never *has* nightmares. For one thing he has that stomach<sup>[11]</sup> and—a still better reason, perhaps—he never sleeps at night. Like the moths and the bats and the burglars and members of Parliament, he makes night his busy day.

And, in other ways, while he is so much like the rest of us worms of the dust, his life differs from that of most people. For instance, he not only works by night while we work by day, and works underground while we work on top, but he takes his vacation in the Winter while we take ours in Summer. In that respect Mr. Earthworm is like the millionaires at Palm Beach; for in Winter he, too, goes in the direction we call south on the map—that is to say *down*.

But, as you say, it takes all kinds of people to make a world; including earthworms and millionaires!

#### HIDE AND SEEK IN THE LIBRARY

Who was that in Mother Goose that went a-fishing "for to catch a whale"? Anyhow, there are fishworms so big that one might suppose they were made for catching whales. How long do you suppose they are, these big fishworms? A foot?

Pshaw! We have fishworms of our own a foot long. Two feet? More. Three feet? More. You look it up in the article on the earthworm in the "Britannica."



And how many kinds of earthworms do you suppose there are? You will be surprised to learn.

Also, you will find that the earthworms have relatives who live in the water all the time.

The article in the "International" tells why these modest neighbors of ours don't come to the surface in the daytime. That will be an interesting thing to know. Don't you think so?

And did you ever count an earthworm's rings? Other scientists have. (All live boys and girls are scientists; they want to *know*.) Try counting the rings of an earthworm and then compare your figures with those given in the article in the "International."

How many hearts do you suppose an earthworm has? You will find in the "International's" article they have a good many of what are sometimes called "hearts," and how different the earthworm's circulation system is from ours.

Does our saliva do for us anything like what it does for the earthworm; and our pancreatic juice?

Compare the earthworm's method of digging his subway with that of the armadillo. How do they differ in the way of using their noses?

Do you know how men dig subways; like those under New York City and Boston, for instance? Books that tell about this phase of human engineering and tell it in a very interesting way are "On the Battle-front of Engineering" ("New York's Culebra Cut") and "Romance of Modern Engineering" ("City Railways"), "Travelers and Traveling" ("How Elevated Roads and Subways Are Built").

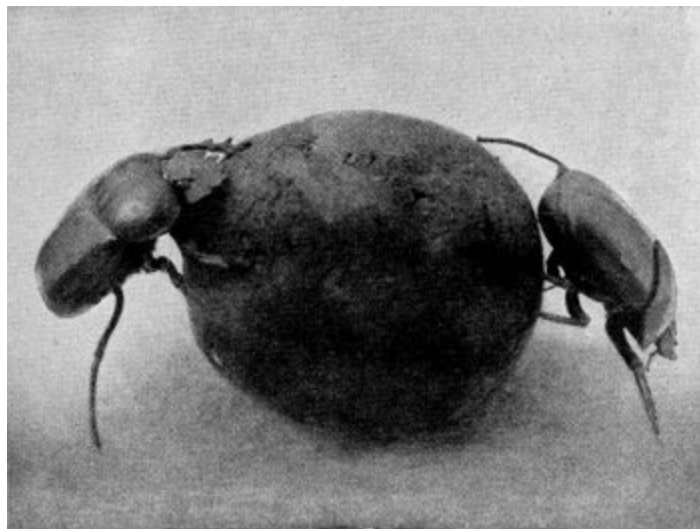
Speaking of the earthworm's wedge and how he uses it, do you know that all of man's complicated machinery is the result of only a few simple mechanical principles combined; and that the wedge is one of the most important? Look up "*wedge*," "*machine*," "*simple machine*," etc., in the dictionary or encyclopædia.

How does the earthworm's method of pushing his way in the world with the end of his nose compare with the way a root works along in the ground? (See [Chapter X](#).)

The earthworm's neat way of disposing of the dirt he casts out reminds me of how the beaver handles dirt when he builds a canal, and the way of the ants in digging their underground homes. (Chapters [VI](#) and [VIII](#).)

We have little brains in our finger-tips just as the earthworm has on the end of his nose. How much do you know about the little brains scattered through our bodies (*Ganglia*)?

You see the simple earthworm is the A, B, C of a lot of things; and even Mr. Darwin's famous book doesn't contain all there is to be learned about him in books and in personal interviews with Mr. Earthworm himself. A farm boy to whom the writer read the story of the earthworm, when asked how he thought the worm could turn in his burrow when it fits him so closely, said, "Why, he turns around in that little room at the end of the hall," thereby solving, as I think, a problem that puzzled Mr. Darwin, and which he left unsolved.



### SINFUL TACTICS OF A SACRED BEETLE

The beetle pushing backward is the owner of the ball and is on his way—as he thinks—to his burrow. The other is altering the direction toward his own burrow. Fabre's book on the Sacred Beetle—the tumblebug of our fields and roadways—tells how the thing came out.

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## CHAPTER VI

(JUNE)

Go to the ant, thou sluggard;  
Consider her ways, and be wise.

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*Proverbs 6:6.*

### THE LITTLE FARMERS WITH SIX FEET

I don't believe I've ever heard anybody say anything against an angleworm; although not many people, even to this day, I'll be bound, realize what a useful citizen the angleworm is.

But now we come to a class of farmers that, as a class, are positively disliked; farmers that nobody has a good word for, that nobody wants for neighbors. The charge against them is that, like the man in the Bible, they are always reaping where they have not sown; always helping themselves to other people's crops—bushels of wheat, bushels of rye, tons of cotton, loads of hay and apples and peaches and plums; and nice garden vegetables; and even the trees in the wood lot. It is estimated, for instance, that the chinch-bug helps himself every year to \$30,000,000 worth of Uncle Sam's grain; while other insects make away with 10 per cent of his hay crop, 20 per cent of mother's garden vegetables, \$10,000,000 worth of father's tobacco; and the Hessian fly sees to it that between 10 and 25 per cent of the farmer's wheat never gets to mill.

"Yes, and sometimes it's 50-50 between the farmer and the fly," said the high school boy, who often spends his vacation with a country cousin.

Then there are insects that injure and destroy forest trees because they like to eat the leaves or the wood itself; and some 300 kinds of insects that make themselves free with other people's orchards.

## I. CONSIDERING THE ANT

But, as I said a few moments ago, it takes all sorts of people to make a world; and as there are good and bad citizens among men, so there are good and bad among insects. Indeed there are so many useful insects that help make or fertilize the soil by grinding up earth and burying things in it, that even this chapter, which is rather long, as you see, can't begin to tell about all of them. So suppose we give our space to a few by way of example, and then look up others in other books in the library.

### AMOUNT OF WORK DONE BY ANTS

First of all let us consider the ways of the ant (as the Bible tells us to). The ant's work may be said to take up where the earthworm leaves off. Mr. Earthworm, as we have seen, is a little fastidious about the kind of land he tills. Among other things, he is inclined to avoid sandy soil, while the ants will be found piling up their pretty cones of sand or clay as well as of black earth. And in some soils the ants do more important work than the worm that helped make Mr. Darwin famous. In the course of a single year they may bring fresh soil to the surface to the average depth of a quarter of an inch over many square miles. This not only helps to keep the farmer's fields fertile by adding fresh, unused earth, but enriches them by burying the vegetation—such as leaves and twigs and branches broken from dead trees by storms—so that it decays. This burying of vegetation is the very thing the good farmer does when he spreads his fields with manure from the barnyard, or when he ploughs under the stubble.



A HEAP OF GRIST FROM AN ANT SOIL MILL

Something of an ant-hill, isn't it? It is a foot high and measures nearly three feet across. You will find such ant hills in the Arkansas Valley in Colorado, where the photograph of this one was taken.

Ants are very glad to do this for the farmer because it isn't any extra trouble for them. Their little heaps of fresh earth are thrown out in connection with the building of their homes. The mining ants dig galleries in clay, building pillars to support the work and covering them with thatches of grass. The red and yellow field ants are the masons. They first raise pillars and then construct arches between them, covering these arches with the loose piles of soil which we know as ant-hills. The carpenter-ants bore their cells in the dead limbs of trees, and the wood dust they make from them hurries on the process of returning these dead limbs to the soil. One kind of carpenter-ant covers its walls with a mixture of sawdust, earth, and spiders' webs. An ant in Australia builds its home of leaves fastened together with a kind of saliva. One kind of ant, whose calling card among scientific people is *Formica fusca*,<sup>[12]</sup> adds new stories to old houses as the colony grows; much as in the growth of cities and hamlets the buildings grow taller with the growth of the town. Just as men do, such ants first build the side walls and then the ceilings. As if these ants are working under contract and must get their job done by a certain time, two groups are employed on the ceiling at the same time, each group working toward the other from the opposite wall and meeting in the middle.



### THE DESERTED VILLAGE UNDER THE STONE

If Oliver Goldsmith had been as much interested in ants as was the French "Homer of the insect," Henri Fabre, he might have written of another kind of "Deserted Village," its "desert walks" and its "mouldering walls." This is a deserted village of ants. The little citizens that built it lived under a stone. When the stone was lifted it took the entire roof off the place.

### THE ANT WHO DIDN'T KNOW HIS TRADE

As you may suppose, this is real architectural engineering and no place for amateurs. I once saw a foolish worker starting a roof from the top of one of the side walls without paying any attention to the fact that the other wall was much higher. The result was he struck the middle of it, instead of joining it at the top. Another ant passing, possibly the supervising architect, saw what was going to happen. So what does he do but stop and tear down the other's work and build the ceiling over again!

"There! *That's* the way to put in a ceiling," he seemed to say. "For goodness sake, where *did* you learn your trade?"

Huber, the famous student of ants, saw two of these wonderful insects do the very same thing.

Sometimes the situation is such that it is necessary to build a very wide ceiling, so wide that it would fall of its own weight unless supported in some way. Then what would you do; that is, if *you* were an ant?

"Why, I'd put up pillars to hold it."

That's exactly what the ants do; they put up pillars; but instead of using steel beams, as men do in this day of steel, the ant architects make pillars of clay—build them up with pellets, little clay bricks which they shape with their mandibles—their jaws.

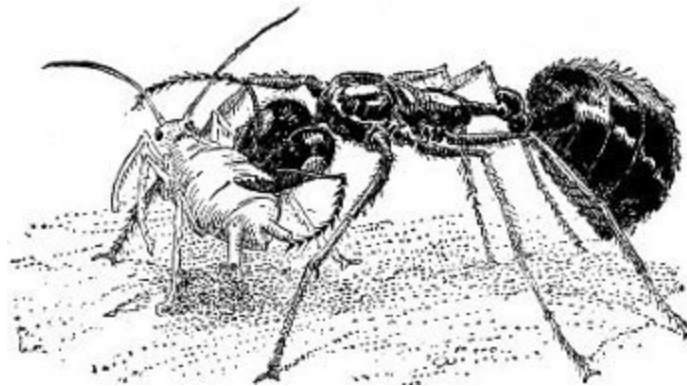
But the ants seem to have some of the methods of steel construction, too; the use of girders and things. Ebrard, a French student of ants, tells how, when a certain roof threatened to fall, some Sir Christopher Wren of the ant world used a blade of grass as a girder, just as Sir Christopher in his day put

in girders to support the roof of Saint Paul's Cathedral, and as men use steel girders to-day. The ant fastened a little mass of earth on the end of a grass stalk growing near to bend it over; then gnawed it a little at the bottom to make it bend still more, and finally fixed it with mud pellets into the roof.

But here's something that will make you smile! You have heard about the lazy man down in Arkansas with the hole in his roof? You remember he never mended it in dry weather because it didn't need it, and when it rained he *couldn't* mend it on account of the rain!

#### RAINY-DAY WORK IN THE ANT WORLD

Well, these *Formica fusca* folks are as different from that Arkansas man as anything you could imagine. First of all, being ants, they are anything but lazy; secondly, they never put off needed work on their roofs on account of rain. In fact, they *choose* the first wet day to do it. As soon as the rain begins they build up a thick terrace on the roof of the old dwelling, carrying in their jaws little piles of finely ground earth which they spread out with their hind legs. Then, by hollowing out this roof, they turn it into a new story. Last of all they put on the ceiling. You see the rain helps them in mixing their clay. There are ants that build up vaulted viaducts or covered ways, and they use clay for that.<sup>[13]</sup> They make the clay by mixing earth with saliva. Some of these viaducts reach out from the house—the ants' house—to their "cow" pasture.



AN ANT CARRYING ONE OF HER COWS



You know about how ants keep cows, little bugs called aphids? The aphids feed on plants, and the clay viaducts protect the ants from their enemies and from the sun in going to and from the pasture; for this particular family of ants doesn't like the sun. They make clay sheds for their cattle, too. Here and there along the clay viaduct are large roomy spaces, cow-sheds, so to speak—where the little honey cows gather when they aren't feeding. Another kind of ant builds earth huts around its cow pastures. The large red ants (*F. rufa*), sometimes called "horse ants," build hills as large as small haycocks.

## **II. THE TERMITES AND THEIR TOWERS OF BABEL**

But speaking of big buildings, did you ever hear of a skyscraper a mile high? Well the home of the six-footed farmer I am going to tell you about now is as much taller than he is as a mile-high skyscraper would be taller than a man. The remarkable little creatures that build these skyscrapers are called "termites." Termites are also known as "white ants." This seems funny when we know that they are neither "ants" nor are they white. The young of the workers are white, to be sure, but the grown-ups are of various colors, and never milky white as they are when young. The termites were first called "white ants" in books of travel because the termites the travellers saw were the young people.

### **HOW TERMITES ARE LIKE THE ANTS**

The termites are really closer relatives of dragon-flies, cockroaches, and crickets than of the ants, but they do look a great deal like an ant, and they have many of the ways of the ants. As in the case of ants, all the members of one community are the children of one queen. The king lives with the queen in a private apartment. Sometimes—as with human royalties—the king and queen will have separate residences, but the termite royalties always live in the same house with their people; they are very democratic.

Some kinds of termites live in rotten trees, which they tunnel into, and that is their contribution to soil-making; while others build great, big solid houses of earth and fibres, mixed. These houses are called "termitariums," and are six, eight, ten, even twenty-five feet high; fully 1,000 times the

length of the worker. Think of a man five feet high, and then multiply by 1,000, and you see you have got nearly a mile!



#### SKYSCRAPERS A MILE HIGH

"Some kinds of termites build great, solid houses of earth and fibres mixed. These houses are six, eight, ten, even twenty-five feet high, fully one thousand times the length of the worker. Think of a man five feet high and then multiply by one thousand, and you see you have got nearly a mile."

These termite skyscrapers aren't much to look at on the outside, but inside they're just fine; they have everything the most particular ant could want. For instance, the termites are right up-to-date in their ideas about fresh air, their houses being well ventilated through windows left in the walls for that purpose. You can see the importance of this fresh-air system when you know there are thousands of termites under the same roof. They also have a sewage system for carrying off the water of the rains. And a fine piece of mechanical engineering the building of it is, too; for these "water-pipes" are the underground passages hollowed out in getting the clay to build the

homes. The termites build their homes with one hand and dig the sewer with the other, so to speak.

#### THE THERMOSTATS FOR THE NURSERIES

The termitarium has as many rooms in it as a big hotel—oh, I don't know *how* many—and they are all built around the chambers of the king and queen. Next to the royal apartments are the pantries, a lot of them, and they are all stored with food. In the upper part of the termitarium are the nurseries—many nurseries—for no one nursery could care for any such numbers of babies as the queen has. Between the nursery and the roof is an air-space, and there are also air-spaces on the sides and beneath. The nursery thus being surrounded by air, the eggs and, when they come along, the babies are protected from changes of temperature. It's the same principle that's employed in making refrigerators and thermos bottles. The rooms in which the eggs are kept are divided by walls made of fragments of wood and gum glued together. This mixture is a bad conductor<sup>[14]</sup> of heat or cold. And so the eggs are kept at an even temperature.

While we cannot see any of the termite skyscrapers in the United States, because we have none of the species of termites that build them, we can see a member of the termite family. This is the common white ant that digs into joists of houses. On the outside of these same joists, and up in the attics of old farmhouses, if there happens to be a broken window-pane, or some other hole through which she can get in, you can see the nest of another tiller of the soil, the wasp. The mason-wasps or mud daubers are the most common. You will find their nests on the rafters of the barn when you go to throw down hay, or when you go into the corn-crib. They have all sorts of fancies—these wasps—about their clay homes and where to build them. Some build on the walls and some in the corners of rafters, others prefer outdoor life. Some want to live alone, others like society. What are known as "social" wasps sometimes build their nests in tiny hollows that they dig in the ground; others fasten their nests to the boughs of trees. The work of these wasps, from the farming standpoint, is useful not alone in grinding the soil, but helping to supply it with humus; for their nests are made of wood fibre, which they tear with their mandibles from gateposts, rail fences, and the bark of trees.



NESTS OF MASON-WASPS

The carpenter-wasp is both a wood-worker and a clay-worker. He cuts tubular nests in wood and divides them by partitions. We think we're pretty smart, we humans, because we are always picking up ideas, but here's a creature, no bigger than the end of your finger, who has picked up an idea from the carpenter-bee, grafted it on his native trade of clay-worker, and made himself as nice and cosey a country place as you'd want to see!

#### **ABOUT THE WASP, THE FOX, AND THE BUMBLEBEE**

Here's another example of the same thing, this spreading of good ideas among the neighbors. It's about the fox, the digger-wasps, and the bumblebee. The fox can dig his own burrow when he has to, but if he finds somebody else's that he can use, he just helps himself—provided, of course, the owner isn't Brer Bear, or some other big fellow that Brer Fox doesn't care to have any words with. In the same way the digger-wasps make their own little burrows if they are obliged to, but prefer to help themselves to ones they find already made, although they don't drive anybody else out. They simply take possession of holes left by field-mice. The bumblebee does the same thing. The bumblebee digs a hole a foot or more deep, carpets it with leaves, and lines it with wax. Leading up to the home is a long, winding tunnel. As Bumblebeeville grows bigger there may be two or

three hundred bees in one nest. As the bumblebee babies keep coming and coming, the burrow has to be dug bigger and bigger, to take care of them.

### III. THE HOUSE THAT MRS. MASON BUILT

But the greatest of bee workers in the soil is the mason-bee. You can get an idea of what a useful citizen the mason-bee is when I tell you that one of the little villages of one species sometimes contains enough clay to make a good load for a team of oxen. Yet for all that, they might have gone on with their work for years and years to come—just as they have for ages in the past—and people wouldn't have thought much about it, if it hadn't been for some boys.

One time, in a village in southern France, a school-teacher, who was getting on in years, took his small class of farmer boys outdoors to study surveying—setting up stakes and things, you know, the way George Washington used to do. It's a stony, barren land—this part of France—and the fields are covered with pebbles. The teacher noticed that often when he sent a boy to plant a stake, he would stoop every once in a while, pick up a pebble and *stick a straw into it*! That's what it looked like! Then he would suck the straw.

Well, to make a long story short,<sup>[15]</sup> these pebbles had on them the little clay cells of the mason-bee. Mrs. Mason-Bee fills these cells with honey, lays an egg in the honey, and when the babies come along—don't you see? In other words, Mother Bee not only puts up their lunch for them, but puts them right into the lunch! This makes it convenient all around; for, like almost all insect mothers, Mrs. Mason-Bee is never there after the babies come.



MASON-BEE CELLS AMONG THE ROCKS

There were so many of these pebbles scattered over the plain, and the bees that were building new homes or repairing old ones flew so straight and so fast between the pebbles and a near-by road that "they looked like trails of smoke," as Fabre expresses it.

Now, you may well wonder why the bees flew clear over to that road to get dirt to build their nests when there was plenty of loose earth right at their own door-steps; right around the pebbles themselves. Isn't that queer?

Well, here's something that sounds stranger still. Mrs. Mason-Bee takes those extra trips because a roadway is so much harder to dig in! It's not because she needs the exercise, goodness knows—this busy Mrs. Mason-Bee—but because the hard earth of the roadway makes the strongest homes; that is, when she finally gets it dug out and worked up. And here's another thing that will seem odd at first; although the soil she thus works over must be dampened before she can plaster it into the walls of her home, she just won't use damp soil to begin with. Nothing will do her but dust, and dust that she herself scrapes from the roadway. The reason of this is that the

moisture already in the soil will not answer at all. She has got to knead the soil carefully and thoroughly with saliva, which acts as a kind of mortar. This saliva, of course, she supplies.

And the dust she works with must be as fine as powder and as dry as a bone. Then it absorbs the saliva, and when it dries it is almost like stone. In fact it's a kind of cement, like that men use for sidewalks and for buildings and bridges.



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FABRE STUDYING THE MASON-BEE

But this wonderful old teacher and his boys<sup>[16]</sup> found that even this isn't all this little house-builder and house-keeper has to think of. She must have dust that is really ground-up stone! So she digs in the roadway where the bits of stone in this stony soil have been ground to powder and then packed hard by the wheels of the farmer's cart and by the hoofs of horses and oxen drawing their heavy loads. But what did Mrs. M. B. do for ground-up stone

in the long ages before man came along with his carts? Mr. Earl Reed, who, beside being the distinguished etcher of "The Dunes," is a close observer of nature in general, tells me he has often seen a mason-bee gathering the pulverized stone at the base of cliffs. Evidently the mills of the wind and rain, that we have read of in previous chapters, had Mrs. B's wants in mind too.

### BEING A MASON-BEE FOR A LITTLE WHILE

Now, just to show you one more thing about Mrs. Mason-Bee as a house-builder—how clever she is—let's try something right here. Let's suppose ourselves—yourself and myself—Mrs. Mason-Bees. We have got a home to build for some baby mason-bees that will be along by and by. Say we already know that we must use this stone dust of the roadway, and that we must make our mortar not with *water* but with *saliva*. Here's the *next* problem:

Shall the mixing be done where the building is going up over there? That's the way human masons do it. But Mrs. Mason-Bee evidently thinks otherwise, for at the very time she is prying up those atoms of dust with so much energy, you notice she is doing her mixing. She rolls and kneads her mortar until she has it in the shape of a ball as big as she can possibly carry. Then "buz-z-z-z!" Away she goes, straight as an arrow, back home, and the mortar is spread where it is needed.

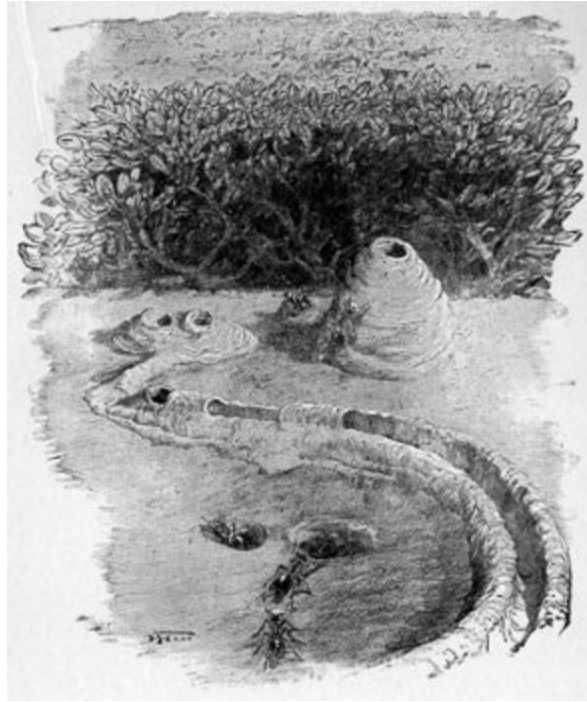
You see, after all, this is the best way. If she didn't turn the dust into mortar before she started, so a good-sized lump of it would stick together, she couldn't carry much of it at a time, and it would be forever and a day before she could get her house built. As it is, the pellets she carries are of the size of small shot; a pretty big load, let me tell you, for a little body no bigger than Mrs. Mason-Bee.

And remember, this goes on all day long from sunrise to sunset. Without a moment's rest, she adds her pellets to the growing walls and then back she goes to the precise spot where she has found the building material that best suits her needs.

In building a nest, the mason-bee, in going to and fro, day after day, travels, on the average, about 275 miles; half the distance across the widest part of



France. All in about five or six weeks, she does this. Then her work is over. She retires to some quiet place under the stones, and dies. As I said, she never sees the babies she has done so much for.



SURFACE MOUNDS OF THE MASON-ANT

There are mason-ants as well as mason-bees. This illustration shows the works thrown up by some mason-ants that Dr. McCook found in a garden path one morning in May.

And although they are so stoutly built, the houses of the mason-bees, like those "cloud-capped towers and gorgeous palaces" that Shakespere speaks of, finally go back to the dust. But while one of these little mothers is building a new home or repairing an old one left by a mother of the previous year, you would suppose the fate of the world hung on it; as indeed the fate of the world of mason-bees does.

Scrape! Scrape! Scrape! With the tips of those little jaws, her mandibles, she makes the stony dust.

Rake! Rake! Rake! With her front feet she gathers and mixes it with the saliva from her mouth.

How eager and excited she gets, how wrapped up in her work as she digs away in the hard-packed mass in the tracks of the roadway! Passing horses and oxen, and the French peasants with their wooden shoes, are almost on her before she will budge. And even then she only flits aside until the danger has passed. Then down she drops and at it again!

But sometimes, the boys and the teacher found, she starts to move too late—so absorbed is she, it would seem, in the thought of that tiny little home over there among the pebbles.

Poor little lady!

### HIDE AND SEEK IN THE LIBRARY

Perhaps nothing in nature is more wonderful than an insect; particularly when you consider that he *is* only an insect! So, of course, whole libraries have been written about insects. Here are a few of the most interesting books dealing with the subject: Beard's "Boy's Book of Bugs, Butterflies and Beetles"; Comstock's "Ways of the Six-Footed"; Crading's "Our Insect Friends and Foes"; Doubleday's "Nature's Garden"; Du Puy's "Trading Bugs with the Nations." This about trading bugs is an article in "Uncle Sam: Wonder Worker," and tells how Uncle Sam "swaps" with other nations to get rid of injurious insects and bring in useful ones.

Grant Allen's "Sextons and Scavengers" ("Nature's Work Shop") tells many curious things about the sexton beetles; how, by tasting bad, they keep birds and things from eating them; why you will always find an even number—never an *odd* number—of sextons at work together; what they use for spades in their digging; why male sextons bury their wives alive, and why there is reason to believe that these weird little insects have a sense of beauty and of music.

The same essay tells about the sacred beetle of the Egyptians, the insect that we know as the "tumblebug"; why first the Egyptians and

then the Greeks regarded this bug as sacred; and why men and women wear imitation beetles for brooches and watch-charms to-day.

But the greatest work on this famous beetle has been written by the famous French observer Fabre, "The Homer of the Insect." You will find this book, "The Sacred Beetle," in any good public library. Among other things Fabre gives a very minute description of the variety of tools used by the beetle; tells how two beetles roll a ball;<sup>[17]</sup> how they dig their holes; how they "play possum," and then (I'm almost ashamed to tell this) rob their partners! How they wipe the dust out of their eyes; about a tumblebug's wheelbarrow; why their underground burrows sometimes have winding ways; why there are fewer beetles in hard times; about their autumn gaieties; their value as weather-prophets, and how Fabre's little son Paul helped him in writing his great book.

Allen's essay, "The Day of the Canker Worm" in "Nature's Work Shop," tells many interesting things about the Cicada, the locust that only comes once in seventeen years;<sup>[18]</sup> about Lady Locust's saw (it looks like a cut-out puzzle); about the clay galleries the locusts build when they come up out of the ground; how many times they have to put on new dresses before they finally look like locusts; why, at one stage of the process, they look like ghosts, and how they blow up their wings as you do a bicycle tire.

(Fabre's book on the sacred beetle also deals, incidentally, with the Cicada.)

Often one thing is named after another from a merely fanciful resemblance, as, for instance, the "sea horse." But the mole cricket really seems to have been patterned on the mole; either that, or both the four-legged and the six-legged moles were patterned after something *else*. Mole crickets are very useful little people to know. You should see how they protect their nest-eggs from the weather and how and why they move their nests up and down with the change of the seasons.

What good to the soil do the insects do that eat up dead-wood? Scott Elliott, in his "Romance of Plant Life," deals with this subject.

The mining bees are very interesting, and some of these days, perhaps millions of years hence, they will be still more interesting, for they are learning to work together, although not to the extent that the bees and ants do. Working together seems to develop the brains of insects just as it does human beings. Thomson's "Biology of the Seasons" tells how the mining bees are learning "team-work."

The tarantula spider is a relation of the six-footed farmers, you should know, although he is not an insect himself. In "Animal Arts and Crafts" in the "Romance of Science" series you will find how, in his digging, he makes little pellets of earth, wraps them up in silk, and then shoots them away, somewhat as a boy shoots a marble.

The same book tells why the trap-door spider usually builds on a slope. It also tells why she puts on the front door soon after beginning her house. (This looks funny, but you wouldn't think it was so funny if *you* were a trap-door spider and you had a certain party for a neighbor, as you will agree when you look it up.)

The door, by the way, has a peculiar edge to make it fit tight. What kind of an edge would *you* put on a door to make it fit tight? (Look at the stopper in the vinegar-cruet and see if it will give you an idea.)

This book also tells about a certain wasp that makes pottery and gets her clay from the very same bank that certain other people depend on for *their* potter's clay. This wasp sings at her work and has three different songs for different parts of the work.

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### THE FIELD MOUSE AND THE FARMER

When we remember how much soil the field mouse worked over, and so made better, long before man's time on earth—to say nothing of what the mice have done since—doesn't it give an added and deeper meaning to the lines of Burns?

"I doubt na, whyles, but thou may thief.  
What then? Poor beastie, thou maun live."

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## CHAPTER VII

(JULY)

Well said, old mole! Canst work i' the earth so  
fast?

—*Shakespeare: "Hamlet."*

### FARMERS WITH FOUR FEET

Before we start this chapter—it's going to be about the farmers with four feet, you see—I want to say something, and that's this: *Don't let anybody tell you moles eat roots.* They don't! They eat the cutworms that do eat the roots. Haven't I been in mole runs often enough to know! Of course, the moles do cut a root here and there occasionally when it happens to be in the way, as they tunnel along, but what does that amount to?

Why, in France they put Mr. Mole in vineyards—on purpose! He's one of the regular hands about the place, just like the hired man.

#### I. MR. MOLE AND HIS RELATIONS

Moles do a lot of good work for the farmer. Not only were they ploughing and ploughing and ploughing the soil—over and over again—thousands of centuries before man came along to plant seed in it, but they are all the time eating, among other things, destructive worms and insects in the soil. They work all over the world, that is to say, in the upper half of it—the Northern Hemisphere; and there's where the biggest half of the land is, if I haven't forgotten my geography.

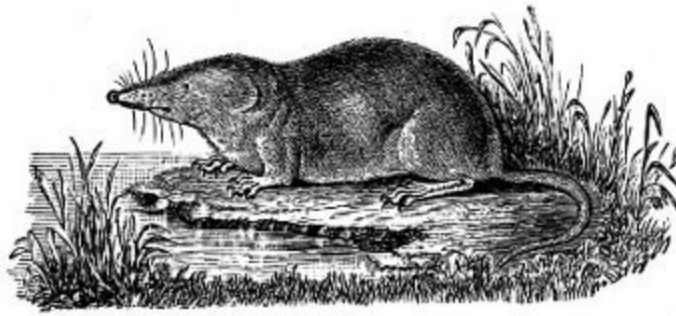
#### WONDERFUL LITTLE MACHINES ON FOUR LEGS

Closely related to the moles are the shrews—quaint little mouse-like creatures with long, pointed heads and noses that they can twist about almost any way in hunting their meals and finding out other things in this big world that concern them. On these funny, long noses they have whiskers like a pussy-cat; and that helps, too, when you want to keep posted on what's going on around you. Like the moles the shrews are found all over the Northern Hemisphere. What is known as the "long-tailed shrew," is the very smallest of our relations among the mammalia. Why, they're no bigger than the end of a man's little finger; and the smallest watch *I* ever heard of was a good deal bigger than that. Yet, inside these wee bodies is as much machinery as it takes to run any other mammal—an elephant, say.



THE COMMON AND THE SHORT-TAILED SHREW

The shrews get around very fast, considering their size; and they're on the go all the time. I never saw such busy-bodies; nosing about in the old leaves and dead grass and under logs and boring into loose loam, punky wood, decayed stumps—anywhere you'd be likely to find a worm, a grub, a beetle, or a slug. Hard workers, these shrews, but so quarrelsome! When two Mr. Shrews meet there's pretty sure to be trouble. They're regular little swashbucklers among themselves; and—the queerest thing, until you know why—they don't seem to be afraid even of cats. Fancy telling Cousin Mouse that! But it isn't because the shrews *wouldn't* be afraid if the cats got after them, but because cats always let shrews alone. They don't taste good!



THE CILIATED SHREW

Shrews are so nimble on their tiny feet and so quick of hearing, they are very hard to catch. And please don't try! You simply *can't* tame them, and in spite of the fact they're so fierce and bold at home—among their own kind—they're easily frightened to death. A shock of fear and that wonderful little heart engine of theirs stops short—never to go again.

#### MR. MOLE'S PAWS AND HOW HE WORKS THEM

But while the shrews can get around so much faster above ground the moles are the most remarkable travellers *under* ground. The mole's paws, you notice, are turned outward, as one's hands are when swimming. In fact he does almost swim through the soft, loose soil—so fast does he move along! His two shovels, with the muscles that work them, weigh as much as all the rest of his body. Why, he has a chest like an athlete! He pierces the soil with his muzzle and then clears it away with his paws. His skull is shaped like a wedge. He has a strong, boring snout and a smooth, round body.

This snout, by the way, has a bone near the tip. You see how handy that would come in, don't you? At the same time, although it's so hard—this snout of his—it's very sensitive, like the fingers of the blind; for Mr. Mole must always be feeling his way along in the dark, you know.





#### SECTION OF MR. MOLE'S CASTLE

This is a cross-section of a mole-hill, showing the central chamber and the rooms leading into it.

The kind of moles you find in Europe live in what seem to be little earthen fortresses, and the tops, sticking above ground, make hillocks. In each of these little forts there is a central chamber; then outside of this, running all the way around, are two galleries, one above the other. The upper gallery has several openings into the central chamber. The galleries are connected by two straight up-and-down shafts. From the lower galleries several passages, usually from eight to ten, lead away to where the moles go out to feed; and if there is a body of water near by—a pond or a creek, say—there's a special tunnel leading to that.

Mr. Mole works hard and he sleeps hard. The big middle room in his home is the bedchamber of Mr. Mole and his family. Usually he sleeps soundly all night, but occasionally, on fine Summer nights, he comes out and enjoys the air.



THE COMMON AND THE STAR-NOSED MOLE

You'd think he'd get awfully dirty, wouldn't you, boring his way along in the ground all the time? But he doesn't. His hair is always as spick and span as if he'd just come out of the barber-shop. Do you know why? It's because he wears his hair pompadoured. It grows straight out from the skin. So you see he can go backward and forward—as he is obliged to do constantly in the day's work—without mussing it up at all. If it lay down, like yours or like pussy-cat's, it would get into an *awful* mess! In France the children call Mr. Mole "The Little Gentleman in the Velvet Coat."

## II. FOUR-FOOTED FARMERS THAT WEAR ARMOR

But, speaking of coats, I want to introduce you to a still more rapid worker in the soil, who wears a coat of mail. He is called the armadillo. There used to be a species of armadillo in western Texas. Whether there are any there still I don't know,<sup>[19]</sup> but go on down to South America and you'll find all you want. The woods are full of them, and so are those vast prairies—the pampas. The plates in the armadillo's coat of mail are not made of steel, of

course, but of bone. These bony plates are each separate from the other on most of his body but made into solid bucklers over the shoulders and the hips. The armadillos have very short, stout legs and very long, strong claws, and how they can dig! They can dig fast in any kind of soil, but in the loose soil of the pampas they dig so fast that if you happen to catch sight of one when out riding and he sees *you*, you'll have to start toward him with your horse on the run if you want to see anything more of him. Before you can get to him and throw yourself from the saddle, he'll have buried himself in the ground. And you can't catch him; not even if you have a spade and dig away with all your might. He'll dig ahead of you, faster—a good deal faster—than you can follow.

#### **MR. ARMADILLO'S REMARKABLE NOSE DRILL**

For all he looks so knightly, so far as his armor is concerned, the armadillo is timid, peaceful, and never looking for trouble with anybody, but once aroused fights fiercely and does much damage with his long hooked claws. His chief diet is ants. These he finds with his nose. He locates them by scent and then bores in after them. You'd think he'd twist it off, that long nose of his; he turns it first one way and then the other, like a gimlet. And so fast!

The armadillo dislikes snakes as much as all true knights disliked dragons. That is, he doesn't like them socially; although he's quite fond of them as a variation in diet. He'll leap on a snake, paying not the slightest attention to his attempts to bite through that coat of mail, and tear him into bits and eat him.

Another armored knight that eats snakes and that other animals seldom eat—much as they'd like to—is the hedgehog. If you were a fox, instead of a boy or girl, I wouldn't have to tell you about how hard it is to serve hedgehog at the family table. One of the earliest things a little fox learns in countries where there are hedgehogs is to let the hedgehog alone.

"Hedgehogs would be very nice—to eat, I mean—if they weren't so ugly about not wanting to be eaten."

We can imagine Mamma Fox saying that to the children. Then she goes on:

"The whole ten inches of a hedgehog—he's about that long—are covered with short, stiff, sharp, gray spines. He's easy to catch—just ambles along, hardly lifting his short legs from the ground. And he goes about at night—just when we foxes are out marketing. That would be so handy, don't you see; but the trouble is about those nasty spines of his. Try to catch him and he rolls up into a ball with all his spines—they're sharp as needles—sticking out everywhere, and every which way. And—well, you simply can't get at him, that's all. So just don't have anything to do with him. It's only a waste of time."

Hedgehogs live in hedges and thickets and in narrow gulches covered with bushes. They do their share of ploughing when nosing about with their pig-like snouts for slugs, snails, and insects, and when they dig places for their home nests. These homes they line with moss, grass, and leaves, and in them spend the long Winter, indifferent to the tempests and the cold.

But there's another place to look for hedgehogs, and you never would guess! In people's kitchens. If you ever go to England you'll find them in many country homes, helping with the work. They're great on cockroaches, and they're perfectly safe from the cat and the dog. Both Puss and Towser know all about those spines, just as well as Mrs. Fox does.

When they've eaten all the cockroaches, give them some cooked vegetables, porridge, or bread and milk, and they'll be perfectly content. They're easy to tame and get very friendly.

In the wild state, besides the insects and things I mentioned, they eat snakes; and poison snakes, too! The poison never seems to bother them at all. Their table manners are interesting, also, when it comes to eating snakes. They always begin at the tail.<sup>[20]</sup> They'd no more think of eating a snake any other way than one would of picking up the wrong fork at a formal dinner.

#### **UNDER THE HEDGEHOG'S WATER-PROOF ROOF**

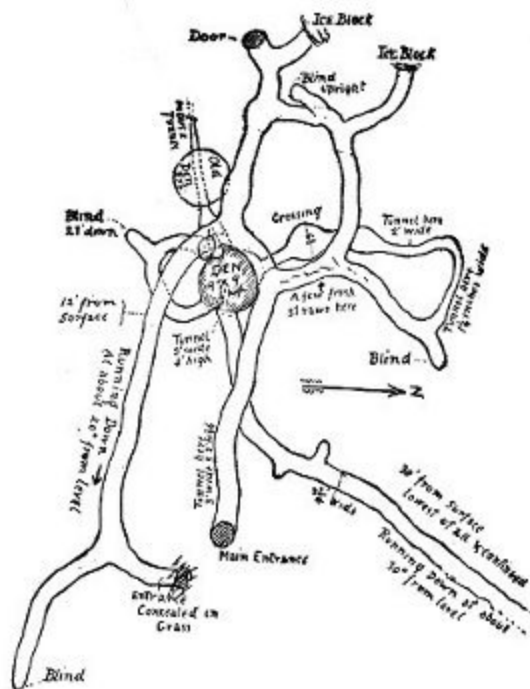
That's one of the things about good manners Mamma Hedgehog teaches the babies, I suppose. Of these she has from two to four, and she makes a curious nest especially for them; a nest with a roof on it that sheds rain like

any other roof. Just as it is with puppies and kittens, the babies are born blind; and not only that, but they can't hear at first, either. While they are young their spines—I don't mean their back-bones, but their other spines—are soft, but they become hard as the babies grow and open their eyes and ears on the world. The muscles on their backs get very thick and strong, so that when they don't want to have anything to do with anybody—say a fox, or a dog, or a weasel—they just pull the proper muscle strings and tie themselves up into a kind of bag made of their own needle-cushion skins, with the needles all sticking out, point up!

### **III. A VISIT TO SOME FARM VILLAGES**

#### **TWELVE LITTLE MARMOTS ALL IN ONE BED**

Next I'd like you to visit with me certain other farmers who remind us of the Middle Ages also; not because they wear armor, like the armadillos and the hedgehogs and the lords of castles, but because they live in farm villages as the farmer peasants used to do around the castles of the lords. Moreover, one reason they live together in this way is for protection—just as it was with the peasants—only among these little democrats there's no overlord business; each one's home is his castle. Another reason for this village arrangement is that it's such a sociable way to live; and they're great society people, these farm villagers. The marmots, for example, the largest and heaviest of the squirrel family, just love company. In their mountain country—they're mountain people, the marmots—they play together, work together, and during the long, cold night of Winter snuggle together in their burrows. Their burrows are close by each other among the rocks. They have both Summer and Winter residences. In Summer they go away up in the mountains, hollow out their burrows and raise their babies. When the snows of late Autumn send them down the mountainsides, twelve or fifteen of them, all working together, pitch in and make a tunnel in the soil among the rocks, enlarging it at the end into a big room. Next they put in a good pile of dry hay, carefully close the front door and lock it up with stones caulked with grass and moss. Then they all cuddle down together, as snug as you please, and stay there until Spring.



#### HIGHWAYS OF GROUND-SQUIRREL TOWN

Almost as crooked as the streets of London town, aren't they? And as hard to find one's way about in—unless, of course, one were a ground-squirrel. This is the burrow of a Richardson ground-squirrel sketched by Thompson Seton, near Whitewater, Manitoba.

Another member of the marmot family who is very fond of good company is the prairie-dog. There may be thousands in a prairie-dog town. Each little prairie-dog home has in front of it a mound something like an Eskimo's hut. The prairie-dogs make these mounds in digging out their burrows. They pile the dirt right at the front door. This may not look neat to us, but you'll see it's just the thing—this dirt pile—when you know what the prairie-dog does with it. He uses it as a watch-tower.

When, from this watch-tower, he spies certain people he doesn't want to meet, you ought to see how quickly he can make for his front door and into the house! The times are still lawless where the prairie-dog lives, and he has to be on the lookout all the while for coyotes, for foxes, for badgers, for the

black-footed ferret and the old gray wolf; to say nothing of hawks and brown owls.

#### SUCH NEAT CHAMBERMAIDS!

The prairie-dogs like sandy or gravelly soil for their homes, and in making them they do a lot of ploughing. And besides they supply this same soil with a great deal of humus—the grass that they use for bedding. They're very particular about changing their beds every day; always clearing out the old bedding and putting in new. They do this along about sundown. You can see them do it right in New York City, for there is a flourishing colony of them at the zoo.



#### THIS MUST BE A PLEASANT DAY

In nice weather the Prairie Dog's front door stands wide open like this, but before a rain he stuffs it tight with grass because, when it *does* rain in the arid regions where he lives, it comes down in bucketfuls!

Mr. Prairie-Dog is about a foot long and as fat as butter. The reason he's called a dog isn't because he is a dog or even looks like one, but because he has a sharp little bark like a very much excited puppy. He thinks he sees something suspicious: "Yap! Yap!"

Or he spies a neighbor down the street: "Yap! Yap! Hello, neighbor! Looks like another fine day, doesn't it?"

"Yap! Yap!" says neighbor. (This "yap" passes for "yes," no doubt—although it isn't quite the way Mr. Webster would say it, perhaps.)

Then maybe a neighbor from away over on the avenue, that he hasn't seen for some time, comes calling—as they're always doing, these neighborly little chaps. Then it's:

"Yap! Yap! Yap! Yap! Why, how *are* you? And what have you been doing? And how are the little folks?"

And so it goes, all day long.

The prairie-dog's native home is on our Western plains, but he has a cousin away off in South America—although he may never have heard of him—called the viscacha.

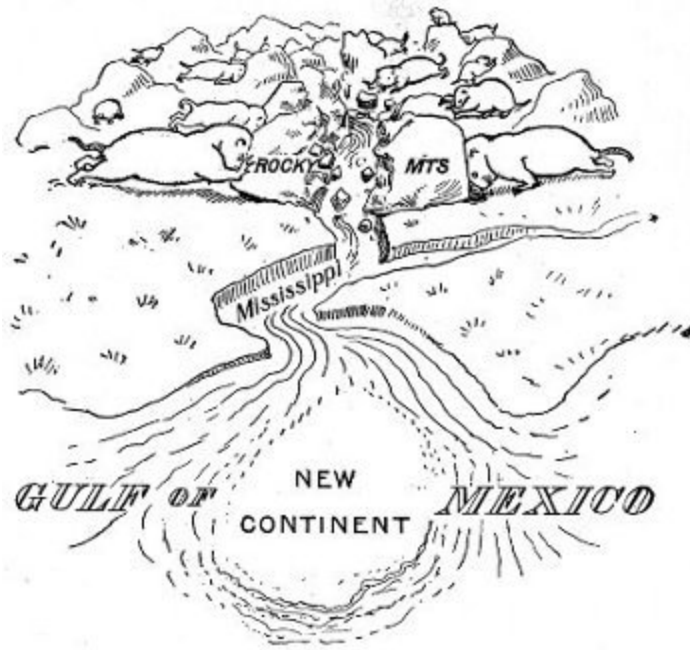
The viscachas live on the great grassy plains of the La Plata in colonies of twenty or more, in villages of deep-chambered burrows with large pit-like entrances grouped close together; so close, in fact, that the whole village makes one large irregular mound, thirty to forty feet in diameter and two to three feet high. These villages being on the level prairie, the viscachas are careful to build them high enough so that floods will not reach them. They make a clear space all around the town. In doing this these little people seem to have two purposes: (1) To make it more difficult for enemies to slip up on them unnoticed, and (2) to furnish a kind of athletic field for the community; for it is in these open spaces that they have their foot-races, wrestling matches, and the like.

If you ever happen down their way, the first thing that will strike you is the enormous size of the entrances to the central burrows. You'd think somebody as big as a bear lived in them. The entrance is four to six feet across and deep enough for a tall man to stand in up to the waist.

Like our prairie-dogs, the viscachas are very sociable, and little paths, the result of neighborly calls, lead from one village to another. They are neighborly indeed; and in the Bible sense. Of course, they like to get together of an evening and talk things over and gossip and all that, but that isn't the end of it. To take an instance: These South American prairie-dogs, like our prairie-dogs up North, are not popular with the cattlemen; and the cattlemen, to get rid of them, bury whole villages with earth. Then



neighbors from distant burrows come—just as soon as the cattlemen go away—and dig them out!



MR. P. GOPHER AS THE MASTER PLOUGHMAN

Thompson Seton calls the pocket-gopher "the master ploughman of the West," and this is how he illustrates the extent of his labors.

Another ploughman besides the prairie-dog and the viscacha, who isn't popular with farmers—although Thompson Seton calls him "The Master Ploughman of the West"—is the pocket-gopher. He has farmed it from Canada to Texas, all through the fertile Mississippi Valley. The reason he has that queer expression on his face—you couldn't help noticing it—is that each cheek has a big outside pocket in it; and, like the big pockets in a small boy's trousers, they're there for business. On each forefoot he has a set of long claws; and dig, you should see him! He's a regular little steam-shovel. He sinks his burrow below the frost-line and into this, stuffed in his two pockets, he carries food to eat when he wakes up during the following Spring, before earth's harvests are ripe.



POCKETS OF THE POCKET-MOUSE

#### **IV. THE HOME OF THE RED FOX**

Another country gentleman, not as popular with his neighbors, I must say, as he might be, but whose people, in the course of the ages, have done a good deal of ploughing, is Brer Fox. I mean particularly the red fox, for the gray fox usually lives in hollow trees or in ready-made houses among the rocks of the mountainside.

##### **THE THREE ROOMS IN THE FOX HOUSE**

The red fox is the cunningest of his tribe. One of the ways he shows his cunning—and also his lack of conscience, in dealings outside the fox family—is in his way of getting a home. Whenever he can find a burrow of a badger, for example, he drives the badger out and then enlarges the place to suit his own needs. For Mr. Fox's residence is quite an affair. Usually it has three rooms; the front room where either Mr. or Mrs. Fox—depending on

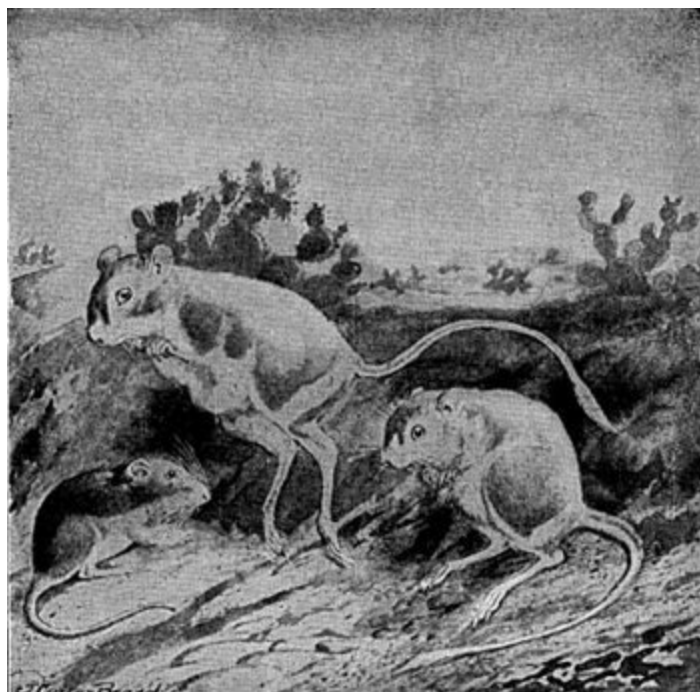
which is going marketing—stops and looks about to see if the coast is clear; back of that the storeroom for food, and behind this the family bedroom and nursery.

Mr. and Mrs. Fox are among the thriftiest folks I know. They not only provide for to-day, but for to-morrow and the day after. For example, when Mr. Fox visits a poultry-yard, he doesn't simply carry off enough for one meal. He keeps catching and carrying off chickens, ducks, or geese—whatever comes handy—all night; working clear up to daybreak. And the fresh meat he thus gets for the family table he buries—each fowl in a separate place—not so very far away from the poultry-yard. Then later he comes and gets this buried treasure and takes it home to be shared with mother and the babies.

Of these babies there are from three to five. Young foxes are very playful and think there's no such sport as chasing each other about in the sunshine, while mother sits in the doorway keeping an eye out for possible danger and watching their antics with a complacent smile, as much as to say: "*Aren't they the little dears!*"

If just one little fox wants to play while his brothers and sisters want to sleep—and that sometimes happens—he goes off by himself and chases his own tail around, just like a kitten.

Little foxes are very nice and polite that way.



THE KANGAROO RAT AND THE POCKET-MOUSE

The kangaroo rat and the pocket-mouse live in the arid regions of the United States. Both have pockets in their cheeks, but the mouse is named for his pockets and the rat for his long kangaroo hind legs.

## V. WORK AND PLAY IN CHIPMUNKVILLE

It isn't often one gets a chance to see little foxes at play, except occasionally in the big city zoos, for foxes are now so scarce; and, besides, their papas and mammas in the wild state are suspicious of human spectators, but there are certain nimble four-legged babies to be found all over the country that play in much the same way.

If, along in July, you should see a certain little body in a lovely striped suit chasing another little body in a striped suit, exactly like it, along the old rail fence or over the boulder wall or across the meadow, ten to one, it will be two baby chipmunks playing tag. When one bites the other's tail—they're always trying to do that in these tag games—it means he's "it," I think. In fact, I'm quite sure, for always, when one little Mr. Chipmunk bites another

little Mr. Chipmunk on the tail, little Mr. Chipmunk No. 2 turns right around and chases little Mr. Chipmunk No. 1, and tries to bite *his* tail.

They keep this up on sunshiny days all through July and along into early August. Then the serious business of life begins. They sober down, these chipmunk children—they were only born last May—and learn to make homes for themselves. You never would think the way they love the sunshine that the homes of all the chipmunks are under the ground, and as dark as can be. But they are. You notice the chipmunks have rather large feet, considering what dainty little creatures they are. These feet, like the feet of the mole, are for digging. The chipmunk digs deep under the roots of trees and stone walls, if there happens to be either handy by, but, so far as I've seen, he's quite contented to make his burrows in the open meadows. The round nest at the end of the burrow is lined with fine grass. It has two entrances, one right opposite the other, like front and back doors. Sometimes there are as many as three doors; four, maybe, in case of a chipmunk of a particularly nervous disposition. All chipmunks are easily frightened and dive into their holes, quick as a wink, when there's any danger; and often when there's really nothing to be scared at at all.

#### WHEN THOSE EXTRA DOORS COME HANDY

But you can't blame them. There are times when it's no fun being a chipmunk, I tell you. The hawks get after you, and the minks and the foxes and the weasels. Those extra doors into the nest are very useful places to dodge into when you're outside and a savage old hawk swoops down on you, or a fox makes a jump at you. And they're just as handy—these extra doors—to run *out* of when a mink or a weasel follows you in. They'll do that, if you're a chipmunk; chase you right into your own house!

When a pair of grown-up chipmunks start housekeeping for themselves—that is to say when they are about ten weeks old—they first dig a little tunnel, almost straight down for several feet. Then they make a hall that runs along horizontally—like anybody's hall—for a few yards. Then, supposing you're Mr. or Mrs. Chipmunk in your new place, after it's all done—you go up a slant—a flight of stairs, you might say, although, of course, there aren't any stairs—and there you are in the family bedroom, the nest.

Not long after the chipmunks stop their outdoor games in the Fall you might think it was because they had the mumps; they go around with their faces all swelled out in such a funny way. The reason is they have their cheeks full of nuts and seeds that they are storing for the Winter. They don't put these stores in the nest—for then where would they sleep, the nest is so small—but in special cellars that they build near the nest, with connecting passages. These cellars, like the nests, are well below frost-line, so that Jack can't get the nuts or nip the noses of the chipmunks while they are asleep.



#### PICTURESQUE HOME OF A CONNECTICUT WOODCHUCK

This is the truly artistic residence of a Connecticut woodchuck which I found in a rocky knoll by the wayside during a summer vacation at Kent and reproduced as well as I could with my fountain-pen. Mr. W. as he often does in digging his burrows, had availed himself of the protection of the roots of a tree. Here there were two projecting roots, forming a curious arch over the doorway, which was tastily decorated by a little overhanging vine, on its way up the knoll, along the stones, and up the foot of the tree.

When Winter finally sets in, the chipmunks get very drowsy and go up to bed. And there they stay until Spring—one great long nap, except that they wake up and stir around occasionally on bright days and if it happens to warm up a little.

"Such sleepyheads!" you say. "And what about all those nuts? I should think they'd be fine for Winter parties."

They would, I dare say. But you know a body doesn't have much of an appetite when he doesn't get any outdoor exercise, and that's why the chipmunks only take a few bites now and then, during the Winter. And, besides, if they ate up everything in the Winter—you know how folks eat at parties—what would they do in the Spring, with no good nuts lying around on the ground, as there are in the Fall; and nothing else to be had that chipmunks care about? So they keep most of the nuts and seeds and things for the great Spring breakfast, and all the other meals, until berries are ripe. The berries they eat until the next nut harvest comes along.

Until then, you see, they haven't much of anything to do but play around and sit in the sun and chat. So why shouldn't they?

### HIDE AND SEEK IN THE LIBRARY

You will find some most readable things about foxes in Burrough's "[Squirrels and Other Fur Bearers](#)"; Comstock's "Pet Book"; Cram's "Little Beasts of Field and Wood"; Wright's "Four-Footed Americans"; Jordan's "Five Tales of Birds and Beasts"; Long's "[Ways of Wood Folk](#)"; and Seton's "[Wild Animals I Have Known](#)."

Comstock's "Pet Book" also tells about the prairie-dog; and Seton, in his "[Wild Animals I Have Known](#)," tells about "The Prairie Dog and His Kin."

It's a very common superstition among English country folk that shrews always drop dead if they attempt to cross a road. How do you suppose such a strange idea ever got started? Allen, in his "Nature's Work Shop," reasons it out, and his reasons seem very plausible. It's a fact that their dead bodies are nearly always found in roadways. You'll also find some interesting information about shrews in Johonnott's "Curious Flyers, Creepers and Swimmers" and Wright's "Four-Footed Americans."

There's some little dispute about squirrels as tree-planters; that is to say as to just how they do it, for there's no question that they *do* plant oaks and other trees. Thoreau, in his "[Walden](#)," gives the squirrel credit for doing an immense amount of tree-planting, but Ernest Ingersoll, in his article on squirrels in "Wild Neighbors," thinks the squirrel leaves

comparatively few acorns or hickory-nuts, and that he doesn't forget where he puts them, as other writers on nature say. "They seem to know precisely the spot," says Mr. Ingersoll, "where each nut is buried, and go directly to it; and I have seen them hundreds of times when the snow was more than a foot deep, wade floundering through it straight to a certain point, dive down, perhaps far out of sight, and in a moment emerge with a nut in their jaws."

But *how* the squirrel knows it's there—that's the mystery! Read what Ingersoll says about it. The whole essay is extremely good reading, and will tell you a number of things to watch out for in squirrels that you perhaps never have noticed.

In Pliny's "Natural History" you will find, among other quaint stories, one to the effect that mountain marmots put away hay in the fall by one animal using itself as a hay-rack—lying on his back with his load clasped close while he is pulled home by the tail. "Animal Arts and Crafts" tells what a simple little thing originated this idea. Many of the peasants of the Alps still believe it.

Hornaday, in his "Two Years in the Jungle," gives an interesting account of how one of the four-footed knights in armor—the pangolin—does himself up in a ball, and how next to impossible it is to "unlock" him.

Ingersoll, in discussing the various uses of tails in "Wild Neighbors," tells how a gerboa kangaroo brings home grass for his nest, done up in a sheaf of which his own little tail is the binder.

An interesting four-footed burrower, when he can't rob a prairie-dog of his hole—or some other body smaller than himself—is the coyote. There is a long talk on the coyote and his ways in "Wild Neighbors." This little book also gives pictures of the different kinds of shrews in the United States, and a lot of detail about them and their little paws and their noses and their tails.

It's a queer thing how systematic and prompt shrews and moles are in business. You can actually set your watch by them, as you will see in the same book.



In the article on the gopher in the "Americana" you will find how the gopher got his name. Can you guess, when I tell you it's from a French word meaning "honeycomb"?

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## CHAPTER VIII

(AUGUST)

'Till he came unto a streamlet  
In the middle of the forest  
To a streamlet still and tranquil  
That had overflowed its margin,  
To a dam made by the beavers,  
To a pond of quiet water,  
Where knee-deep the trees were standing,  
Where the water-lilies floated,  
Where the rushes waved and whispered.

—*Longfellow: "Hiawatha."*

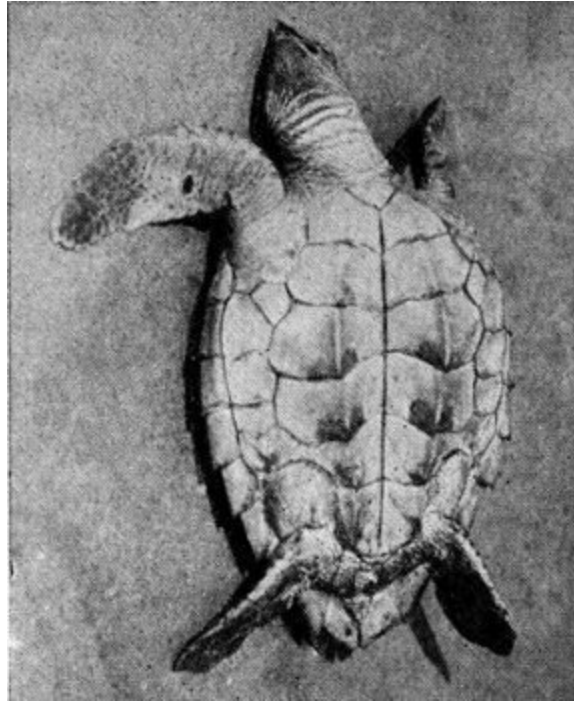
### WATER FARMERS WHO HELP MAKE LAND

As we all spend more or less time in the water in August I thought it would be a good idea to take as the subject of this chapter the lives of the water farmers. Some of these—the crayfish and the turtle, for example—you know well, and everybody has heard of the beaver family, but they will all bear closer acquaintance. I know, for I've spent a good deal of time among them.

#### I. THE TURTLE PEOPLE

Every boy who has tramped along creeks and ponds knows the mud-turtle. We ought to call him a tortoise, perhaps, but the name turtle is more common. I don't know why; perhaps because it's a little easier to say. Strictly speaking, the name "turtle" is applied to the members of the family that have flippers, and spend nearly all their time in the water; while the tortoises are the ones that have feet and put in much of their time on land.

(And then, of course, there are the tortoises of fables that run races with hares, and so teach us not to be too confident of ourselves because we think we are cleverer than some other people.)



A HAWKSBILL TURTLE

The common box-turtle of the United States you'll meet in the woods in the evening and early morning, wandering about looking for something to eat. He spends practically all his time on land in Summer; and in the Winter, all his time in bed. As soon as cold weather comes on he digs a hole in the ground, or scoops out a place under some brush, and turns in.

But the box-turtle—he's really a tortoise—is what some of his relatives would call a "landlubber," no doubt, for many of the tortoises who live in the sea rarely leave it; as if they had half a mind to go back and be only flipper people, as the ancestors of both the turtles and the tortoises must have been; since all life is supposed to have begun in the sea.

All the tortoises of temperate regions dig in for the Winter, but one Southern member of the family makes his home in a dugout throughout the year. He's called the "gopher" turtle. The gopher turtles are natives of Florida, and live in pairs in burrows. Other members of the turtle tribe do not pair, but there's one time in their lives when both land and water turtles dig into the soil and that's when they are laying their eggs. The females scoop out hollows with their hind legs, kicking up the dirt, first with one leg and then with the other. But they're as careful of the dirt they dig out as a beaver is when he digs a canal. They scrape it up in a little ridge all around the hole.

What for? Just watch.

#### **HOW MOTHER TURTLE "TAMPS" HER NEST**

As soon as she has finished laying her eggs, Mother Turtle carefully scrapes this dirt back over them and tamps it down, much as a foundryman tamps the sand in a mould. You can guess what she uses for a tamper—the under side of her shell, raising and lowering herself on her legs like a Boy Scout taking his morning setting-up exercises in a Summer camp. After that she doesn't pay any more attention to her eggs. She leaves the sun to do her hatching for her. Both land and sea turtles—or, more properly speaking, the tortoises and the turtles—hatch their young in this way. The sea-turtles scramble up out of the water on their flippers, much as a seal does in climbing on a rock, and make their way back from the shore, great crowds of them, at nesting-time, to some stretch of sand, and there lay their eggs. This march of the mother turtles always takes place at night. When the young are hatched they dig their way up through the sand and make for the sea.

## **II. THE CRAB FAMILY**

Another one of the water people who help make land and one that everybody knows, is the crayfish. Every small boy is afraid Mr. Crayfish will catch his little big toe sooner or later, when he goes swimming; although I never heard of a crayfish that did. But they never worry about *their* toes—the crayfish don't. When they lose a whole foot even—as they

often do—it grows right out again. The science people say this is because they belong to a low order in the animal world, but I think it would come in right handy for any of us—this way of regrowing not toe-nails alone, but toes and all—don't you?

The crayfish, as you may know, love to burrow in the mud, for you are always coming across their little mud towers along the margins of the brooks. Related to the crayfish are the crabs. Mother Nature seems to have been very fond of crabs—she has made them after so many different patterns and scattered them all over the world; in the deep sea, along the shallows of its shores, and on land. Those you are most apt to meet must have more or less business on land, for the shape of their legs shows that they are formed for walking rather than swimming. But go far out to sea and you'll find crabs with paddles on all four pairs of legs, like banks of oars; while others, living on the borders of the sea, have paddles only on the last pair.



#### SOUTH SEA ISLAND AND COCOANUT COLUMBUS

Here we are on an island of the Southern Seas—the home of a colony of cocoanut crabs. One of the members of the colony is

climbing a tree to get a nut. "And who has a better right?" says he. "This tree," he might continue, "is the descendant of a nut that some of my ancestors sailed upon to this island; for a cocoanut, dropping into the water from a tree near some far shore, often carries on it the crab who had started to eat it. Then a current of the sea carries the nut and its passenger to some other island. Later cocoanut Santa Marias and their Columbuses reach the island in the same way, and so it becomes populated with both cocoanuts and crabs—which makes it very nice for the crabs!"

One of the big families of crabs live on land most of the time and make burrows in which they live. These have legs specially fitted for digging. Like most of the crab family, the land-crab earns its living at night and, except in rainy weather, seldom leaves its burrow by day. Like small boys, these crabs seem to love to play in the rain. The fact is they do this to keep their gills wet; for, although they spend most of their time on land, crabs breathe with their gills, like fish; and while some of them—as the mountain crab of the West Indies—live quite a distance back from the sea, they must have some moisture for their gills, and this they get, in part, in their damp cellars—the burrows.

But it's queer, isn't it, what different ways people have of looking at things? Take land crabs and turtles, for example. Turtles, when they lay their eggs, think the only thing is to get clear away from the water and put their eggs in an incubator, as we saw them do a few pages back. The land-crabs evidently think just the opposite; for no matter how far they may live away from the sea—one, two, even three miles sometimes—nothing will do but they must go to the water to lay their eggs. In April and May you'll see them swarming down by hundreds and thousands. And they'll climb right over you if you don't get out of their way!

"This is my busy day and I can't stop for anything," says Mrs. Crab.

Besides the work they do for the soil in grinding and mixing it, the crab people, like all the crustaceans, help a lot by adding lime to it, and that's one of the very best things you can do to soil, you know. They add this lime when they change their clothes; that is, when they moult or cast their shells. The shell they take off as if it were indeed a dress. They "unbutton" it down the back. Sometimes, in trying to get out of the legs of the suit, they leave

not only the leg covering but the leg itself. That leg is good for the soil, too, of course, and the loss of a leg doesn't bother a crab so very much. He just grows a new one, that's all!

These shells—particularly the shells of the largest species of crabs—not only contain a great deal of lime but carbon and phosphorus, also, and these are splendid soil stuff, too. In the smaller kinds of crabs—of crustaceans, generally—these shells are mostly chitin, the stuff that the coverings of insects is made of.

The crustaceans, by the way, are closely related to the insects. You may *suspect* this by comparing their shapes, but then you'll see there isn't any doubt about it when I tell you that in getting born from the egg, the crabs and their kin don't come out dressed in their final shape, but change after they are born, first into one shape and then into another, just as insects do. Each shape, as it comes along, looks funnier than the rest; that is, it looks funny to us, but not, naturally, to the crabs. It must seem just the thing to them, for they always dress the same way and look as solemn about it as a man does when he wears a monocle. In fact, they do something almost as funny as wearing a monocle. For many of them carry their eyes about, not on the end of a cord, to be sure, but on the end of a stick. These "sticks" are called foot stalks. And they're not a bad idea either—for a crab. By moving them around the crabs can keep much better posted on what is going on about them than they could otherwise; particularly as a crab always moves sidewise or backward. What good a monocle does, though, nobody knows.

### **III. THE STRANGER THAT MADE LONDON LAUGH**

But if we can hardly look a crab in the eye and keep a straight face, what would we do if we met a duck-billed mole? We'd laugh right out! I'm sure of it, for that's what even the men of science did when they saw the first one that came to England. This strange foreigner—it came to London all the way from Australia—had a body like a mole. But you couldn't call it a mole. For one thing, it had a bill like a duck. Yet no more could you call it a duck; for, besides having a body like a mole, it had a tail like a beaver. Still I'm afraid the beavers wouldn't have owned it—hospitable as they are—

even if they could have overlooked that bill. For—can you believe it?—this duck-billed, mole-bodied, beaver-tailed creature lays eggs!



#### THE ANIMAL X FROM THE ANTIPODES

A mole's body, a duck's bill, a beaver's tail, this strange citizen of that land of strange animals, Australia, lays eggs like a bird and suckles its young like a pussy-cat! Do you wonder that the wise men of London laughed at the idea that there is any such creature—even when they were looking right at one?

Yet the ducks just couldn't take it into their families either, for what else do you think it does? It suckles its young, like a pussy-cat! Talk about your sensations; it made the hit of the season—this Animal X from the Antipodes. The learned men of London town, they looked him up and they looked him down, and they came to the same conclusion, at first, that the old gentleman did when he saw the dromedary. They said: "They *ain't* no such animal!" (Only, of course, being learned men, they used good grammar.)

They really did say that in effect, and you can't blame them; for, as if to complete the joke, the first member of the duck-billed mole family to move in scientific society came in like a Christmas turkey; in other words, he was a stuffed specimen. So the men of science said he wasn't *real* at all; that he was just made up of the parts of *other* animals. But being true men of science, after all, they finally began looking up the stranger's record among



his neighbors back in Australia, and they found there actually are living creatures in that land of strange creatures, just like that specimen, and that they live in burrows which they dig in the banks of the streams.



COUSIN ECHIDNA

The echidna—you can see one in the New York Zoo—is closely related to our duck-billed friend and is also a native of Australia. It uses that long, tapering nose and those claws to burrow for the ants on which it lives.

Still the scientists didn't know what to call this paradox of the animal kingdom; so they named him just that—paradoxicus, *Ornythoryncus paradoxicus*. A little Greek boy, without having to look it up in a dictionary, would have told us that "ornythoryncus" means "bird-billed"; for it's like those Greek picture words that always told their own story to the little Greeks. As for "paradox" if you don't know what that means, look it up in the dictionary and then look at the *Ornythoryncus paradoxicus*, and you'll understand.

#### IV. THE BEAVERS

Of course you wouldn't like to be a duck-billed mole—nobody would, but I always thought it would be rather nice to be a beaver. The beaver is, in many ways, the most remarkable of all the water people that help make the lands that give us bread.



#### BEAVERS AT WORK AND AT PLAY

Whether he's working because he is more industrious than those beavers in the water or because it's recess time with them, the young beaver gnawing the tree seems to be having quite as good a time practising his profession as the others do in playing about.

But it is not alone for the amount of work he does that I admire Mr. Beaver so much; it is for his intelligent, not to say brilliant, way of doing it. Suppose, for instance, you had to build a house out in the water, the way our great, great-grandparents, the lake-dwellers, did, to protect yourself from enemies and for other reasons. And then suppose you didn't have any *tools*; nothing but a pair of paws and a set of teeth. Could you do it?

Another thing: The lake-dwellers had plenty of water to build in; plenty, but not too much. The beavers don't have this advantage. They usually build in the water of flowing streams, and they have to make their *own* lakes. How would you do it; even if you had tools? But remember, being a beaver, you've got nothing to use but two honest paws and a set of teeth. It was with these Mr. Beaver did it all—with his teeth, his paws, and his head; the inside of his head, I mean—his brain. Take the matter of water arrangements. He gets the water to lie quietly and at just the right depth by

building his dam across the stream. This dam not only provides him with water of just the right depth to protect his front door from enemies and to keep rushing torrents from carrying his house away, but the spreading out of the original stream bed into a pond helps in gathering the Fall harvest of trees, since it brings the trees nearer to the water's edge, and water transportation among beavers, as among men, is always cheapest.

Although dams are usually built of trees which the beavers cut down themselves, they also use cobblestones where trees are scarce; for Mr. Beaver is a very thrifty soul; he doesn't waste material nor time nor effort. Many books about beavers say they cut the trees so they will fall across the stream, but Mills says, in his book on the beaver, written after many years of patient observation, that beavers don't seem to care how the tree falls, just so it doesn't fall on *them*! Not but what they *could* cut trees to fall in the water if they thought best; for just watch them build a dam and see how clever they are; cleverer, possibly, than some of us.



BEAVERS AT WORK ON A DAM

See how many of the features of the building of a beaver dam, as described in our story of these wise little people, you can make out in this picture.

Let's see. Say you've got your trees up to where the dam is to be; now how are you going to set them in building the dam?

## SEE IF YOU'RE AS CLEVER AS MR. BEAVER

"Right across the dam," you would say, wouldn't you? That is what most people have said when I have asked them that question; for that is the way men do it. But remember, if you built the dam as men build dams you would have to drive stakes or do something to keep the logs from washing away. Years ago, when writers used to theorize a great deal on how things were done, instead of getting outdoors and watching patiently to see how they actually *were* done, it was said that Mr. Beaver in building his dam did really drive stakes and that he did it with that big tail of his. But what Mr. Mills found was that the beaver lays his trees lengthwise of the stream. You see why that is, don't you? When the trees are laid lengthwise, the water, instead of striking them broadside, strikes only the end and so there is less likelihood of their being carried away.

Another thing, two things, about the trees in the dam—in fact four:

1. It wouldn't do, you see, to lay the trees broadside to the stream, but what position could we give them that would help still further in keeping the water from carrying them away?
2. Shall we use trees with the branches still on them or trees trimmed down like sticks of cord-wood? (What kind do you see in the picture of the beaver dam?)
3. Or shall we use both trimmed and untrimmed trees? If so, why? And how?
4. If we use untrimmed trees, which end shall we put up-stream? The butt or the tip?



SECTION OF A BEAVER DAM

You can see that there was a sufficient flow of water in the stream from which this sketch of a section of a beaver dam was taken; otherwise the dam would have been plastered with mud to conserve the supply. The longest slope, of course, was up-stream—a fundamental principle in beaver bridge engineering.

In building his dam the beaver uses, for the most part, slender green poles trimmed and cut in lengths; but mixed with these are small untrimmed trees which he places with the butt end up-stream, and propped with mud and sticks so that the up end will be a foot or so higher than the down end. In this way, you see, the branches are made to resist the push of the waters against the butt end; while, if they were placed the other way, the current would have a pulling purchase on the butt end. The raising of the ends also lessens the pushing force of the water as it doesn't strike the butt of the tree "full on," as it would otherwise do. And the branches not only help to hold the trees in place, but, together, form a kind of foundation on which to pile and intermix the trimmed poles.

The timbers, being cut green, become water-soaked. This makes them heavier and so causes them to sink and helps to hold them in place; while the branches and twigs of the untrimmed trees form a kind of basketwork that catches the sediment and drift of the stream, and so the dam lets less and less water through. The upside stream is plastered by the beavers with mud in cases where the flow of water in the stream is meagre. Otherwise it is left unplastered. You see Mr. Beaver's idea is not to make the dam absolutely water-tight, for then it would be running over all the time and so be worn away. What he wants is a dam that will let the water through slowly and at the same time keep a proper level.



#### BEAVER HOME WITHOUT TIME LOCK

Here is a beaver home as it looks before the time lock is put on in the Fall.

Mr. Beaver's chief purpose in building these dams seems to be to keep his front-door yard full of water. This may look like a funny idea at first, but in this, as in other things, Mr. Beaver shows he has a very wise head on his shoulders; for one peculiarity of his life is that he is obliged to come and go through the cellar door. As he doesn't want any of his enemies—the wolf, the coyote, and all that class of people—to use this door, he keeps it under water. And in winter-time, when he goes out to the wood-pile to get something to eat, the water must be deep enough so that the pond doesn't freeze solid to the bottom.



A BEAVER HOME WITH TIME LOCK

Here, as it looks after being made secure against hungry wolves and the Winter winds.

As for those professional highwaymen, the wolves and coyotes, that are so much bigger than he is, Mr. Beaver keeps out of their way in Summer, when they don't bother much about him, anyway, as he sticks so close to the water and is hard to catch. In the Winter, when they get hungry and desperate and would break into his house, if they could, he makes it practically burglar-proof, by putting on a time lock; a lock that just won't open, even to a wolf's sharp claws, until Spring.

And in the simplest way.

Just before Winter sets in Mr. Beaver plasters the outside of his house with mud, and the mud freezes as hard as a stone. But sometimes, even among the beavers, there are shiftless characters, like that Arkansas man who just *wouldn't* look after his roof. These careless beavers don't plaster their roofs. But then, just see what happens! Some hungry wolf comes along and breaks through and has a nice fat beaver for supper, maybe. And maybe not; for,

even in that case, if Mr. Beaver wakes up in time, he dives down through the cellar door and into the tunnel and out under the ice.

"Aha! You got fooled that time, didn't you? You mean old thing!" (Can't you almost hear him say it?)

In putting the mud coating on their houses or dams the beavers carry it in their fore paws. Sometimes, in a very steep place, they climb up the roof with three feet and hold the mud with one. When they have delivered the mud they use these same little paws to pat it down—not their trowel-like tails, as one would naturally suppose.

#### **THAT MYSTERY ABOUT THE BEAVER'S TAIL**

Then what *do* they do with those tails? Well, for one thing, they sometimes use them to carry mud by curling them between their legs and holding the mud against their bodies. Perhaps they resort to this way of carrying mud where they have such a steep climb up the roof they need all four legs to climb with; or it may be just an individual fancy of some beavers. For, being really *thinkers* and not mere machines, acting entirely on what is called instinct, different beavers have different ways of doing things. The beaver's tail is also very useful in swimming, and Mr. Beaver is a great swimmer. You should see him. He swims mostly with his hind feet and tail, holding his fore paws against his breast as a squirrel does when he's sitting up looking at you. His tail he uses as one uses an oar in sculling, turning it slightly on edge as he works it back and forth.

But he has two other important uses for this big tail, as we shall now see; for the beavers of this colony we are watching, having put up their dam and built their big house, are now ready for the Fall harvest that is to provide for the long Winter. The beavers are strict vegetarians. Their diet consists of the tender bark of young trees and roots dug from the bottom and along the banks of the ponds in which they live.

"But, for mercy's sake, where are they going to get the tender bark of trees in the dead of Winter, when all the trees are frozen solid and the beavers can't get from under the ice anyhow?"



Well, Mr. Beaver has thought out just how to do it and we didn't. That's the beauty of being a beaver. What he does is to cut down small trees, trim them, divide them into lengths, and then heap them up in a great pile at his door, under the water.

By the time they are three years old beavers feel grown-up; as, indeed, they are in size, although, like certain other young people I could name, they have a great deal yet to learn. At this age they choose their mates and either settle down in the home colony or go away somewhere else.

School takes up with the beavers in September. All through September and October the harvest is gathered and preparations made for the long Winter. The baby beavers of the Spring, who by this time are four or five months old, take part in the harvesting; at least they play at it. They don't do much, but they learn a great deal. Now let's all be little beavers for a few minutes and see what we can learn. We are out in the harvest-field—the woods—with father, and he's going to cut down a tree for the Winter food-pile. Watch him.

He picks out a young tree something less than six inches thick. Then he looks up as if he wanted to see what kind of a day it was going to be; although the fact is he never bothers his head about the weather. What he is really looking up for is to see if the top of the tree he is going to chop down is likely to get tangled in the tops of other trees when it falls. (All beavers, I should add, don't take this precaution; only the older and wiser ones.) After this inspection he either cuts the tree in two with his long sharp chisel teeth so that it will fall clear of the tangling branches of other trees, or, if he sees he can't prevent this, he moves away to another tree.

Just before the tree is ready to fall he thumps the ground several times with his tail to warn other beavers working near by. They all scamper as fast as their fat bodies and short legs will let them. If they are near water, as they usually are—they "plunk" into it. After the tree falls the limbs are cut off, the trunk gnawed into sections four to six feet long, depending on the size of the trunk, the distance from the water, and the number of beavers that are going to help move it. Although, as a rule, only one beaver works on a tree in cutting it down, they all pitch in and help in getting the sections home;

dragging them across the ground and into the pond or into one of their wonderful canals.

### THE BEAVERS AND THEIR PANAMA CANALS

The beavers knew all about digging canals long before the days of Colonel Goethals. They dug them for much the same reason we dug the great Panama Canal, to save time and expense in moving freight and for protection from possible enemies. On land the beaver is easy prey for wolves and such, but once in the water he can laugh at them. These canals not only enable him to haul his wood easily and safely, but are just the things to dive into when somebody is after you. Another purpose of the canals is to fill ponds where water is getting low; or to make a pond where there isn't any at all, as in a dry ravine.

Whether you look at them from the standpoint of their intelligence and good habits, or their usefulness, beavers are the most interesting of all our little four-legged brothers of field or wood, and it is pleasing to know that many States have passed laws to protect them.



SUN BATH AFTER THE SWIM

Boys, after an hour or so in the "ole swimmin' hole," like to take a sun bath. That's what these young beavers are doing on a nice grassy spot by the pond.

And besides he is such a good fellow, Mr. Beaver is; peaceable, industrious, dependable, and with the best heart in the world! Why, do you know what they do—the beavers—when neighbors get burned out by forest-fires or their houses broken into by a mean old wolf or coyote or anything? Take them right in, children and all!

If you were a little beaver you'd have from two to four twin brothers and sisters to start with, and then two to four more for each of the remaining two years before you left home to make your own way in the world. You'd be born with your eyes open and not like a puppy or kitten. And, what do you think, *in less than two weeks* you could go swimming. Mother would be right with you in case anything happened. Then when you were tired swimming you'd climb up on top of the house and rest and doze in the sun; take your afternoon nap just like any other baby.



LITTLE BEAVERS IN THEIR HOME

But maybe it wouldn't be your own mamma that would be with you; for lots of sad things happen to beaver people, and when one little beaver's mother dies another mother beaver will take care of him, and all his brothers and sisters besides! Mr. Mills tells in that most interesting book of his about

how one day a mother beaver was killed by a hunter who thought he didn't have anything better to do than kill poor little beavers; and the very next evening a lady beaver, who *already* had four babies of her own, travelled a quarter of a mile with them to the house of her dead neighbor and stayed there and brought all the little orphans up!

## HIDE AND SEEK IN THE LIBRARY

The crayfish is a thing you've got to take seriously if you want to get the most out of it. Huxley says that a thorough study of a crayfish is almost a whole course in zoology. Think of going to school to a crayfish! But you'd enjoy it, I'm sure. For just look—and these are only a few of the interesting things you will find in Huxley's famous book on "The Crayfish":

How they swim backward (no doubt you know this already), and how they walk on the bottom of the water.

Why they seem to know the points of the compass—for they prefer rivers that run north and south.

Why they are most active toward evening.

Where they spend the winter.

Why they eat their old clothes.

How early in the spring you may expect to find them.

When they hatch their eggs and how the mother crayfish uses her tail for a nursery.

In what respect they resemble moths.

How they chew their meals with their feet and work their jaws like a camel from side to side—only more so!

How they grow by fits and starts, and what this has to do with the way they change their clothes.

How you can tell the age of a crayfish. (You don't do it by looking at its teeth. You couldn't see its teeth anyway, because they are in its stomach.)

And all this in less than the first fifty pages of a book, which has more than 350.

One of the most famous of the crab family, not only on account of his part in agriculture, but because of his funny ways, is the robber-crab. You should read about the wild life of adventure some of these crabs lead—regular Robinson Crusoes who get wrecked on islands far away from home and build houses there and shift for themselves in many ingenious ways, just as the human Robinson Crusoe did. Kingsley's "[Madam How and Lady Why](#)," has some interesting pages about them; and so has Darwin's "Voyage Around the World."

Of the many things that have been written about beavers the following are among the most interesting: The story of the beaver in "Stories of Adventure," edited by Edward Everett Hale; "The Forest Engineer," by T. W. Higginson, in Johonnott's "Glimpses of the Animal World"; "How the Beaver Builds His House," in "The Animal Story Book," edited by Lang; "The Builders," in Lang's "Ways of Wood Folks"; and "The House in the Water," by Roberts.

The most interesting book of all on beavers, however, is "The Beaver World," by Mills, referred to in this chapter. I have not told you one-half of the remarkable things you will find about them in this book.

One of the most curious is about how a beaver sometimes gets his breath in the winter time. He may have to travel quite a distance under the ice, and one good breath has to last him to the end of the journey.

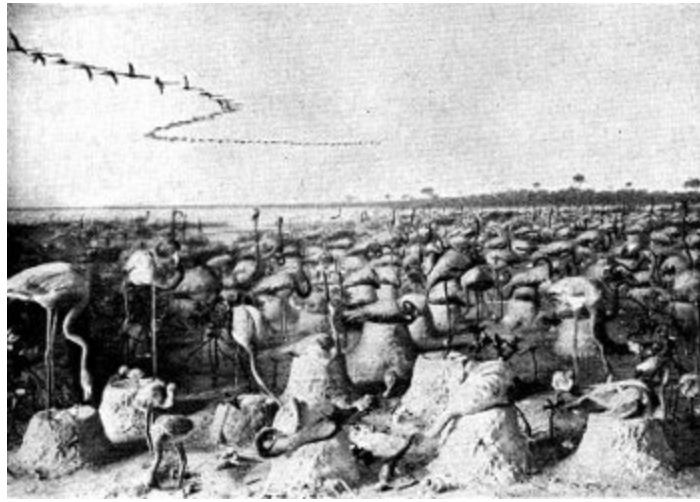
"But does he hold his breath all this time? How can he?"

He can't. He just uses the same breath over again. See how he does it. The Mills book tells.

Look up the muskrat and compare his ways with those of the beaver.

In the "Country Life Reader" you will find a graphic description of one of the perils of life for the beavers and their cousins the muskrats; namely in attacks by the great horned owl.

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#### CITY LIFE AMONG THE FLAMINGOES

We don't have to go to Florida to get this bird's-eye view of a flamingo city. It is one of the habitat groups in the American Museum of Natural History in New York, and reproduces perfectly the architecture and the social life of these interesting people.

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## CHAPTER IX

(SEPTEMBER)

On the housetop, one by one  
Flock the synagogue of swallows  
Met to vote that Autumn's gone.

—Gautier:

"*Life.*"

### FARMERS WHO WEAR FEATHERS

Sh! Go easy! Pretend you're a horse or a cow.<sup>[21]</sup> We've gone south with the swallows—it's September you see—and those queer birds over there are flamingoes. The flamingoes are a shy lot; I don't know why. I can't think it's on account of their looks; for there's the kiwi, the hornbill, and sakes alive—the puffins! *They* all have funny noses, too, but none of them are particularly shy, and you can walk right up to a Papa Puffin almost. Whatever the reason is, the flamingoes are very easily frightened and they're particularly suspicious of human beings. Yet we've simply got to meet them and have them in this chapter, for they are among the most interesting of the feathered workers of the soil. They just live in mud; build those tower-like nests out of it, walk about in it, and get their meals by scooping up mud and muddy water from the marshes where they live, on the borders of lakes and seas. They strain out the little creatures wiggling about in these scooped-up mouthfuls.

#### I. FEATHERED FARMERS WITH QUEER NOSES

"What a funny nose! What happened to it?"

I knew you'd say that. Everybody does. But just watch now and see. That flamingo over there, stalking about on his stilt-like legs, sticks his long neck



down to the muddy water, turns that funny nose upside down and——

"Why, of all things, is he going to stand on his head?"

#### **WHY FLAMINGOES HAVE SUCH FUNNY NOSES**

No, not that. Don't you see, he's getting his dinner? After that crooked scoop bill—for that's what it really is, a scoop—is filled, the water strains out through ridges along the edge of the bill and what's left is his food.

That picture looks as if it had a tremendous lot of flamingoes in it, doesn't it? It has. It's quite a town, Flamingoburg is. Although flamingoes are so wary about meeting two-legged people without feathers—that is, human beings—they're very sociable among themselves and there may be a thousand, even two thousand, pair in a single flamingo city, such as Doctor Chapman studied in the Bahama Islands some years ago.

Their nests are cupped-out hollows in little towers of dried mud raised a foot or so to keep high tides from swamping them. They scrape up the mud with that shovel-like bill. After the conical-tower nest is made, the mud piled up and patted into shape with her bill and feet, Mother Flamingo lays one or two eggs—and then she goes to setting. You notice there's just one little chick in the nest in the lower left-hand corner of the picture, and just one egg in the nest near by.

With such a low stool to sit on you wonder what the mother bird does with her long legs. In some pictures in children's nature books of not so many years ago you'll find her represented as sitting on the nest with her legs hanging down the sides—but you see that couldn't be; the nest isn't tall enough. What she really does is to fold her legs under her body; just once, of course, at the joint. But they're so long that, even when folded, they reach out beyond her tail. While setting, the lady birds reach around with their long necks shovelling up things to eat and gossiping, more or less, with the neighbors; for the nests, you notice, are very close together. Sometimes two of them will reach across the narrow alley that separates the residence of Mrs. Flamingo Smith from Mrs. Flamingo Jones, take each other playfully by the bill and hold together for a while. Maybe this is their way of saying "Good morning," or "How do you do?"



#### THE TOILETTE

You'd expect a lady wearing so many nice feathers to be particularly careful about her dress, wouldn't you?



#### A LITTLE NAP

Queer notion, sleeping on one leg like that, isn't it? But then flamingoes *are* queer!



#### A TOUCH OF RHEUMATISM

Of course flamingoes don't go around like that even in zoos. This is the artist's joking way of telling that in our northern climate they are subject to rheumatism. And the keepers actually do oil their legs.

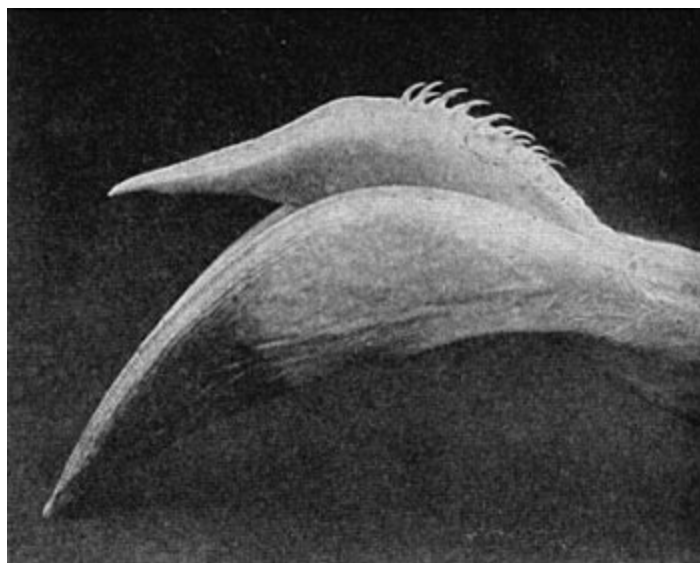
FLAMINGO SOCIETY NOTES FROM THE ZOO

You'd hardly think it—with those long legs of theirs—but the flamingoes swim beautifully. With their long necks drawn back—the way swans do it, you know—they are very graceful, and a flock of them floating about is one of the loveliest sights in the world. They look like a big, fleecy, pink cloud resting right on the surface of the water. You can now find only a few flamingoes in Florida, where there used to be so many; but go on south into Central and South America and there are thousands of them. They are still fairly numerous in countries bordering the Mediterranean and the Indian Ocean. In Persia they are called "red geese." And the name isn't so far wrong as you'd think. You notice that, unlike those stilt-walkers, the herons, the flamingoes have webbed feet. Like geese and ducks, also, they have those rows of tooth-like ridges on the edges of their bills. It is these "teeth" that, coming together, act as strainers.

But a queer thing about their bills, besides the funny-way they have of crooking down all of a sudden, is that the upper bill is smaller and fits down into the lower. Stranger still, the birds can raise and lower this upper bill like the cover of a coffee-pot.

They can move the under bill a little, too, but not to amount to anything; so you see there was even more to the upside-downness of that bill than there seemed to be at first. The whole arrangement looks odd to us, but it works out beautifully for the birds. When they turn their heads upside down they can stir the ooze to various depths, as required, by using the upper bill as a ploughshare and setting it at different angles.

Although they've borrowed some ideas from both the goose and the heron families, the flamingoes are so different from either they are put into a family by themselves, the *Phænicopteridæ*. This family name is from two Greek words meaning "red-winged." If you want to be formal in speaking of or to a goose you must refer to her family as the *Anserinæ* which is Latin for "geese."



#### WHERE THE FLAMINGO KEEPS ITS TEETH

While teeth, like those of the *Hesperornis*, went out of fashion ages ago, the flamingoes have substitutes for teeth which answer their purposes much better. They have little horny spines on their bills and on their tongues. These spines serve as fences to prevent the escape of the minute creatures which the flamingo scoops up with its bill. You notice the spines on the tongue are pointed backward toward the throat; and that's a help—to the flamingo, I mean, for once on that tongue there's no turning back.

#### A LATE BIRD, BUT HE GETS THE WORM

Another of the long-nosed earth workers, as curious in his make-up as the flamingoes, is the kiwi of New Zealand. Like the flamingo, the kiwi uses his queer bill to get his living out of the soil. You've heard the saying "it's the early bird that gets the worm"; but while this is true of most birds it doesn't apply to the kiwis. Although they live on worms, as does Mr. Early Bird of the proverb, they do their feeding by night.

And such a funny thing for a bird to do, the kiwis go about with their noses to the ground like a dog smelling after a rat. The reason they do this is that their nostrils are situated, not next to their heads, as in most birds, but at the end of the bill—and on purpose; for they locate their suppers, the worms in

the earth, by the sense of smell, although most birds have a very poor sense of smell. Just after sunset, you'll see the kiwis moving about softly (as if they were afraid of scaring away the worms!), and with the tips of their bills against the ground.

"Sniff! Sniff!" (You actually can hear them sniff.)

There, he's found one! His bill is not only long, but bends rather easily and that's why, perhaps, he's able to follow up so closely the hints he gets from his nose as to the location of worms, for he usually brings the worm out whole, and not all pulled apart as the robins do it sometimes. He works in soft earth, where most worms are found, and generally drives his bill in up to his forehead. If all goes well he pulls it right out with the worm at the end; but if there is any likelihood of an accident, the kiwi gently moves his head and neck to and fro until he has the soil loosened up and so clears the way. Once the worm is fairly out of the ground, he throws up his head with a jerk and swallows it whole.

Because they roam about so much at night, the kiwis sleep much of the day. You'll find them in thickets or in among the forested hills, where they make their homes. Sometimes, however, you'll see one standing, leaning on his long bill, like a street-idler propping himself up with his cane. If you disturb him, he yawns, as if to say:

"Oh, these bores! Why can't they let a fellow alone?"

But don't you go too far and annoy him or he'll get real peevish and strike at you with his foot.

Both Mr. and Mrs. Kiwi drill the earth every day—or rather every night—in their search for worms, but Lady Kiwi does all the excavating when it comes to making the nest. This she does by digging a tunnel, generally under the roots of a tree fern. There she lays two eggs and then her family cares are practically over for the time being, since it is the male kiwi who does most of the setting.



#### MR. HORNBILL LOCKS THE DOOR

In Africa, Southern Asia, and the East Indies live the Hornbills. After the nest is built and the eggs laid in the hollow of some big tree like that, Mrs. Hornbill begins to set; and Mr. Hornbill, to protect her from enemies, walls up the nest with mud—all but that hole through which she puts her bill and gets food from the devoted father and husband.

Other long-nosed tunnel diggers you must have seen many a time when you've been fishing, for they are fishers, too—Mr. and Mrs. Kingfisher. Their home is at the end of a tunnel in the banks of the stream where they do their fishing.

While we're visiting them and making a study of their household arrangements, it's a good thing for us that we're not kingfishers ourselves; for if there's anything that makes the kingfishers mad it's to have other kingfishers fooling around their place or even coming into their front yard. Each pair of kingfishers lays claim to the part of the creek in the neighborhood of their nest, as their fishing preserve, and woe betide any other kingfisher that trespasses!

Human fishermen and hunters give it out sometimes that kingfishers eat big fish that might otherwise be caught with a hook or a seine, but the fact is these birds catch only minnows and little shallow-water fish.

In digging the tunnels for their nests the two birds work together, and these tunnels are sometimes fifteen feet long. So you see that with kingfishers scattered around the world as they are—some 200 species in all—they must have done an enormous amount of ploughing in the course of time; to say nothing of what they have done in the way of enriching the soil with fish-bones, one of the very best of all fertilizers.

The kingfisher's nest wouldn't be at all attractive to some birds—the swallows, for example, who are so particular about having feather-beds. It has just a hard-earth floor like the cabins of the American pioneers, but the little kingfishers are perfectly contented and happy; for their meals are very plentiful, fairly regular, and the fish are always fresh.

#### **FISHING DAYS AND OTHER DAYS**

But some days even the kingfishers don't have fish for dinner. Instead they serve crayfish and frogs. This is on cloudy days, or when the wind is stiff and the water rough. On such days even the keen eyes of the kingfisher can't see a fish or make out exactly where the fish is when he does see one. But on clear, quiet days, you should see him fish. He often dives from a perch fifty feet or more above the creek and strikes the water so hard you'd think it would knock the breath out of him. But up he comes with his fish, nearly every time!

Of course he misses occasionally, but just think of seeing a fish that far away—under the water, mind you; and not a big fish, but a little minnow, only two or three inches long.

#### **II. UNDER THE OVEN-BIRD'S FRIENDLY ROOF**

Another great little farmer is the oven-bird. We can't afford to miss him and his wife for anything; and although we have to go to South America to meet them, we'll do it. So here we are! The oven-birds build a nest of clay mixed with some hair or grass or real fine little roots. This nest, when it's all done

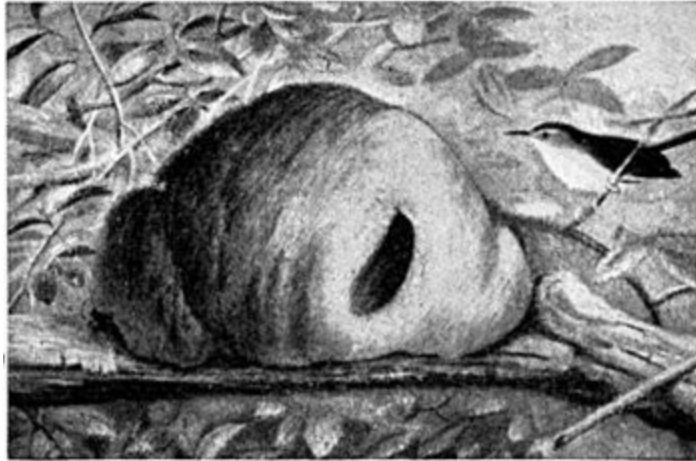
—it takes a good while to build it—is so big you'd hardly believe it was the home of so small a bird. It's a dome-shaped affair, like a Dutch oven. In the United States we have what we call an "oven-bird," too—one of the water-thrushes; but as its dome-shaped nest is made of grass and leaves and has no clay in it, we will not include this bird among the feathered farmers. The oven-bird of South America knows how to build its dome of clay without any scaffolding, which isn't easy.

#### OVEN-BIRD DOORS AND THE FRIENDLY ROAD

While the big flamingoes are so shy, the little oven-birds don't care who sees them—provided they can see *him* first. This is possibly because they want to keep an eye on any suspicious movements; for they make it an invariable rule to build so that their front doors will face the road. But really I think they do this, not because they are suspicious, but because they want to be neighborly and arrange their homes so they can sit on their front stoop and watch the crowd go by. They not only have their doors where they can see what's going on, but they nearly always build near the country road or the village street, and in the most conspicuous place they can find, instead of staying off by themselves in those vast, lonesome woods of Brazil where they lived before man came.

When a nest is to be built the oven-bird picks up the first likely-looking root fibre, or a horsehair, or a hair from an old cow's tail, carries it to some pond or puddle and, with this binding material, works bits of mud into a little ball about the size of a filbert. Then he flies with this pellet to the place where the nest is going up. With clay balls like this laid down and then worked together, the two birds make the floor of their little house. On the outer edge of the floor they build up the walls. These walls they gradually incline inward, just as the Eskimos build their snow-block huts, until they form a dome with a little hole in it. The last little ball they bring goes to fill that little hole and then the house is done, so far as the walls and roof are concerned. Next, a front door is cut through the wall that faces the road.





#### THE FRIENDLY DOOR THAT FACES THE ROAD

Oven-birds make it a rule to build their adobe homes so that the front door will face the road. And they nearly always build near the road or the village street. Neighborly little creatures!

From the front door a partition is built reaching nearly to the back of the house, shutting off the front room from the family bedroom. After the eggs are laid Papa Oven-bird stays in the front room—or thereabouts—while mamma sets in the back room. The object of the little partition seems to be to protect mother and the eggs and, when they come, the babies from wind and rain. When the four or five baby birds arrive both papa and mamma put in most of their time, of course, feeding them.

The nests of the oven-birds weigh eight or nine pounds. The work of these little feathered farmers and their wives reminds us in more ways than one of that of Mrs. Mason-Bee,<sup>[22]</sup> but they evidently have quite different notions about housekeeping; for, although their residences are so big, the oven-birds would evidently rather build than clean house, while with Mrs. Bee it's just the other way. The nests of the oven-birds are so thick and strong they often stand for two or three years in spite of the rains; but the birds build a new nest every year, nevertheless.

### III. THE MOUND-BUILDERS

Another class of birds that have a fancy for big dome-like nests are the mound-birds. We find them in Australia, the Philippines, and the islands of the South Seas. Their scientific nickname is *Megapodidae*, the "big-footed." It's with their big feet that they pile immense heaps of leaves, twigs, and rotten wood over their eggs.

And what for, do you suppose?

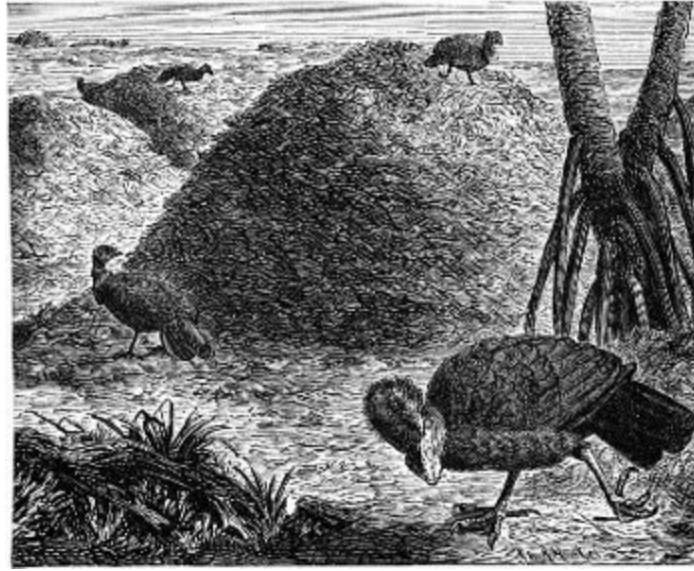
To hatch them! This heap of material not only absorbs the heat of the sun, but, in decaying, makes heat of its own. These mounds, of course, contribute tons and tons of fertilizer to the soil, but what interests the birds is that these warm heaps hatch their eggs. It's a kind of an incubator system, you see. As it is with many tens of thousands of our own little chickens, these days, the baby megapodes are born orphans. That heap of dead sticks, leaves, and earth is all the mother they ever know. As soon as the mother birds have laid their eggs in the mounds and covered them up, they go off gossiping with other lady megapodes, and don't bother their heads any more about their babies.

#### **WHY LITTLE BIG FOOT NEVER SAYS "MAMMA"**

But it really doesn't seem to matter. It's more of a question of sentiment than anything else, for the babies get on very well by themselves. When the time comes they not only make their own way out of the shell, as all birds do, but they work their way up through the rubbish-heap and run off at once into the woods to hunt something to eat.

It's all right, after all, I suppose; but if *I* were a little mound-builder's baby, I'd rather have a mamma that would stay around and go places with me, wouldn't you?

There's one nice thing about these mamma mound-builders, though; they're so neighborly and sociable. It's like a regular old-fashioned quilting party to see them build a nest. The birds look like turkeys, and one of the species is called the "brush turkey," but they are no bigger than an ordinary chicken—than a rather small chicken, in fact. When I tell you, then, that these mounds of theirs are often six feet high and twelve feet across in the widest part, the middle, you can see it takes good team-work to put them up.



#### BRUSH TURKEYS BUILDING THEIR INCUBATORS

It's like an old-fashioned quilting party—the co-operative mound building of the brush turkeys. The text tells you about that back kick of theirs.

So a lot of the lady mound-builders get together in woodsy places, where there's plenty of leaves and twigs lying around and together build a mound. One will run forward a little way, rake up and grasp a handful of sticks and leaves—I mean to say a footful—and kick it backward. The motion is much like that of an old hen scratching. Then another bird gathers a footful; then another, and soon they are all throwing the rubbish toward the same pile; all as busy as a sewing-circle, but—curiously enough—nobody saying a word! Before the mounds are quite done, they all begin laying their eggs in them; as many as forty or fifty, before they are through.

Some species frequent scrubby jungles along the sea. These scratch a slanting hole in the sandy soil about three feet deep and lay their eggs on the bottom, loosely covering up the mouth of the hole with a collection of sticks, shells, and seaweed. The natives say these birds, before they leave, go carefully over the footprints leading to this treasure-house, scratch them out and make tracks leading in various directions away from the nest. And

all species lay their eggs at night. You see why, don't you? They're just that cautious.

#### SUCH AN EGG FROM SUCH A BIRD

But if you should find one of their nests full of brick-red eggs you'd never guess who laid them, they're so big! Away back in 1673, an English missionary to China who had stopped off at the Philippines, on his way, wrote a little book when he got back home about where he had been and what he had seen, and he just couldn't get over the wonder of the mound-builders. Among other things he says, in one place in his book:

"There is a very singular bird called Tabon. What I and very many more admired<sup>[23]</sup> is that being in body no bigger than an ordinary chicken, it lays an egg larger than a goose's."

"So," he adds, "the egg is bigger than the bird itself!"

#### IV. THE SWALLOWS

To make the acquaintance of either the mound-builders or those dear little oven-birds—*aren't* they dear?—we must be travellers, of course, for with their short wings neither the mound-builders nor the oven-birds ever could come all the way up here to see us. But another feathered farmer—and, like the oven-bird, a clay-worker and most neighborly—everybody knows; the swallow. Like Kim, the swallow is the little friend of all the world.

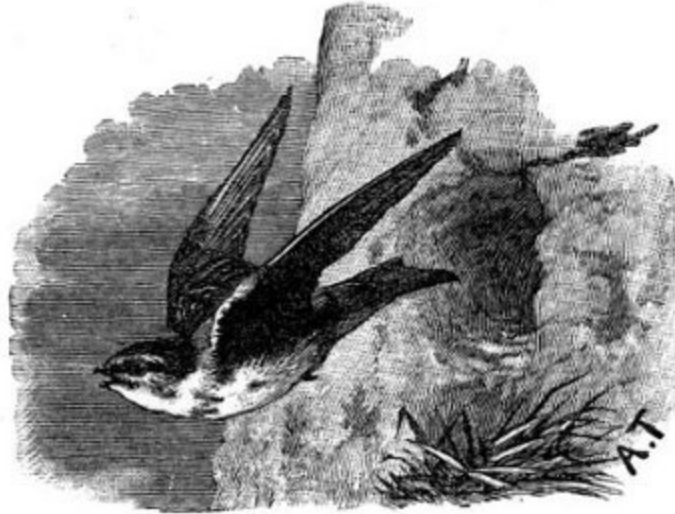
Swallows of one kind and another are found everywhere—almost everywhere that people can live; usually where people *do* live. And if all the soil they've helped pulverize and mix—even since the days when the swallows built under the eaves and rafters of the ark—was spread out, it would easily make another Egypt, I do believe!

But, speaking of the way swallows take to human society, do you know where our barn-swallows came from? They were originally cliff-dwellers away out West. The early explorers found enormous collections of their nests plastered all over the perpendicular cliffs and along the bluffs. Just as soon, however, as the country settled up and men put up barns these little

cliff-dwellers, deserting rocks and bluffs, began building their bottle-shaped nests under the eaves. The swallows live on insects—including squash-bugs, stink-bugs, shield-bugs, and jumping plant-lice; and that's supposed to be one of the reasons for the curious fact that they left their ancient family seats—they found so many more insects about the barns and the farmer's fields and the gardens and the orchards.

#### **TINY SOIL MILLS OF THE BABY SWALLOW**

Haven't you often watched them and listened to them, diving and chattering around the barn in their busy season; that is to say, in the spring and summer time? Then the air is full of insects and is fairly woven with their darting wings. Some keep busy picking up the insects that are always hovering about in a barnyard, while others dash away to some near-by marsh or to the meadow or to the creek. Over the grain-fields they go, over the meadows and back again straight to the nest where downy babies are cheeping for them. The parents feed them, stop and chatter a moment, and then off they go. Follow that one down to the marsh. See how she flies high, round and round in circles, and then swoops for an insect. She missed him! Then she wheels, darts up—darts down—to right—to left. There, she's got him! Then off like an arrow to the nest. The soft-bodied insects are chosen and chewed up for the babies, while the parents eat the tougher ones. And to help digestion they give the babies little bits of gravel, although they don't use it themselves. So, in grinding up this gravel the baby birds help make soil before they are old enough to do any nest-building.



THE SAND MARTIN AND HIS HOME IN THE BANK

You've noticed, of course, that all the swallows about a barn don't build under the eaves. Some build under the rafters inside the barn. That isn't just a matter of taste; it's family tradition. The eave-builders are descendants of the cliff-swallows, while the birds known to bird students as "barn" swallows build under the rafters.

But they don't take to the fine, new modern barns—all spick and span—the barn-swallows don't. If there's an old gray barn with doors that never shut quite snug, a board off here and there, and several panes in the cob-webbed windows broken out——

"Oh, just the thing!" say Mr. and Mrs. Swallow, and they turn their backs on the new barn and proceed to build their cute little nests of clay among the rafters of that old tumbled-down affair. In their preference for the old gray barns, the swallows are like the artists, the painters that Mr. Dooley told about. He was talking about artists to his friend, Mr. Hennessey:

"I don't mane the kind of painther that paints yer fine new barn," said Mr. Dooley. "I mane the kind of painther that makes a pitcher of yer *old* barn and wants to charge ye more'n the barn itself is worth."

#### WHY ARTISTS AND SWALLOWS PREFER OLD BARNs

The reason the artists prefer old barns is that they look better in pictures, but the reason the barn-swallow shows the same taste is that, with windows that have panes in them and doors that shut tight you'd no sooner start to build a nest than, coming back with a pellet of clay, or bringing a feather for the little feather-bed, you'd be liable to find the door shut and you could no more get in until chore time than you could open the time-lock in the First National Bank. And suppose there were babies and you'd just *got* to get back—you see it wouldn't do at all!

But both the barn-swallows and the old gray barns will be seen only in pictures before long, if things keep on; what with these new barns and the cats always trying to catch the few swallows there are left—when you're swooping low to catch a squash-bug, say—and those hateful sparrows that tear your nest to pieces. And for several years swallows were killed by thousands to make ornaments for women's hats until this shameful business was stopped by law!

On the Pacific Coast, if you're out there even as early as March, you'll see a purplish-bronze swallow, with bronze-green markings. These swallows make a specialty of orchard insects and that's why, perhaps, they build under the eaves of the farmhouse rather than the barn. But, like the rest of the swallow family, they think nothing quite so nice as a bed of feathers to raise babies in, and they know as well as the cliff-swallows and the barn-swallow that a barnyard is a great place for feathers.

And besides, there's a man out there, in one place, that keeps a supply of feathers just to give away when the swallows are nesting. Watch him, over on the hillside. He takes a little bunch of feathers and throws them up into the air from his open hand. A swallow skims by and catches one of these feathers before it touches the ground. But soon the word passes along:

"Here's that nice man with the feathers!"

And, pretty soon, there are a half-dozen in the game. They flit closer and closer to that generous hand, seizing the feathers almost the moment they are in the air. Then one, bolder than the rest, snatches a feather right from the man's thumb and finger. The little rogue!

By the way, do you know who that man is? It's Mr. W. L. Finley, State Ornithologist of Oregon. "Our little brothers of the air," as Olive Thorne Miller calls the birds, are getting to be so much appreciated, not only as the friends of man, but for their beauty and the usefulness of their lives, that both our State and national governments have laws to protect them, and such men as Mr. Finley are employed to look after their interests.

Of course, he doesn't *have* to furnish feather-beds for the baby swallows—he just does!



### HIDE AND SEEK IN THE LIBRARY

If you want to get better acquainted with ostriches you should read Olive Thorne Miller's "African Nine Feet High," in "Little Folks in Feathers and Fur." Carpenter deals with the ostrich in his "How the World is Clothed" and in his "Geographic Reader on Africa"; Johonnott's "Neighbors with Wings and Fins" gives a chapter to "Giants of Desert and Plain," among which you may be sure he includes the ostrich.

Allen, in writing about "Some Strange Nurseries" ("Nature's Work Shop"), tells why it is Papa Ostrich has most to do with the hatching of



the eggs when the sun is not on the job.

Lucas, in his "Animals of the Past," speaks of ostriches and crocodiles as the nearest living relatives of—guess what—the dinosaurs! (Yet look at the dinosaur in "The Strange Adventures of a Pebble" and see if you can't make out a good deal of the ostrich and the crocodile in him.)

But, speaking of Papa Ostrich's parental duties, did you know that it's *Mr.* Puffin, and not *Mrs.* Puffin, who digs the family burrow? Arabella Buckley's "Morals of Science" tells that and many other interesting things about devoted husbands among the birds, including how Papa Nightingale feeds Mamma Nightingale.

In the "Children's Hour," Volume 7, page 310, you will find an interesting article about the puffins of Iceland.

"The Romance of Animal Arts and Crafts" tells about one of the feathered clay-workers, the nuthatch of Syria, and why he makes his nest look like a rock. These nuthatches love to build so well that they often make nests that they never use; and they even help put up nests for their neighbors!

This book also gives interesting details about the hornbill, and how and why he walls up his mate in her nest in the hollow of a tree. Father Hornbill, of course, gets all the meals for Mother Hornbill, while she's setting. She simply *can't* get out, and you should see him by the time the babies are old enough to leave the nest. He's worn to a shadow!

Rooks, it seems, do a little digging under certain circumstances. Selous tells about it in his "Bird Life Glimpses." In this book you will find a delightful description of martins building. It almost makes you want to *be* a martin. It also tells about the work of the sand martins. You will hardly believe how fast they work. The house-martin's nest is more elaborate than the swallow's. This book tells why the house-martins begin work so early in the morning, and why they have to delay their nest-building if the weather is either too wet or too dry.

White, in his famous "Natural History of Selbourne," tells how worried he was because certain swallows just *would* build facing southeast and southwest.

Birds, besides being workers of the soil, are great sowers of seeds. Darwin tells how he reared eighty seedlings from a single little clod on a bird's foot. What do you suppose he did that for? You just look it up in the index to his "[Origin of Species](#)."

Doesn't it seem funny that one of the little farmer birds—a burrower—should go into partnership with a lizard? There is one in New Zealand that does that very thing. He is called the titi. What the titi does for the lizard is to provide him with a home in his burrow, but what do you suppose the lizard does in return to pay for his lodging? Read about it in Ingersoll's "Wit of the Wild," in the chapter on "Animal Partnerships."

Do you know why the phœbe bird so often uses moss in building her nest? And how the phœbes that make green nests keep them green? And how Mrs. P. puts a stone roof on her house? You will find all about it in "Wit of the Wild."

The same chapter, "The Phœbe at Home," tells why the phœbe bird took to building under bridges, and why she builds in a carriage shed instead of a barn, as the barn-swallow does.

"Bird Life," by Chapman, is a guide to the study of our common birds. The beauty about this book is that it has seventy-five full-page plates in the natural colors, with brief descriptions, so that all you have to do is to bring the *mind* picture of the bird you have seen alongside the picture in the book, and there's the answer! Nobody has written more delightful books on birds than Olive Thorne Miller. "[Little Brothers of the Air](#)" is one of them. You couldn't keep your hands off a book with a name like that, could you? Then there is her "Children's Book of Birds," "True Bird Stories," illustrated by Louis Agassiz Fuertes, and "Little Folks in Feathers and Fur," which, as you can see, goes outside the bird family. John Burroughs's "[Wake Robin](#)" deals not with robins alone, but with birds and bird habits in general.

But the greatest book about birds—the wonder of the bird and his relations to the whole animal world—is very properly called "The Bird," by C. William Beebe, who is at the head of the bird department of the great New York Zoo. Among other things it tells:

How Nature practised drawing—so to speak—for years before she could finally make a proper bird. (If you have ever tried to draw a bird from memory and realized what a bad job you made out of it, you will sympathize with her.) How they know that the earliest birds Nature made, as well as being very homely, weren't at all smart; not to be mentioned in the same breath with clever Jim Crow, for example. How "a bird's swaddling clothes and his first full-dress are cut from the same piece," the very words of the book. About certain birds that have one set of wings to play in and a new set for flying, like a child wearing jumpers to save his nice clothes! About the world of interesting things you can discover with the bones of a boiled chicken.

And so on for nearly five hundred pages, and as many illustrations; the most striking collection of pictures explaining birds that I ever saw.



THE END OF A BUSY SEASON

"And there's the corn and the pumpkins and the carrots and the turnips and the potatoes in the root cellar and the jelly in the jelly-glasses—we helped make them all."

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# CHAPTER X

(OCTOBER)

It is hardly an exaggeration to say that the tip of a root acts like the brain of the lower animals.

—*Darwin.*

## THE BUSY FINGERS OF THE ROOTS

This has been a very busy season for Mr. Root and his family. It always is, and you can imagine they're all glad when Fall comes and they can lay by for the Winter.

"There's your apple crop, I helped make that," Mr. Root might say. "And there's the corn and the wheat in the granary, and the rye and the oats and the barley; and the hay in the mow; and the pumpkins and the carrots, and the turnips, and the potatoes in the root cellar; and the jelly in the jelly-glasses, and the jam, and the preserves—we helped make them all.

"And we've been working for you almost since the world began; almost, but not quite—for the earliest plants, the Lichens, for example—didn't have true roots.

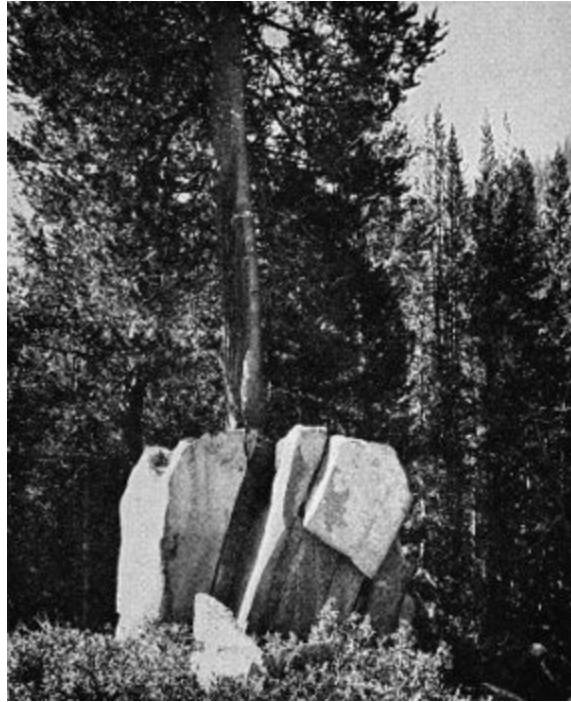
"Yes, and—well, I don't want to say anything—Mr. Lichen has been a good neighbor—but he never did amount to much; never could. No plant can amount to much without roots. But with roots and a good start a plant can do almost anything—raise flowers and fruit and nuts, and help grow trees so tall you can hardly see the tops of them. And, it isn't alone what we do for the plants we belong to, but for the soil, for other plants and roots that come after we're dead and gone. For them we even split up rocks, and so start these rocks on their way to becoming soil."

### I. ALL IN THE DAY'S WORK

It's a fact. Roots do split rocks. Hundreds of times I've been in the cracks of rocks that were split in that way. I mean right when the splitting was going on. This happened oftenest where trees grew on the stony flanks of mountains. Seeds of the pines, say, dropped in crevices by the wind, sprout in the soil they find there, and then, as these shoots grow up into trees, the enlarged roots, in their search for more soil, thrust themselves deeper and deeper into the original lodging-place, and so split even big rocks. The tap-roots do the heaviest part of this pioneer work. After the older and larger roots have broken up the rock, the smaller roots and fibres, feeling their way about among the stones, enter the smaller openings and by their growth divide the rock again and again.

But it's a lot of hard work for little return, so far as these early settlers are concerned; just a bare living. All these rock fragments, in the course of the years, become soil, but the amount of decay is small in the lifetime of the tree that does the breaking.

A root, as you doubtless know, tapers. This enables it to enter a rock crevice like a wedge. As it pushes its way in farther and farther it is growing bigger and bigger, and it is this steady pressure that breaks the rock. Even the tiny root of a bean grows with a force of several pounds, and the power exerted by the growth of big roots is something tremendous. At Amherst Agricultural College, one time, they harnessed up a squash to see how hard it could push by growing. From a force of sixty pounds, when it was a mere baby, what do you suppose its push amounted to when it had reached full squashhood in October? Nearly 5,000 pounds; over two tons!



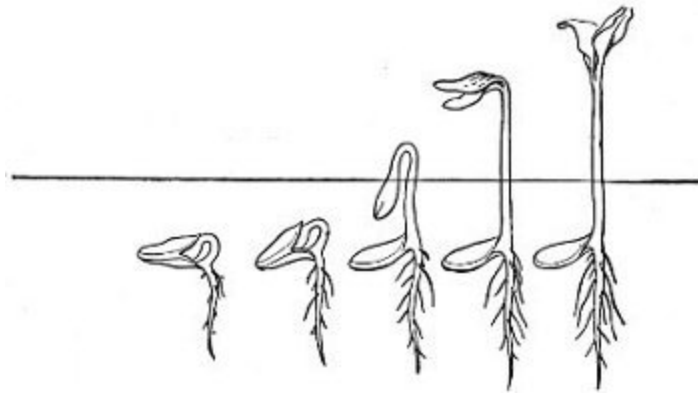
#### HOW A LITTLE ROOT SPLIT A GRANITE BLOCK

The little winged seed from which this pine-tree grew was carried by the wind one day into a tiny crack in that big granite block. As the treelet grew the tap root split the rock, penetrated to the earth below and fed the trunk until it became, as you see, a tree 40 feet high and 18 inches in diameter!

But don't think because roots can and do split rocks, if need be, that they go about looking for such hard work. On the contrary. In travelling through the soil they always choose the easiest route, the softest spots. They use their brains as well as their muscles, and what they do with these brains is almost unbelievable.

Yet the roots are such modest, retiring folks, always hiding, that it was a long time before the wise men—the science people—found out what all they do. It took a lot of science people and the wisest—including the great Darwin—to get the story, and they haven't got it all yet, as you will see. It was Darwin who first thought of having Mr. Root write out his autobiography—or part of it—the story of his travels; for he does travel, not

only forward—as everybody knows—but around and around. A regular globe-trotter!



#### WHY BABY PLANTS BACK INTO THE WORLD

Most plants back into the world out of the seed like that. Why? To protect their tender first leaves. Suppose you were taking some very valuable thing, easily injured—baby brother, say—through a swinging door and you had to use both hands to carry him. You wouldn't open the door by pushing that dear, little tender head of his against it, would you? You'd open it by backing through.

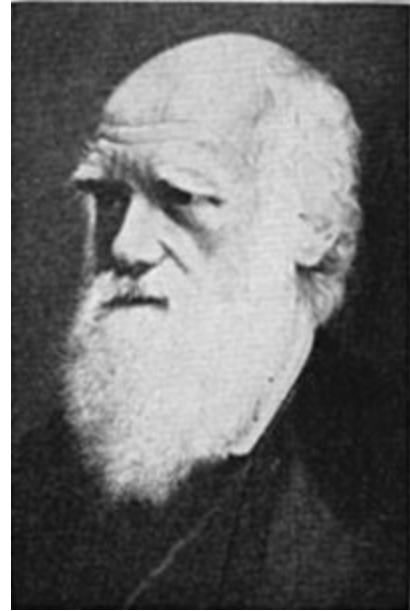
Mr. Darwin was a wonderful hand at that sort of thing—getting nature people to tell their stories. He was an inventor, like Mr. Edison; only, instead of inventing telephones for human beings to talk with, he invented ways of talking for nature people. You saw how he fixed it so that the earthworms could tell what they knew about geometry and botany. Well, in the case of the roots, what did he do one day but take a piece of glass, smoke it all over with lampblack—you'd have thought he was going to look at an eclipse—and then set it so that Mr. Root could use it as a kind of writing-desk. In a hitching, jerky sort of way roots turn round and round as they grow forward. In the ground, to be sure, a root can't move as freely nor as fast as it did out in the open and over this smooth glass, but it does turn, slowly, little by little. The very first change in a growing seed is the putting out of a tiny root, and from the first this root feels its way along, like one

trying to find something in a dark room. Thus it searches out the most mellow soil and also any little cracks down which it can pass.

"Here's a fine opening for a live young chap," we can imagine one of these roots saying when it comes to an empty earthworm's burrow or a vacancy left by some other little root that has decayed and gone away. Roots always help themselves, when they can, to ready-made openings, and it is this round-and-round motion that enables them to find these openings.

But even this isn't all. A root not only moves forward and bends down—so that it may always keep under cover and away from the light—but it has a kind of rocking motion, swinging back and forth, like a winding river between its banks, and for a somewhat similar reason.

"It's looking for a soft spot!" says the high school boy, "just as the river does."



CHARLES DARWIN

The great naturalist.

#### NO HIT-OR-MISS METHODS FOR MR. ROOT

Exactly. But not in the sense that this phrase is used in slang. The root has certain work to do, and it does it in the quickest and best way. It can get food more quickly out of mellow soil than out of hard, and so it constantly hunts it up. I mean just that—*hunts it up*. For it isn't by aimless rocking back and forth that roots just *happen* upon the mellow places. It's the other way around; it's from a careful feeling along for the mellow places that the rocking motion results.

"But how on earth do the roots do this? What makes them do it?"

That's what any live boy would ask, wouldn't he? So you may be sure that's what the science people asked, and this is the answer:

The roots, like all parts of the plant—like all parts of boys and girls and grown people, for the matter of that—are made up of little cells. Well, these



cells, first on one side of the root and then the other, enlarge, and so pump in an extra flow of sap. Now, as we know, the sap contains food for the plant, just as blood contains food for our bodies; and more food means more growth. So the side of the root where the cells first swell out grows fastest and thus pushes the root over on the opposite side. Then the cells on this opposite side swell, and the root is turned in the other direction again. So it goes—right and left, up and down. And when these two motions—the up and down and right and left—are put together, don't you see what you get? The round-and-round motion!

Precisely the same thing happened right now when you turned your finger round and round to imitate the motion of the root. (I saw you!) The muscles that did the work swelled up first on one side and then on the other, just as they do when you bend your elbow, when you walk, when you breathe, when you laugh.

And more than that: You know how tired you get if you keep using one set of muscles all the time—in sawing fire-wood, for example. Yet you can play ball by the hour and never think of being tired until it's all over; because, for one thing, you are constantly bringing new muscles into action as you go to bat, as you strike, as you run bases. It's the same way with the roots, it seems. For the theory is that after the cells on one side have swelled, they rest; then the cells on the other side get to work.

"But what starts the movement?" you may say. "The idea of moving my arms and legs starts in my brain."

#### **WHERE MR. ROOT KEEPS HIS BRAINS**

Just so again. The root has a brain, too, or what answers for a brain. And the root's brain, is in its head; at least in the vicinity of its nose—that is to say, its tip. It's the tip that first finds out which side of the road is best, and passes the word back to the part of the root just behind it to bend this way or that. It's also the tip that feels the pull of gravity and knows that it's the business of roots to keep under cover. And Mr. Root just *will* have it that way! You can't change his mind. Mr. Darwin tried it and he couldn't; although he finally changed human people's minds a lot.

This is how he tried it on a root. He took a bean with a little root that had just started out into the world. He cut off the tip and then set the bean so that the root stuck straight up. It continued to grow that way for some little time. Finally, however, a new tip had formed. Then there was a general waking up, as if the tip said to the rest of the root:

"Here, here, this will never do! Where are you going? You must bend *down*!"

Anyhow that's what the root proceeded to do. One side seemed to stop growing, almost, while the other side grew rapidly and so the bending was done.

"Did you ever! But how does the tip send back word?"

"Don't ask me!" says the science man; say all the science men, even to this day. "We don't know yet just *how* it's done. But we're studying these things all the time, and we'll know more about it by and by. Meanwhile, perhaps you'll tell *us* why you say 'ouch' and pull your finger away when you touch something hot."

"Oh," you reply, "I say 'ouch' because it hurts; and teacher and the Physiology say my arm pulls my hand away because my head tells it to."

"Yes, but how does the head make the arm do the pulling? What's the connection?" says the science man.

Well, I guess we'll have to tell him we don't know, won't we?

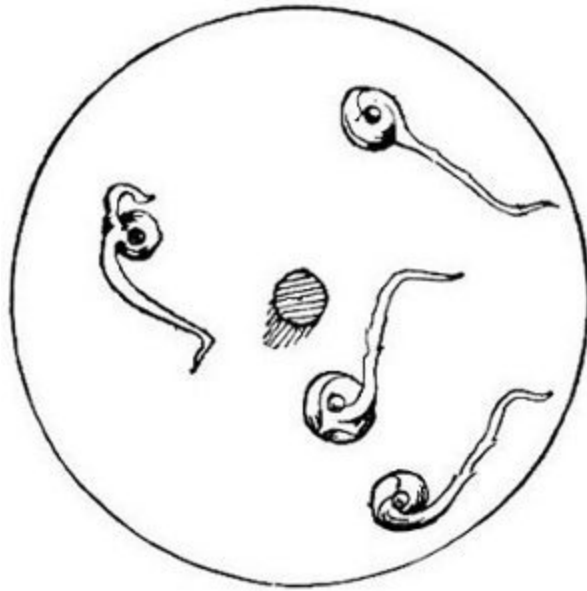
But all the root's brains aren't in the tip, any more than all *our* brains are in our heads. Scattered through our bodies, you know, are *little* brains, the ganglia, that control different parts of the body. So it is with roots. For instance, a root at a short distance from the tip, is sensitive to the touch of hard objects in such a way that it bends toward them instead of turning



WHERE MR. ROOT  
WEARS HIS CAP

A root wears its cap right where you do—over its brain department; that is to say, the tip. It is called the "root cap" and protects the tip from injury.

away, as the tip does. The result is that when a root comes to a pebble, say, under ground, the sides of the root press close up to the sides of the pebble—turn around corners sharply, by the shortest route—and so get over the obstruction as soon as possible and resume their course in the soil.



#### BUT THEY COULDN'T CHANGE ITS MIND

Some sprouting seedlings were attached to a disk like that, and when the roots started to grow down, the disk was turned to make them point upwards. But, no Sir! The roots just *wouldn't* grow upward. They turned downward. Every time!

And different parts of a plant's root system respond in different ways to the pull of gravity, and some don't respond at all. The tap-root, for example, which always grows down, has roots growing out from it horizontally. They just won't grow any other way, and yet this is also supposed to be due to the influence of gravity. Then, from these horizontal roots, grow out a third set, and they don't seem to pay any attention whatever to gravity. They grow out in all directions—every which way—so that if there is a bit to eat anywhere in the neighborhood they are reasonably sure to find it. You see it works out all right.

When a plant first begins to peep into the world out of that wonder box we call the seed, it's the root, as we know, that does the peeping; it comes first. And its first business is to get a firm hold in the soil. So a lot of fine hairlike fibres grow right and left and all around and take a firm grip. There is an acid in the root that dissolves whatever the root touches that has any food in it—including pebbles and old bones—and so makes a kind of sticky stuff that hardens. In this way these fibrous roots not only get good meals for themselves and the rest of the plant, but they hold the plant firmly in the soil, against the strain of the winds. They also give the tap-root something to brace its back against, as it were, while it pushes down for water, for the moisture in the damper portion of the soil beneath.

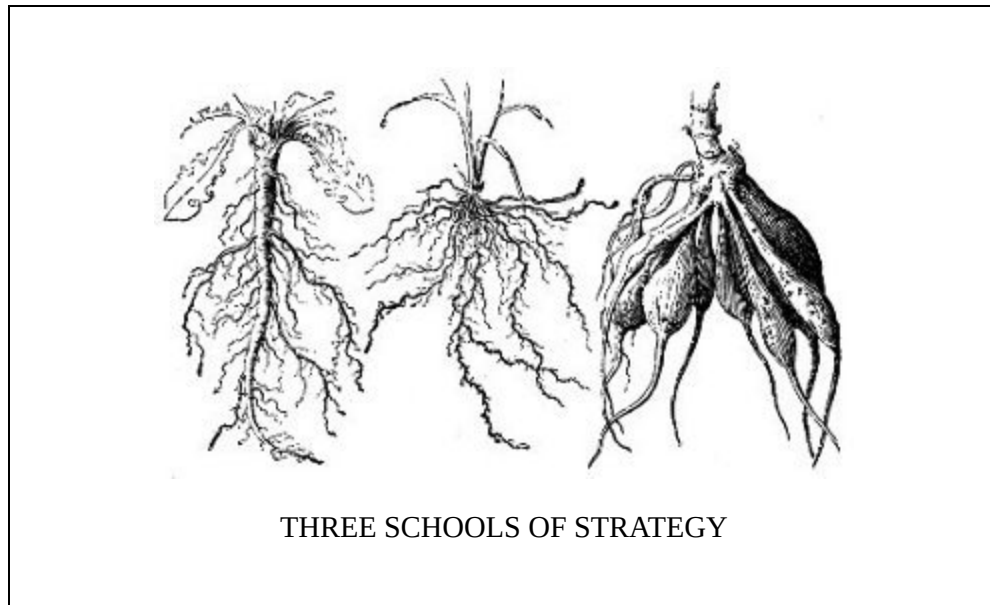
As you may have noticed, a seed merely lying loose on the ground is lifted up by its first little root in its effort to poke its nose into the soil. But Nature makes provisions for covering seeds up. They are covered by the castings of the earthworms, the dirt thrown out by burrowing animals and scratching birds. Some seeds fall into cracks where the ground is very dry and others are washed into them by the rains; while these as well as seeds lying on the surface are covered by the washings of the rain. Then come the roots that grip the soil.

Always growing just back of the tip, are thousands of root-hairs, as fine as down. These get food from the soil. They soon disappear from the older parts of the root, so that it stops gathering food itself and puts in all its time passing along to the stem and leaves the food gathered by the finer and younger roots. This is why plants are so apt to wilt if you aren't careful when transplanting them; the root-hairs get broken off. For the same reason, corn, after it grows tall, is not ploughed deeply. The fine roots reach out between the rows and the ploughshare would cut them off.

## **II. MR. ROOT'S PRESENCE OF MIND**

All these things and more the roots do in their daily work—in the ordinary course of business. And it's wonderful enough. Don't you think so? But there are even stranger things to tell; things that would almost make us believe roots have what in human beings we call "presence of mind." That is to say, the faculty of thinking just what to do when something happens

that one isn't looking for; when the house takes fire, for example, or the baby upsets the ink.



#### **A ROOT'S WAY OF CROSSING A ROAD**

Take the case of tree roots crossing a country road for a drink of water. They do it just as you or I would, I'll be bound. Just suppose you and I were roots of a big tree that wanted to reach the moist bank of a stream, and there was a hard road-bed between. We can't go over the top, and the road-bed is so hard we can't go straight through on our natural level so we'll just stoop down and go under, won't we? That's exactly what the roots do. They dip down until they get under the hard-packed soil, and then up they come again on the other side and into the moist bank they started for.

The roots of each kind of plant or tree have their natural level; that's one reason, as we know, why so many different kinds of plants—grass, trees, bushes, and things—get on so well together in the fields and woods. The tree roots that we have just seen crossing the road only went down below their natural level because they had to, as if the tip said:

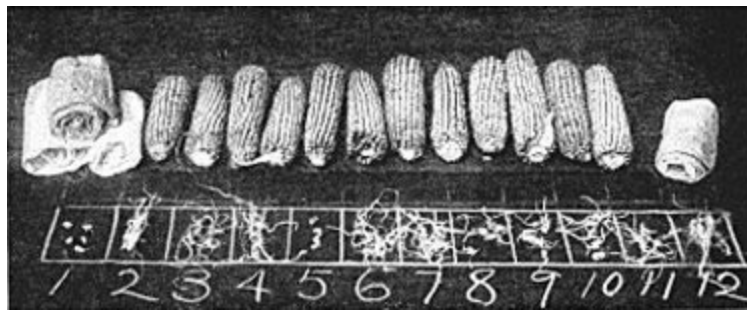
"This soil is too hard. We can never get through. Bend down! Bend down!"

So the roots bent down until they came to softer soil, then forward, but always working up toward their natural level, and so it was at their natural

level they came out on the other side.

#### A ROOT'S STRANGE ADVENTURE WITH A SHOE

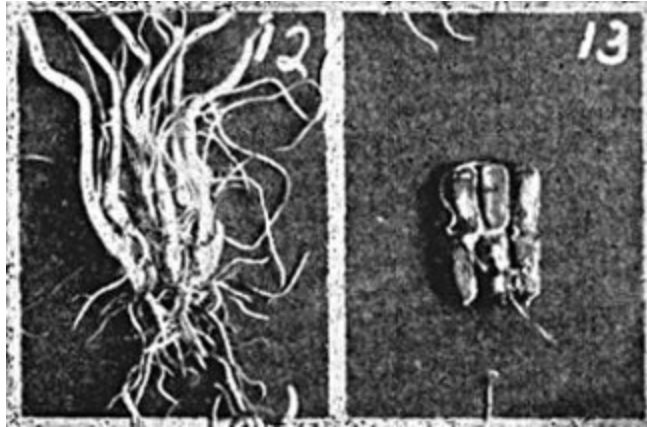
But here's an example of "presence of mind," that nobody has accounted for. A good-sized root, working along through the soil, like Little Brother Mole, to earn its board and keep, came right up against the sole of somebody's old shoe that had got buried in the soil. In the sole were a lot of holes where the stitches used to be. The root divided into many parts, and many of these smaller roots found their way through the stitch holes. Then, coming out on the other side, these little roots got together and travelled on, side by side!



#### HOW THE RAG BABIES TELL THE FORTUNE OF THE SEED CORN

In what is popularly called "the Rag Baby Test" the seed corn is placed on squares marked on cloth with numbers corresponding to the numbered ears. Then they are rolled up in one of those moistened rags until they sprout.

Isn't that a story for you? But there's no accounting for it. As we have seen, the men of science know a little bit about how a root manages to turn round and round and away from the light and so on, but what kind of machinery or process is it that could tell the root if it would split up into little threads it could get through the stitch holes in that old boot? You can't imagine; at least, nobody so far has thought how it was done. But it's all true. We'll find the story and a lot of other things about the ways of roots in one of the books we'll get acquainted with when we come to the "Hide and Seek."



© International Harvester Company

### THIS IS THE ANSWER

The seed from Ear No. 12 came out beautifully, didn't it? That from Ear No. 13 looks as if they were superstitious in Corn Land; but of course it was the fault of the seed and not of the number.

Here's another example of the same thing; what we have called "presence of mind," resourcefulness, invention. This example is even more striking, if possible, because, for one thing, it is a case where roots still more completely altered their habits to save a tree struggling for its life on a stony mountain cliff. Maeterlinck tells about it in his picturesque and dramatic style. The subject—the hero, as it were—of this story was a laurel-tree growing on some cliff above a chasm at the bottom of which ran a mountain torrent.

"It was easy to see in its twisted and, so to say, writhing trunk, the whole drama of its hard and tenacious life. The young stem had started from a vertical plane, so that its top, instead of rising toward the sky, bent down over the gulf. It was obliged, therefore, notwithstanding the weight of its branches, stubbornly to bend its disconcerted trunk into the form of an elbow close to the rock, and thus, like a swimmer who throws back his head, by means of an incessant will, to hold the heavy leaves straight up into the sky."

This bent arm, in course of time, struggling with wind and storm, grew so that it swelled out in knots and cords, like muscles upholding a terrific burden. But the strain finally proved too much. The tree began to crack at the elbow and decay set in.

"The leafy dome grew heavier, while a hidden canker gnawed deeper into the tragic arm that supported it in space. Then, obeying I know not what order of instinct, two stout roots, issuing from the trunk at some considerable distance above the elbow, grew out and moored it to the granite wall."

As if the roots, naturally so afraid of light, had heard a frantic call for help and, regardless of everything, had come to the rescue.

To be sure, certain roots—the prop-roots of corn-stalks, for instance, as you have noticed—habitually reach from above ground down into the soil, and serve to brace the tall stem swaying in the winds, but trees usually have no such roots and no such habits. Yet, here a tree seems suddenly to have learned, somehow, that elsewhere in the land of plants this thing is done. But how did it learn it? Did the brownies or the gnomes tell it; or was it some of the spirits of the wind that go everywhere and see everything? It might have been the same wind sprites that carry the seeds of the laurel and the pine so far up the mountain flanks. Or it might have been the dryads, those beautiful creatures of the wood the Greeks knew so much about.

I tell you there are some mighty queer things going on in the plant world, and perhaps Bud was right!

"Some peoples thinks they ain't no Fairies *now*,  
No more yet! But they *is*, I bet!"

### HIDE AND SEEK IN THE LIBRARY

And, what is more, real live fairies have been found right down in the world of roots! The science people call them "Bacteria," but what of that? The thing about a fairy that makes it a fairy is that it is always changing something into something else. Isn't that right? Well, that's exactly what is done by the bacteria on the roots of certain kinds of



plants—clover roots, for one; and the roots of beans, peas, peanuts, and alfalfa. These plants belong to the legume family, and if you will look up the word *Legumes* you will find out all about these fairy factories on the roots.

Among other things you'll learn how small these fairies are. Why, 100,000 of the bacteria that live on clover roots, marching single file, wouldn't much more than reach across this typed page.<sup>[24]</sup> And in their little "villages" on one system of clover roots there are so many that all of them put together would make a city as big as London or New York; if the bacteria were as big as people, I mean.

Of course you have to take a microscope to see them—a very powerful microscope—and even then some kinds of bacteria you can't see until you put colored clothes on them. (Every high school boy who has worked in the "lab" knows how this is done.)

And when you finally see them, a strange thing happens. You've hardly got your eye on a little Mr. Bacteria before he's two!

"What's this! What's this!" you say. "Am I seeing double?"

You look again and he's *four*! But don't be alarmed, you aren't seeing double; it's just the little Mr. Bacterias multiplying by division. How they multiply by division is one of the interesting things you can learn by looking them up.

But it's a good thing that the bacteria people in the little nitrogen factories on the clover roots can get more farm-hands in this way, for they have a lot to do, and their work is one of the most interesting things that goes on about the place.

The article in the "Country Life Reader" on "The Smallest Plant on the Farm" will tell you how important these nitrogen farmers are.

You would hardly believe how great their work is, they're so quiet about it. Do you know what a human nitrogen factory is like? Well, for one thing, it's the *noisiest* place in the world. Men, as do the bacteria, capture the nitrogen out of the air, but they do it by keeping up continual thunder and rain storms in big barrels. You will find one of

these factories described in an article in *St. Nicholas*, Volume 45, page 1137.

But what a fuss these human factories make! Why, in growing-time, out in the clover field, where the loudest sound you hear is the drone of the bumblebee among the blossoms, the little bacteria people down among the roots are making nitrogen so much cheaper than the big noisy factories that it only costs the farmer about one-fifth as much as the storm-barrel nitrogen. And yet, of course, it often pays to buy the artificial nitrogen, too.

There are many more striking things about the habits of roots than I have had room to tell about here, which you will find in such books as Elliot's "Romance of Plant Life," Coulter's "Plant Studies," Coulter's "First Book of Botany," Allen's "Story of the Plants," Chase's "Buds, Stems and Roots," Atkinson's "First Studies of Plant Life," Darwin's "[Power of Movement in Plants](#)," France's "Germs of Mind in Plants," Gray's "How Plants Behave," Carpenter's "Vegetable Physiology," Detmer's "Plant Physiology," and Parsons's "Plants and Their Children."



#### THANKSGIVING DINNER OF THE DORMICE

They don't sit at the dinner table like that, to be sure, but along in the Fall and up to nearly the time of our Thanksgiving dinners, the dormice eat unusually heavy meals and put fat on

their little bones to help them through the long, cold, and barren months of winter.

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# CHAPTER XI

(NOVEMBER)

All-cheering plenty, with her flowing horn  
Led yellow Autumn, wreathed with nodding corn.

—Burns: "*Brigs of Ayr*."

There's silence in the harvest field,  
And blackness in the mountain glen,  
And clouds that will not pass away  
From the hill tops for many a day;  
And stillness round the homes of men.

—Mary Howitt: "*Winter*."

## THE AUTUMN STORES AND THE LONG WINTER NIGHT

When the caveman was still living from hand to mouth; before he had even got as far as his first crooked stick for a plough, and when Mrs. Cave couldn't have canned a bean or a berry to save her life, even if she had had the cans, a certain little farmer already knew how to get root crops in the Fall and clean them and cut them and put them away in his little barn under the ground for Winter use.

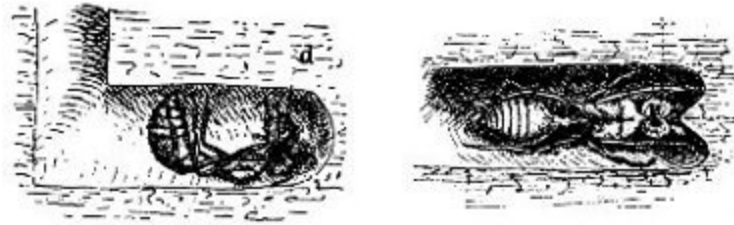
Several of these forehanded folk we have already met—the beaver and the chipmunk, among others—but since we are now at the end of the harvest year I thought we might spend this evening—the last but one, I am sorry to say, that we shall be together—in a little chat about these thrifty brothers of the wild, and how some of them are going to spend the long Winter that begins in the Autumn and lasts until Spring.

### I. LITTLE GRANARIES UNDER THE GROUND

I was going to begin by saying that one of the most *fore*-handed of them all has *six* feet, but as that would be almost as bad as a pun, I decided not to. You would have known, of course, that by people with six feet I meant the insects.

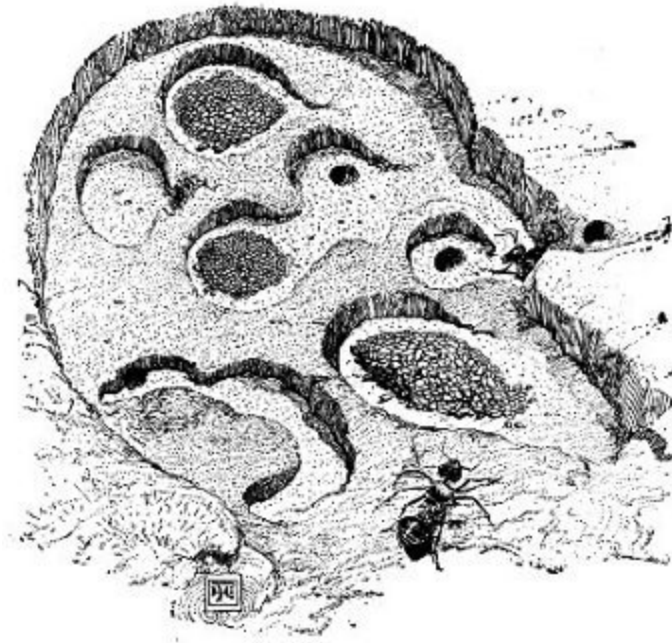
#### ANTS THAT THRESH AND STORE

Among the six-legged farmers, you may be sure, there have always been many who took thought for the morrow—the ants, for example. One can believe almost anything of ants. If that sluggard had gone to the ant, as wise King Solomon told him to, and learned all their ways, he would have found, among other things, how one species harvests the seeds of the plant known as the "shepherd's-purse," by twisting off the pods with its hind legs. These members of the ant family store grains of oats, nettle, and other plants. They pick up all the seeds they can find that the Autumn winds have already threshed for them, but they're not the least like that lazy man who wouldn't have the corn that was offered by kind neighbors to keep him from starving, because it wasn't shelled. If they don't find enough seeds on the ground when it comes time to think about the Winter stores they climb up and gather in the seeds themselves. On the shepherd's-purse, for example, the ant climbs up, selects a well-filled pod which is not sufficiently dried to have had its seeds threshed out by the winds, takes the pod in its little jaws and then—watch him—turns round and round on his hind legs until he twists it off! Then with it he carefully moves down the stem, like a baggageman carrying a big trunk from the third apartment; only the baggageman carries the trunk in front of him or on his shoulders, while the ant backs his way down. Sometimes two ants work together, one twisting, the other cutting away the fibres with its teeth. Sometimes they drop the pods to companions waiting below, and these other helpers never run off with it, but carry it to the common granary; for ants always play fair.



HOW THE ANTS WORK IN DIGGING OUT THEIR GRANARIES

And they have granaries, these ant farmers—hundreds of them, made just for that, each about the size of father's watch.



THE INSIDE OF THE GRANARY

Underneath the dome of the ant house you see in the previous picture, are flat chambers like these, connected by galleries, in which the grain is stored. One is prepared not to be surprised at anything about ants, but listen to this: The Agricultural Ants not only gather and store this grain, but they actually plant and cultivate it. They sow it before the wet season in the Fall, keep it weeded, and gather it in June of the following year. Seems incredible, doesn't it? But I'm only telling you what McCook, an

ant student, recognized everywhere as a reliable observer, saw these six-footed Texas farmers actually do.

Now here's a thing; you stow away a lot of seeds in a little hill where, of course, there's moisture, and what's going to happen? Those seeds are going to sprout and grow and spoil, and this, of course, destroys their value as food. Then what are you going to do? Of course, a human farmer would put his grains in a dry granary where they couldn't sprout, but you see the ants haven't any granary of that sort; nothing but those little holes in the moist ground. Just what they do to these seeds has not been discovered. They do something that keeps them from either spoiling or sprouting. But, when they get ready for these seeds to grow, they let them grow; not so that they can raise a crop, but for the same reason that the Chinaman lets the barley sprout that he uses in making chop-suey; so that it will be nice and soft to eat. This growing digests the starch in the seeds into sugar. When the sprouts have grown as far as the ants want them to, they gnaw the stalk a little, and cut off the roots with their mandibles. When this sugar-making has gone on long enough the ants bring all the plants out into the sun and let them lie there until they are nice and dry. Then they put them in their barns, and as long as Winter lasts they live on this sweet flour, grinding it in their mouth mills as they go along.

Why, it's like living on cookies, almost! Only the ants have been used to this steady diet of sweets for ages, and it doesn't hurt *their* little stomachs as it would ours.



#### CLEANING UP AFTER THE DAY'S WORK

While the Agricultural Ants don't take a bath after the day's work they do the next best thing. They give each other a kind of massage, and they evidently find it very enjoyable. You know how the cat loves to be stroked, dogs and horses to be patted, and little pigs to have their backs scratched. The ants below are giving each other a massage (left, abdomen; right, legs and sides). The lady above who seems to be braiding her back hair, is cleaning her antennæ.

This particular kind of a farming ant is called the Attabara, but there's another kind more wonderful still. If we want to call on them by their scientific names—these remarkable little creatures I'm going to tell about now—we'll have to go to Texas and ask if the *Pogononymex barbatus* family are at home.

"Oh, to be sure," says the gentleman who first introduced them to scientific society,<sup>[25]</sup> "just come with me."

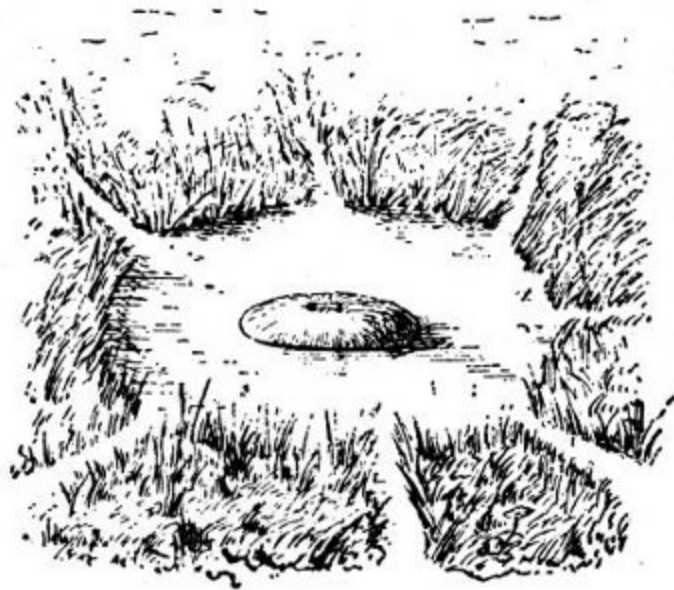
So he takes us over into Texas and shows us the ants at work. They destroy every plant on their little farms except that known as ant-rice. Compared to the size of the ants themselves, these grain-fields are giant forests, far



bigger than the Sequoia Forests of California. The ants watch for rain at harvest-time as anxiously as a farmer, and on the first sunny day, they do their cutting and hurry the grain into the barn. Then on later sunny days, they bring it out to dry before finally storing it away.

"Well," you say, "is there anything left that these farmers *don't* do?"

I can't think of anything except the planting. One observer says that they do actually plant the seeds, and Doctor McCook says, he wouldn't be surprised if they did, but he never saw them do it.



#### THE OLD HOME PLACE

This is the farm of some Agricultural Ants in Texas. See the granary and the roads leading to it? They collect and store the seeds of a plant which from this fact is called "ant-rice." It looks like oats and tastes like rice. All plants growing around the nest—which is also called the granary—the ants cut away, so clearing a space for 10 or 12 feet. Roads 5 inches broad near the nest, but narrowing as they recede, are made for hundreds of feet in different directions.

In tropical America there is a species of ant that raises "mushrooms"; at least a kind of fungus that passes for mushrooms with the ants. They don't exactly set the mushrooms out, but they save time by planting both the mushrooms and the leaves that make them as one and the same job. This is how they do it. They climb the trees, cut circular pieces of leaf with their scissor-like jaws and carry them back to low, wide mounds in the neighborhood of which they allow nothing to grow; the purpose being, as it is supposed, to ventilate the galleries of their homes by keeping a clear space about the mound.

#### HOW THE ANTS RAISE MUSHROOMS

The leaves are used as a fertilizer on which grow a small species of mushrooms. The leaves are first left out to be dampened by the rain, and are carried into the ants' cellars before they are quite dry. In very dry weather the ants work only during the cool of the day and at night. Occasionally inexperienced ants bring in grass or unsuitable leaves, but these are carried out and thrown away by older members of the family. But you see how valuable all these leaves are to the soil.



ANTS CARRYING LEAVES FOR THE MUSHROOM CELLAR

You'd never guess what the ants are going to do with those leaves! Read what it says on this page about these six-legged epicures.

## MR. HAMSTER'S THRESHING HARVESTER

Of course, we always expect the ants to do extraordinary things, but one of those four-legged farmers I mentioned in the beginning of the chapter anticipated the principle of the very latest type of threshing-machine. It's a fact. This remarkable little animal threshing-machine is called the hamster. He is found in Europe east of the Rhine and in certain portions of Asia. He does both his cutting and threshing in his field; something the Gauls did in the days of the Romans in a crude way, but which men of our day have only got to doing in recent years. He pulls down the wheat ear, cuts it off between his teeth, and then threshes it by drawing the heads through his mouth. The grain falls right into sacks as fast as it is threshed; just as it does in those huge, combined reapers and threshers that you see on our big wheat farms. Mr. Hamster's sacks are his cheek-pouches, one on each side. When these are filled, this little threshing-machine turns itself into an auto, a commercial truck, and off it goes with its load of wheat to the little barn hidden in the ground. These cheek-pouches, by the way, reach from the hamster's cheeks clear back to his shoulders, and both of these pouches will together hold something like a thousand grains of wheat. He empties them by holding his paws tight against the side of his face and then pushing forward. Rather a clever unloading device, too; don't you think so? Just as good for Mr. Hamster's purposes as the endless-chain system at the Buffalo grain elevator that Mr. Kipling admired so much.

And in the mere matter of the amount of grain handled, the work of the hamster is not to be laughed at. The peasant farmers are very glad to find a hamster granary, which, of course, they promptly take possession of by due process of law:

"The good old rule, the simple plan  
That they shall take who have the power,  
And they shall hold who can."

One of Mr. Hamster's neighbors, the field-rat of Hungary and Asia, stores his grain right in the house—the place where he lives with his family. Mr. Hamster, however, has his barns separate from his home. Sometimes he has one, sometimes two; and the older members of the community may have four or five.

## II. MR. VOLE AND HIS ROOT CELLAR

The farmer I mentioned at the beginning of this chapter, who is so thrifty about his root crops and so neat, belongs to the Vole family. He lives away over in Siberia and his full name is *Arvicola economus*. In gathering his crop of roots, he first digs a little trench around them and lays them bare. Then he cleans them off nicely so as not to fill his storehouse with dirt; cuts them up in sizes convenient for carrying, and then hauls them home and piles them up in little cellars made specially for them.

He only takes one piece at a time, walking along backward and pulling it after him with his teeth. He travels long distances in this fashion, going around tufts of grass, stones, and logs that lie in the way. When he gets home, he backs in the front door and into the living-room, and then into the barns which are back of the living-room. There are several of these and they are at the end of a long crooked passage.

Some of the Vole family make a specialty of wheat. One species of these wheat harvesters used to be common in Greece. He made such a nuisance of himself—from the Greek farmer's standpoint—that the Greeks had a special god to get after him; Apollo Myoktonos, "Apollo, Destroyer of Mice."<sup>[26]</sup> For the vole is just a kind of field-mouse. The runs of these wheat-harvesting voles are eight to twelve inches below the ground, and are connected with the surface by vertical holes. The end of the run is enlarged into a big room for the nest, and there are special rooms leading from the main runway that are used for the storing of the grain. These voles do their harvesting in the evening. Standing on their hind legs and holding to the stock with their little paws as a beaver clasps a tree, they cut off the wheat head with their teeth. They work very fast.

### HOW DID THESE FARMERS LEARN TO STORE?

Neither the voles nor any other of these interesting farmers and warehousemen used to get much credit for what they did. The fact that they helped themselves to some of the good things of earth annoyed Man, of course, and then, when it came to the matter of intelligence, conceited Mr. Man said: "Oh, *that's* just *instinct*." But nowadays when scientists have begun to study to find out what "instinct" really is, it is thought that man's

brother animals, although they are born with more knowledge of how to do things—with more of what we call "instinct"—have also learned by experience just as man did. It is argued that the storing habit was forced on animals wherever the climate cut off the food-supply for a time—either because it was too cold or too hot. The idea of putting something by for a rainy day appealed particularly to the burrowers because they are a timid lot. Not being able to defend themselves very well against their enemies they were obliged to pack up what they could and hurry to some hidden eating-place. That is where the cheek-pouches, which many of them have, come in handy. They are also very industrious, and as the seeds and nuts on which they lived began to ripen, they just couldn't resist the impulse to gather and gather and gather more than they could possibly eat at the time. So, as a result of this habit, food piled up in their underground homes. Then, as they were kept indoors by cold weather or by their enemies, they took to eating more and more from the pantry shelf, and thus the members of the family that were the busiest and, therefore, had the most to eat would naturally survive and leave children of a similar disposition, while the less thrifty would die off.

### **III. THE LONG WINTER SLEEP**

Some of these forehanded people, instead of putting their Winter supply of food in the ground, put it on their bones. That is to say, before turning in for the Winter, they get as fat as can be and then live on this fat until Spring. A great advantage of this system of storage is that it is particularly pleasant work—you eat and eat and enjoy your meals, that's all. Another advantage is that you can't be robbed of your store as easily as the hamster, for example, frequently is. You carry it right with you wherever you go.

There are a lot of curious things about this hibernation. Not only will warmth arouse the sleepers but also extreme cold, and after the extreme cold may come another sleep from which the sleepers never awaken; in other words, too much cold kills them. So the object of burying one's self as the ground-hog does, or under the snow as rabbits do, or in hollow caves and trees as Brer Bear does, is to keep from getting too cold. Sometimes two or more "bunk" together, as little pigs do on cold March days. The body of each helps to keep his bedfellows warm.

### IT'S THE COLD THAT MAKES ONE DROWSY

It is the cold itself that seems to make hibernating animals feel sleepy; just as it does human beings. At a moderate temperature, say 45 or 50 degrees, dormice and hedgehogs will wake up, eat something, and then go to sleep again. The dormouse usually wakes in every twenty-four hours, while the hedgehog's Winter naps are two or three days long. Hunger seems to be the cause of their waking, just as it is with babies. The little dormouse, as the air grows colder, gradually dozes off, and his breathing is very deep and slow. As the temperature rises, he begins to take shorter and more rapid breaths and gradually wakes up. Then, if he is in his own little home under the ground, he feeds on the nuts and other foods that he stored in Autumn and drops off again. He sleeps from five to seven months, depending on the weather.

Moles and shrews, so far as observation goes, don't hibernate. The moles simply dig deeper, and there they find worms and insects that are buried away from the reach of frost. The shrews hunt spiders and hundred-legged worms and larvæ in holes and crannies of the soil or beneath leaves of ground plants and old logs.



## LITTLE HEDGEHOG IN MAN'S HAND

A queer thing is that the hedgehog, which belongs to the same family as the shrew and the mole, is dead to the world all Winter. Like all complete hibernators he stops breathing entirely. The reason for this difference between the hedgehog and the mole is that the mole doesn't need to go to sleep, because he digs below the frost-line. As for the shrews, they have little bodies and are very active, and so get themselves food and keep warm, while the hedgehog is so much bigger and slower that, when there is so little to eat and it is so cold, he would either freeze or starve to death if he went about looking for food. He finds it cheaper to turn in and sleep than to work.



A HEDGEHOG AND HER BABIES

None of the tree-squirrels seem to take any unusually long naps in the Winter. We often see them around on pleasant days in the parks and in the woods. They run out, get a few nuts from their stores, and then back again to their nests, but the chipmunks and the gophers, who are closely related to

the squirrels, stay from late Autumn to Spring in their burrows, where they have plenty of food stowed away, and they sleep most of the time. In the home of four chipmunks was found a pint of wheat, a quart of nuts, a peck of acorns, and two quarts of buckwheat, besides a lot of corn and grass seed; all to feed four fat chipmunks. So, with such plentiful supplies, it is not surprising that after their long Winter sleep the chipmunks are as sleek as can be and as fat as butter, while Mr. Bear comes out in the Spring lean and with his hair all mussed up and as hungry as—well, as hungry as a bear!

All the bear family, except the polar bears, retire to caves or some sheltered spot under a ledge of a rock or the roots of a big tree. Among the polar bears the rule seems to be that it's Mamma Bear only who goes to bed for the Winter. She is careful to put on enough fat not only for herself, but so that the babies that come along in the Spring will have plenty of milk. She is buried by snow that drifts on her and her breath melts a funnel up to the fresh air.

#### **IV. MR. GROUND-HOG AND HIS SHADOW**

The woodchuck, like the bear, is a "meat-packer." People talk about him more or less in February. His other name is "ground-hog" and his shadow is quite as famous as he is. But is there anything in that old weather saw? Well, yes and no. You see, it's like this: Mr. Ground-Hog goes to bed very early in the Fall—long before the cold weather sets in—and so he is up very early the next Spring; long before the snow is all gone and, as it is with the other all-Winter sleepers, a little extra warmth may wake him up. Along toward morning, you know, we all begin to stir around in our beds and get half awake. So in addition to the fact that it is nearly daybreak for him—that is to say, Springtime—let there come along a bright, warm day in February—the second is as good as any other—and Mr. Ground-Hog is likely to come out of his hole. And, if he does, of course he will see his shadow, after which there is likely to be quite a lot of cold weather.

#### **HOW WEATHER AVERAGES UP**



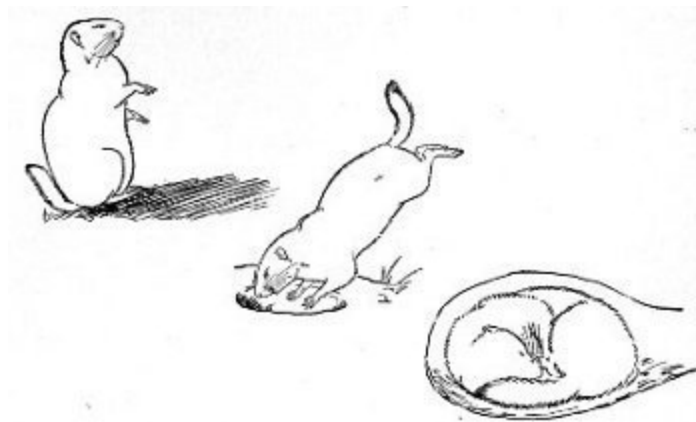
Not that his shadow makes any difference, but the point is that if you have much warm weather *early* in February you are likely to have colder weather *later* and running on into March. It's just the law of averages, that's all. You see it running through the year—this averaging up of weather; it just sways back and forth like a pendulum. Take it in any storm of rain or snow; first the clear sky, then the clouds, then the downfall, and after that the clear sky again. Take any month as a whole, or a year as a whole, and it's the same way; you get about so much rain, so much sunshine, so much heat and cold. The United States Weather Bureau went to work once and, from the records, classified the storms for the last thirty years, and they found that about fifteen storms each year start over the region of the West Gulf States, twelve begin over the mountains of Colorado, forty cross the country from the North Pacific by way of Washington and Oregon; and so on, just about so many from each region each year.



*The Last Snow, by Lippincott*

And records and old diaries, going back a hundred years, show that the longer the period you examine for weather facts, the closer the average. The weather for one ten-year period will be almost as much like any other ten-year period, as the peas in a pea shell are like each other. Coming back to the subject of February weather, we find in the diary of an old resident of Philadelphia in 1779: "The Winter was mild, and particularly the month of February, when trees were in bloom." He doesn't say anything about the

ground-hog, but there is this to be said of the sharper changes of February and March, that at this season the earth is getting more and more warmed up and yet the cold winds from the North don't like to go; so there is a constant wrestling-match, and it is the wrestling of the winds one way and another that brings the changes of the weather. So if the South Winds get the best of it early in February, the North Winds, with their cold weather, are likely to win later in the month, and vice versa. Moreover, if you believe in the ground-hog proverb you are apt to *notice* the warm days (or cold days, as the case may be) for the next six weeks after February 2, and you *won't* notice so much the weather that doesn't fit your proverb! It's a way we all have; *seeing* the things that go to prove what we believe and *overlooking* the things that don't.



MR. GROUND-HOG AND HIS SHADOW

"But is there anything in the old weather saw? Well, yes and no. Mr. Ground-Hog goes to bed early in the Fall and is up early next Spring. Let there come a bright, warm day in February—the second is as good as any—and Mr. G.-H. is likely to come out and see his shadow. And if you have warm weather early in February you are likely to have colder weather later. It's the law of averages, that's all."

HIDE AND SEEK IN THE LIBRARY

I don't care what it says in "Alice in Wonderland," dormice never drink tea; although dormice have been at table with people ever since the days of the Romans. Dormice are still eaten in some parts of Europe, and the Romans used to keep them as part of their live stock. The European dormouse is really a little squirrel. Varro's "Roman Farm Management" (of which you are apt to find a good translation in the public library) tells how the Romans put their dormice in clay jars specially made, "with paths contrived on the side and a hollow to hold their food."

Crocodiles and other tropical animals take very long naps during the hottest weather. Hartwig's "Harmonies of Nature" tells about an officer who was asleep in a tent in the tropics, when his bed moved under him, and he found it was because a crocodile, in the earth beneath, was just waking up! Imagine what the dried-up ponds and streams of the llanos of South America must look like when the rainy season comes on, after the dry spell, with crocodiles asleep just under the surface everywhere. Doctor Hartwig's book tells.

But the most remarkable case of drying up that ever I heard of was that of the Egyptian snail in the British Museum, that Woodward tells about in his "Manual of the Mollusca." This snail was sent to England, simply as a shell, in 1846. Never dreaming there was anybody at home, they glued him to a piece of cardboard, marked it *Helix Desertorum*, and there he stuck until March 7, 1850, when somebody discovered a certain thing that indicated that there *was* somebody "at home," and that he was alive. They gave him a warm bath and he opened his four eyes on the world!

In his "Animal and Vegetable Hedgehogs" ("Nature's Work Shop") Grant Allen tells why the hedgehog works at night and sleeps in the daytime.

How he fastens on his winter overcoat of leaves, using his spines for pins, and how funny it makes him look.

How Mother Nature manages to have breakfast ready for him in the Spring just when he is ready for *it*.

How hedgehogs use their spines when they want to get down from a high bank or precipice real quickly.

How their eyes tell how smart they are; for a hedgehog is smart.

You will also find interesting things about hibernation in Gould's "Mother Nature's Children" and Richard's "Four Feet, Two Feet and No Feet."

In one of his essays on nature topics—"Seven Year Sleepers"—Grant Allen tells how the toad goes to bed in an earthenware pot, which he makes for himself, and how this habit may have helped start the story that live toads are found inside of stones.

Ingersoll, in that delightful book I have already referred to several times, "The Wit of the Wild," calls the pikas "the haymakers of the snow peaks." In his article on these interesting little creatures, he tells why you may often be looking right at one and still not see it; why the pikas gather bouquets and why they always lay them out in the hot sun; why their harvest season only lasts about two weeks, and why, although they usually go to bed at sunset, they work far into the night in harvest time.

"The Country Life Reader" has a good story of a woodchuck named "Tommy." Among other things it tells about the variety of residences a woodchuck has; and why animals that work at night, as all woodchucks do, have an unusually keen sense of smell. Can you guess why? The reason is simple enough.

Here's a clever bit of verse about the woodchuck by his other name, that I came across in some newspaper:

"The festive ground-hog wakes to-day,  
And with reluctant roll,  
He waddles up his sinuous way  
And pops forth from his hole.  
He rubs his little blinking eyes,  
So heavy from long sleep,  
That he may read the tell-tale skies—

Which is it—wake or sleep?"

Ingersoll's "Nature's Calendar" tells why Brer Bear stays up all winter when there is plenty of food, but goes to bed if food is scarce; how he uses roots of a fallen tree to help when he is digging his winter house; how he makes his bed and what he uses for the purpose; how the winds help him put on his roof, and how he locks himself in so tight that he can't get out until spring, even if he wants to.



"IT MUST BE BRER BEAR!"

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## CHAPTER XII

(DECEMBER)

While man exclaims "See all things for my use!"  
"See man for mine!" replies the pampered goose.

—*Pope: "Essay on Man."*

### THE BROTHERHOOD OF THE DUST

But whether they store it in their little barns, like the chipmunk, or on their bones, like Brer Bear, these farmers deserve more friendly understanding than they usually get from that two-legged farmer, Mr. Man.

Just think of the ages upon ages that they have been at work, these humble brothers of ours, and their ancestors—making the soil that gives us food—and yet after all this Mr. Man comes along and says:

"Get out of my fields!"

#### I. THE LORD OF CREATION

"Oh, but—please Mr. Man—we were here *first!*"

Was that the dormouse speaking? Anyhow, whoever it was, I think he was more than half right, don't you? Mr. Man, when he complains of these people, is apt not only to forget what he owes to them but in claiming that what they eat is wasted, to forget what a waster he is himself—wasting the soil and wasting the trees and everything.

#### BRER BEAR GIVES MR. MAN A PIECE OF HIS MIND

"Now just don't you overdo this Lord-of-Creation business, Mr. Man," says a deep, growly voice. (It must be Brer Bear!) "Other people have rights as well as you! And if you'd tend to your work half as well as they've attended to theirs, for ages before you were born, this would be a better world to live in; a good deal better, and there'd be a lot more of the good things of life to go around.

"And now that you've waked me up I'm going to tell you something else. You human beings are not only a hard lot, but a stupid lot. You think you're mighty smart, don't you, with your bear-traps and your shooting machines that you shoot each other with, as well as shooting the rest of us! But do you know what *I* think? I think if some of us—the bears or the beavers or the ants, for example—had had half your chance they'd have been twice as smart; and then we bears might have gone around shooting at you, the way Mr. Beard showed once in one of those funny pictures of his."



#### HUNTING THAT DOESN'T HURT

Hunting with a gun is great sport. But now you know from my story what good the animals do in the world you may not like so well to kill them. And there is a new kind of hunting that is just as much fun—with a camera. This picture shows a boy in ambush, ready to shoot, by pressing a bulb; for the bird in the tree is exactly in front of the shutter of the camera.

You see, Brer Bear has a good tongue in his head as well as a wise old head on his shoulders, and I must say he's entirely right when he makes the statement that human beings aren't anywhere near as bright, according to the chance they've had, as the bears and the beavers and the ants and the

bees, and many others that could be named. Why, do you know that in the whole history of the human race there have been only a few really bright people, like Mr. Shakespere and Mr. Kipling, Mr. Archimedes and Mr. Edison. It was such men as these—not over two thousand or three thousand out of the millions upon millions of human beings who have lived on the earth—that raised the rest up from the Stone Age to where they are to-day.

"Into the coarse dough of humanity an infrequent genius has put some enchanted yeast."

That's the way a recent English writer puts it. And then he goes on to say that if snakes and beasts of prey had been as clever as the bees and ants and beavers, men would have been exterminated. They could have saved themselves only by getting on with their education, climbing up the grades, a good deal faster than they have done.

He says it—this Englishman—almost in the very words of Brer Bear. And we can imagine Brer Bear going on, taking up where the Englishman leaves off.

"In other words," says Brer Bear, "it was because the bees and ants and beavers went on minding their own business, neither hurting you nor giving any pointers to the wolves and the lions and the snakes, that you're still here, Mr. Lord Man! That's part of the story of how you got to be lord of creation. Now listen to the rest of it:[\[27\]](#)

"The cave-dwellings of men were stolen from cave-lions and cave-bears; their pit-dwellings were copied from the holes and tunnels burrowed by many animals; and in their lake-dwellings they collected hints from five sources: natural bridges, the platforms built by apes, the habits of waterfowl, the beaver's dam and lodge, and the nests of birds. In the round hut, which was made with branches and wattle-and-daub, stick nests were united to the plaster work of rock martins. Yes, a good workman in the construction of mud walls does no more than rock martins have done in all the ages of their nest-building.

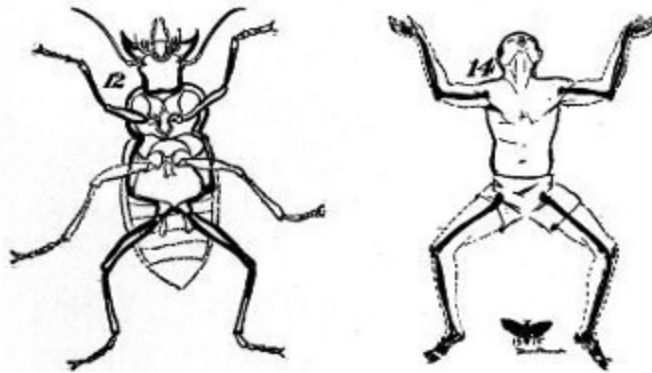
"Suppose primitive man cut down a tree with his flint axe, choosing one that grew aslant over a chasm or across a river; or suppose he piled



stepping-stones together in the middle of a waterway, and then used this pier as a support for two tree trunks, whose far ends rested on the bank sides. Neither of these ideas has more mother wit than that which has enabled ants to bore tunnels under running water, and to make bridges by clinging to each other in a suspension chain of their wee, brave bodies."

#### HOW MAN HELPED HIMSELF TO OTHER PEOPLE'S IDEAS

So you see that isn't just Mr. Bear's way of putting it; there are human beings who think a good deal as he does. Myself, I agree with Brer Bear and Brer Brangyn.<sup>[28]</sup> For man certainly, take him by and large, doesn't always set a good example to his fellow animals, either in making the best of his *opportunities* or in giving his humble brothers a square deal.



From "Bugs, Butterflies and Beetles," by Dan Beard. By permission of J. B. Lippincott

#### IF BEETLES WERE AS BIG AS BOYS

Our six-footed brothers are wonderfully strong in proportion to their size, and it would go hard with us if beetles, for example, were as big as boys.

Do you know what I felt like saying, back there in [Chapter IX](#), when we were speaking of kingfishers, and how certain parties had given it out that kingfishers eat big fish that otherwise might be caught with a hook or a seine? This is what I *felt* like saying:

"What if they do? Who's got a better right?"

Then they'd say—these men—I suppose:

"Why, we have; we're sportsmen!"

"Oh, yes," I'd say, "you're the kind of sportsman that's so afraid somebody else will see and kill something before you do; particularly if that somebody is itself a wild creature that has to earn its living that way and only takes what it needs for its family!"

And they're so good-natured about it, most of these country cousins of ours, that we walked right in on and ordered out, Cousin Woodchuck, for instance.

"The woodchuck can no more see the propriety of fencing off—though he admits that stone walls are fine refuges, in case he has to run for it—a space of the very best fodder than the British peasant can see the right of shutting him out of a grove where there are wild rabbits, or forbidding him to fish in certain streams. So he climbs over, or digs under, or creeps through, the fence, and makes a path or a playground for himself amid the timothy and the clover, and laughs, as he listens from a hole in the wall or under a stump, to hear the farmer using language which is good Saxon but bad morals, and the dog barking himself into a fit."[\[29\]](#)

## II. THE SCHOOL OF THE WOODS AND FIELDS

I don't mean to say, mind you, that the farmer hasn't any rights in his own fields, and that he should turn everything over to the woodchuck and the rest, but I do mean to say that our wild kinsmen have rights and that there is a lot more to be got out of them than their flesh or their hides or the pleasure of killing them.

For one thing, the ant and the angleworm, the birds and the woodchucks, the little lichens and the big trees, the winds and the rains, are all teachers in the Great School of Out-of-Doors, and in this school you can learn almost everything there is to be learned. It's really a university. Nature study, as

you call it in the grades, besides all the facts it teaches you, trains the eye to see, and the ear to listen, and the brain to reason, and the heart to feel.

### STORY OF THE LONDON BANKER AND HIS ANTS

Once there was a London banker who used to go around with—what do you think—in his pockets? Money? Yes, I suppose so; but what else? You'll never guess—ants! He was a lot more interested in ants than he was in money; and so, while the business world knew him as a big banker, all the scientific world knew him as a great naturalist. He wrote not only nature books but other books, including one on "[The Pleasures of Life](#)," and among life's greatest pleasures he placed the "friendship," as he puts it, of things in Nature. He said he never went into the woods but he found himself welcomed by a glad company of friends, every one with something interesting to tell. And, in speaking of the wide-spread growth of interest in Nature in recent years, he said:

"The study of natural history indeed, seems destined to replace the loss of what is, not very happily, I think, termed 'sport.'"

And isn't it curious, when one comes to think of it, why a man should take pleasure in seeing a beautiful deer fall dead with a bullet in its heart? You'd think there would be so much more pleasure in seeing him run—the very poetry of motion. Or, why should a boy want to kill a little bird? You'd think it would have been so much greater pleasure to study its flight or to listen to the happy notes pour out from that "little breast that will throb with song no more."



SIR JOHN LUBBOCK

The great London banker who carried ants in his pocket.

### WHY MAN KILLS AND CALLS IT "SPORT"

Among other animals that this banker naturalist studied was man himself; man when he was even more of an animal than he is to-day, and he came to the conclusion that this curious killing instinct is a survival of the long ages when man had to earn his living by the chase.

"Deep in the gloom of a fireless cave  
When the night fell o'er the plain  
And the moon hung red o'er the river bed,  
He mumbled the bones of the slain.

Loud he howled through the moonlit wastes,  
Loud answered his kith and kin;  
From west and east to the crimson feast  
The clan came trooping in.  
O'er joint and gristle and padded hoof,  
They fought and clawed and tore."[\[30\]](#)

Not a very pretty picture, is it? Yet it's true. But, fortunately, so is this one of the happiest hours of the caveman's grandchild.

"Oh, for boyhood's painless play,  
Sleep that wakes in laughing day,  
Health that mocks the doctor's rules,  
Knowledge never learned of schools:  
Of the wild bee's morning chase,  
Of the wild flower's time and place;  
Flight of fowl, and habitude  
Of the tenants of the wood;  
How the tortoise bears his shell,  
How the woodchuck digs his cell  
And the ground-mole sinks his well.

Of the black wasp's cunning way,  
Mason of his walls of clay  
And the architectural plans  
Of gray hornet artisans.  
For, eschewing books and tasks,

Nature answers all he asks."<sup>[31]</sup>

Some boy wrote to John Burroughs once, and asked how to become a naturalist. In his reply, Burroughs said:

"I have spent seventy-seven years in the world, and they have all been contented and happy years. I am certain that my greatest source of happiness has been my love of nature; my love of the farm, of the birds, the animals, the flowers, and all open-air things.

"You can begin to be a naturalist right where you are, in any place, in any season."<sup>[32]</sup>



WHOSE AUTOGRAPH IS  
THIS?

If you're a boy scout  
you will probably  
recognize this  
autograph in the snow.  
If not look it up in the  
Boy Scout Handbook.

It is the wholesomest, most inspiring reading in all the world, this Book of Nature. And there is simply no end to it. Just see what all we've been led into merely in following out the story of a grain of dust; and even then, I've only dipped into it here and there, as you can see by the hints of things to be looked up in the library. If we had gone into all the highways and byways of the subject—for it's all one continued story, from the making of the planets, circling in the fields of space, to the making of the little dust grains that are whirled along in the winds of March—if we followed the story all through we would have to have learned professors to teach us Astronomy, Geology, Chemistry, Zoology, with its subdivisions of Paleontology, Ornithology, Entomology, and so on; a whole college faculty sitting on a grain of dust!

### III. THE WORLD BROTHERHOOD

An obvious thing in Nature is what is called "the struggle for existence"; animals and plants fighting among themselves and against enemies of their species in the universal struggle for food. What is not so obvious, is how the whole world of things works together toward the common good.

#### **HOW THE LICHENS AND THE VOLCANOES WORK TOGETHER**

For example, working with those quiet little people, the lichens, is one of the biggest and noisiest things in the world—the volcano. The volcanoes not only pour into the air vast quantities of carbon-gas, which is the breath of life to plants, but help the lichens and the rest of the soil-makers with their work in other ways. And as the volcanoes help the lichens get their breath, the lichens forward the world service of the volcanoes by turning their lava into soil; in course of time, hiding the most desolate of these black iron wastes under a rich garment of green. It is thus the dead lava comes to life, and it is the very smallest of the lichen family that starts the process.



*Courtesy of the Northern Pacific Railway*

#### **HOW THE DEAD LAVA COMES TO LIFE**

Lava, after it has been converted into soil, by the agents of decay, makes the richest land in the world. This picture shows a vineyard on the fertile plains overlooked by Mt. Ranier, which is

an extinct volcano. In the days when Mt. Rainer was being built these plains were covered with molten lava.

Among the two principal gases of the air there is a working brotherhood; just as there is between the plants and the animals in their great breath exchange. The oxygen in the air makes a specialty of crumbling up rock containing iron. It rusts this iron into dust; while the CO<sub>2</sub>, as the High School Boy calls what I have called carbon, for short, goes after the rocks that contain lime, potash, and soda.

Working with both these gases is the frost that, with its prying fingers, enlarges the cracks in stones, and so allows the gases of the water and the air to reach in farther than they could otherwise do.

Every Winter, with its frost and its storing up of moisture in the great snow-fields of the mountains, is a benefit to the lands and their people, but the Ice Age, "The Winter that Lasted All Summer,"<sup>[33]</sup> not only worked wonders in other ways, but was of far greater benefit to the soil because it was so much more of a Winter.

Mr. Shakespere, in his day, didn't know anything about an Ice Age, but Brer Bear might have quoted certain lines of his, just the same:

"Blow, blow, thou winter wind,  
Thou art not so unkind  
As man's ingratitude.

Freeze, freeze, thou bitter sky,  
Thou dost not bite so nigh  
As benefits forgot."<sup>[34]</sup>



*Courtesy of the Northern Pacific Railway*

ASTER GROWING IN VOLCANIC ASH ON MT. RANIER

### **THE GREAT PLOUGHS OF THE ICE AGES**

With all the work the other agencies do in changing the rock into soil, and fertilizing and refreshing it with additions from the subsoil, there still remains an important thing to be done, and that is to mix the soil from different kinds of rock. This is still done constantly by the winds and flowing waters, but every so often, apparently, there needs to be a deeper, wider stirring and mixing. This the great ice ploughs and glacial rivers of the Ice Ages did. And they do it every so often, probably; for there was more than one Ice Age in the past, and, as Nature's processes do not change, it is more than likely there will be more ice ages and more deep ploughing and redistribution of the soil in the future. As you will see, if you take the trouble to look it up in "The Strange Adventures of a Pebble," it is thought we may now be in the springtime of one of those vaster changes which bring Springs lasting for ages, followed by long Summers and Autumns, and by the age-long Winters and the big glaciers and all.





#### HOW THE MOUNTAINS FEED THE PLAINS

"The elevations of the earth's surface provide for it a perpetual renovation. The higher mountains suffer their summits to be broken into fragments and to be cast down in sheets of massy rock, full of every substance necessary for the nourishment of plants, and each filtering thread of summer rain is bearing its own appointed burden of earth to be thrown down on the dingles below."

The glaciers, moving over thousands of miles and often meeting and dumping their loads together on vast fields, did the very same thing for everybody that England does for herself to-day in bringing different kinds of fertilizers from all over the world to enrich her farms. I'm very glad to speak of this because the author of the story of the pebble may have left a bad impression of the glaciers—"The Old Men of the Mountain"—as farmers, by what he said about their carrying off the original farm lands of New England, and leaving a lot of pebbles and boulders instead. While these pebbles have not produced what you would call a brilliant performer among soils, they have made a good, steady soil that in New England has helped greatly in growing farm boys into famous men, while the pebbles of Wisconsin have been of immense service to her famous cows. In the counties in Wisconsin where there are plenty of pebbles scattered through the soil, the production of cheese and butter is something like 50 per cent greater than it is in regions where there are comparatively few pebbles. [\[35\]](#)



*From Tarr and Martin's "College Physiography." By permission of the Macmillan Company*

#### GOOD CROPS FROM NEW ENGLAND'S STONY FIELDS

While the stones, big and little, with which the fields of New England are so richly supplied have not produced what you would call a brilliant performer among soils, they have made a good steady soil that can turn its hand to almost anything, and that has helped greatly in growing farm boys into famous men. In building those stone fences, for example, the boys learned that it always pays to do your work well. A hundred years is merely the tick of a watch in the life of a fence like that!

The soils of New England are like the New Englander himself, they can turn their hands to almost anything; raise any kind of crop suited to the climate, while richer soils are often not so versatile. The reason is that these pebbles were originally gathered by the glaciers from widely separated river-beds, and so contain all varieties of rock with every kind of plant food in them. It takes a long, long time to make soil out of bed-rock, but in the case of soils in which there are a great many pebbles it is different; and you can see why. On a great mass of rock there is comparatively little surface for the air and other pioneer soil-makers to get at, and so decay is slow; while the same amount of rock broken up into pebbles presents a great deal of surface for decay.

If you will examine with a glass—an ordinary hand-glass will do—one of these decaying pebbles lying embedded in the grass you can trace on it a number of wrinkly lines—sometimes even a network. These are the marks, the "finger-prints," of little roots. Little roots, as we have seen, are very

wise. They always know what they are about, and the fact that they cling to the pebbles in this way means that they are getting food out of them.

And that's right where the cows of Wisconsin come in. The rootlets of the grasses get a steady supply of food from the decaying surfaces of these pebbles scattered through the pastures, and then pass it on to the cows.



#### HOW PEBBLES HELP FEED THE COWS

You'll think I'm joking at first, but it's the truth: *Pebbles are good for cows*. Otherwise how are you going to account for the fact that in the counties in Wisconsin where there are plenty of pebbles the production of cheese and butter is something like 50 per cent greater than it is in regions where there are comparatively few pebbles? Examine, with a hand-glass, the "finger prints" of the little roots on a decaying pebble, and see if you can't guess why. Then read the explanation in this chapter.

#### TEAMWORK BETWEEN MOUNTAINS AND PEBBLES

But now, going from little things to big things again, notice how the mountains and the pebbles are linked together in this chain of service. The mountains, too, continually feed the plains. Ruskin, in speaking of this great service, says:

"The elevations of the earth's surface provide for it a perpetual renovation. The higher mountains suffer their summits to be broken into fragments, and to be cast down in sheets of massy rock, full of

every substance necessary for the nourishment of plants. These fallen fragments are again broken by frost and ground by torrents into various conditions of sand and clay—materials which are distributed perpetually by the streams farther and farther from the mountain's base. Every shower which swells the rivulets enables their waters to carry certain portions of earth into new positions, and exposes new banks of ground to be mined in their turn. The turbid foaming of the angry water—the tearing down of bank and rock along the flanks of its fury—these are no disturbances of the kind course of nature; they are beneficent operations of laws necessary to the existence of man, and to the beauty of the earth; ... and each filtering thread of summer rain which trickles through the short turf of the uplands is bearing its own appointed burden of earth to be thrown down on some new natural garden in the dingles below."



THE MILL OF THE EARTHWORM AND THE EARTH MILLS OF THE SEA

"From the gizzard mills of the earthworm to the great earth mills of the sea, all are—most evidently—parts of one great system." (In the picture on the left an earthworm has been laid open to show its grinding apparatus.)

So we find a wonderful variety of things working together in making and feeding the soil that feeds the world: mountains and pebbles, volcanoes and lichens, the breath of the living and the bones of the dead; the sun, the winds, the sea, the rains; the farmers with four feet, the farmers with six feet; the swallow building her nest under the eaves, the earthworms burrowing under our feet, each bent on its own affairs, to be sure, but at the same time each helping to carry on the great business of the universe. From the little gizzard mills of the earthworm to the great earth mills of the sea, that renew the soil for the ages yet to come, all are—most evidently—parts of one great system; are together helping to work out great purposes in the advance of men and things; purposes which require that

"While the earth remaineth, summer and winter, seed-time and harvest, shall not cease."

### HIDE AND SEEK IN THE LIBRARY

As I said, most people not only think that they're smarter than their fellow animals, but when you point out to them how clever some of these other animals are, they say: "Oh, *that's* just instinct!" As if animals don't think and learn by experience, and all, just as we do! You look up "instinct" in the encyclopædia, and you'll see. Then read Long's "Wood Folk at School."

There's really a lot more fun in shooting animals with a camera than with a shotgun or a rifle. Did you ever try it? "Hunting with a Camera" in "The Scientific American Boy at School," by Bond, will tell you how to get the best results. Other good pointers on animal photography will be found in Verrill's "Boy Collector's Hand Book" ("Photographing Wild Things") and in "[On the Trail](#)," by A. B. and Lina Beard.

And if you ever feel like killing a bird "just for fun," read in the diary of "Opal" about the farmer boy who shot the little girl's pet crow; it was "only a crow," he said, and he wanted to see if he could hit it. That will cure you, I think. The diary of "Opal" reads like a fairy-tale, but it's all true, and although it was written—every word of it—by a little girl of seven, it is one of the most remarkable books that anybody ever wrote. The crow's name, by the way, was "Lars Porsina of Clusium." The little girl used to give her pets names like that.

Don't forget what the great naturalist, Agassiz, said about the pencil being "the best eye"; that is to say, you can get a more accurate knowledge of things and come nearer to seeing them as they really are, by drawing them. Drawing, in the best schools, is a part of Nature Study, and when you get so that you can draw fairly well—as everybody can with practice—you will find there is even more of a thrill in thus *creating* forms—out of nothing, as you might say—than there is in taking photographs. The pencil is a magician's wand! As an example and inspiration for taking your pencil and sketch-book into the fields, get "Eye Spy," by Gibson, and, of course, Seton's animal books. I do believe Seton drew his pictures with those simple, expressive outlines so that young folks could redraw them. The difference between redrawing a drawing and simply looking at it, is a lot like the difference between *reading* a book and merely glancing at the print.

You are sure to be interested in Sir John Lubbock's book on "Ants, Bees and Wasps," and you will find a world of interesting things about the earlier animal days of man in his "Origin of Civilization" and "Pre-Historic Times."

And who do you suppose had most to do with teaching men they were really brothers, and so bringing them up to the civilized life we know to-day? Mother! (See Drummond's "Ascent of Man," or Chapter XII of "The Strange Adventures of a Pebble," where the whole marvellous story of evolution is told in simple form.)

If Nature Study proves half as delightful and profitable to you as I am sure it will, the following list of books will be very useful in building

up your library on the subject, and in selecting books from the public library:

"[Among the Farmyard People](#)," by Clara D. Pierson, deals with various things you probably never noticed about chickens and pigs, and other domestic animals. "[Among the Meadow People](#)," by the same author, tells about birds and insects. You can see what her "[Among the Pond People](#)" tells about—tadpoles, frogs, and so on. Really, it's a perfect fairy-land, an old pond is! "Among the Moths and Butterflies," by Julia P. Ballard, is about fairies, too, as the title shows.

For children of the seventh to eighth grades, and up, Hornaday's "American Natural History" will be a delight, and it has loads of pictures which, as in all well-illustrated scientific books, are as valuable as the text. You know who Hornaday is, don't you? He is the man at the head of the great Zoo in New York City.

Margaret W. Morley's "The Bee People" is worthy of its subject, and that's about the highest praise you could give to a book about bees, I think. Then don't forget, when you are in the library, to look up her "Grasshopper Land." The grasshopper book also treats of the grasshopper's cousins, which include the crickets and the katydids; yes, and the "walking sticks"; and the "praying mantis." (Did you know that whether you spell this weird little creature's first name, "praying," with an "e" or an "a" you'd be correct?)

Every boy and girl, of course, is supposed to know about Ernest Thompson Seton's books, but for fear some of them don't, I'll mention a few that it simply wouldn't do to miss. "[Animal Heroes](#)" gives the history of a cat, a dog, a pigeon, a lynx, two wolves and a reindeer; "Krag and Johnny Bear" is made up from his larger book, "Lives of the Hunted"; "Lobo, Rag and Vixen" is from his "[Wild Animals I Have Known](#)," and "[The Trail of the Sandhill Stag](#)."

John Burroughs is very different from Seton and Long, but the older you get the better you will like him. His is one of the great names in the study of Nature's pages at first hand and, as literature, ranks with the work of Thoreau. Get his "[Birds, Bees and Other Papers](#)," "[Squirrels and Other Fur-bearers](#)."

Darwin, one of the greatest men in the whole history of science—the man whose name is most prominently identified with the greatest discovery in science, the principle of evolution—how do you suppose he started out? Just by looking around! Read about it in "What Mr. Darwin Saw in His Voyage around the World."

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# INDEX

(For numerous practical suggestions as to the use of an index the reader is referred to the preface to the index in the author's "Strange Adventures of a Pebble.")

Africa, one country where the Hornbills live, [169](#)

Ants, their interesting habits in relation to the history of the soil, [94](#);

ants that thresh and store, [205](#), [213](#);

how they clean up after the day's work, [208](#)

Aphids, how they supply the ants with honey, [99](#)

Armadillo, a four-footed farmer who wears armor;

how fast he can dig, [120](#);

the funny gimlet nose that helps him travel so fast under the ground, [121](#)

Asia, one of the countries where the Hornbills live, [169](#);

home of a farmer who stores grain for the winter, [212](#)

Australia, home of that animal paradox, the Duck-billed Mole, [144](#);

and of birds that hatch their babies with an incubator, [174](#)

Bears, how they go into winter quarters, [216](#), [219](#)

Beavers, their work and their wisdom, [148](#)

Bees. (See Mason Bee and Bumblebee.)

Beetle, Sacred (Tumble Bug), sinful tactics of, [92](#)

Birds, their ancestors among the ancient monsters, [24](#);

service of the Moas in ploughing and in grinding up rock, [28](#);

other farmers who wear feathers, [162](#)

Bumblebees, their homes under the ground, [104](#)

Caveman, what he learned from his fellow animals, [228](#)

Central America, a good place to look for Flamingoes, [166](#)

Chipmunks, work and play in Chipmunkville, [131](#);  
why they have large feet for such little people, [132](#);  
inside the Chipmunk's home, [132](#);  
why they have several front doors, [133](#);  
how they spend the winter, [218](#)

Clouds, how dust helps make them, [56](#);  
and shape them, [57](#)

Colorado, once the home of prehistoric monsters, [27](#)

Corn, how the "rag babies" tell the fortune of the seed, [199](#)

Crabs, water farmers who help make land, [140](#)

Crayfish, their habits and their service in helping get land ready for the farmer, [140](#)

Crustaceans, their relation to insects, [143](#)

Cuvier, Baron, the famous paleontologist, and his adventure with a "monster," [34](#)

Dandelions, flying machines of, [51](#)

Darwin, Charles, on the importance of earthworms in the history of human civilization, [75](#);  
what he said about the intelligence of roots and why he said it (the whole chapter is about that), [186](#);  
how he taught roots to write their autobiographies, [190](#)

Deserts, plant pioneers in, [8](#);  
rich in plant food, [59](#);

how irrigation transforms them, [72](#)

Dormice, their Thanksgiving dinners and their long winter naps, [204](#), [217](#)

Duck-billed Mole, the Animal X that lays eggs like a bird and yet suckles its young like a pussycat, [144](#)

Dust, how it helps the rain come down, [56](#)

Earthworms, great importance of their work in pulverizing and fertilizing the soil, [75](#);

their habits and remarkable intelligence, [75](#);

how the great sea and the little earthworms work together, [242](#)

East Indies, home of some of the Hornbills, [169](#)

Electricity, how it helps in the shaping of the clouds, [57](#)

Elephants, their ancestors among the prehistoric monsters, [27](#);

elephants as ploughmen, [28](#)

Fabre, Henri, his study of the Mason Bee and how his schoolboys helped him, [108](#)

Farms, abandoned, how Nature restores them, [16](#)

Fish, monster fish of other days, [23](#)

Flamingoes, habits of some feathered farmers with queer noses, [162](#)

Florida, one place where you may find flamingoes, [166](#)

Fox, home life and habits, [128](#)

Frost, Jack, how he helps convert rock into soil, [43](#);

how he makes stones "walk" and in other ways co-operates with the river mills in making soil, [60](#)

Geese, their relation to the flamingoes, [166](#)

Groundhog. (See Woodchuck.)

Hamster, a four-footed farmer who uses a threshing-machine, [210](#)

Hedgehogs, why they are so unpopular as food, [121](#);  
their homes and how they do their ploughing, [122](#);  
pictures of baby hedgehogs, [216](#), [217](#);  
why they go into winter quarters, [216](#), [218](#)

Hibernation, "The Autumn Stores and the Long Winter Night," [204](#)

Hornbills, why Mr. Hornbill shuts his wife up in their home in a hollow tree, [169](#)

Hungary, home of the field rat, a farmer who stores grain for the winter, [212](#)

Ice Ages, how the glaciers ploughed and mixed the soil, [237](#)

Insects, their service in pulverizing and fertilizing the soil, [92](#);  
damage done by injurious insects, [93](#);  
relation of insects to crustaceans, [143](#)

Kangaroo rat, [131](#)

Kingfishers, their tunnel homes in the bank and how their fishing habits help enrich the soil, [171](#)

Kiwi, a late bird that nevertheless gets the worm, [167](#)

Lichens, first of the soil makers—how they helped Columbus discover the world by discovering it first, [1](#);  
how the volcanoes and the lichens work together, [235](#)

Lizards, reign of the lizard family in the days of the prehistoric monsters, [25](#)

Lubbock, Sir John, the great London banker who carried ants in his pocket—what he had to say about the pleasures of Nature Study, [231](#)

Maeterlinck, on the presence of mind of a tree and its heroic struggle against adverse circumstances, [200](#)

Marmots, their farm villages, [124](#)

Mason-Bees. The house that Mrs. Mason-Bee built and its relation to the story of the soil, [104](#)

Moles, their work as ploughmen, [115](#);  
how they do their tunnelling, [117](#);  
Mr. Mole's castle under the ground, [118](#);  
how he keeps his hair so sleek, [119](#);  
where he spends the winter, [218](#)

Monsters, prehistoric, what they looked like, their habits and how they help the farmers of to-day with their farming, [20](#)

Mosses, as soil makers, [8](#)

Mound-Birds, how they build their incubators;  
other interesting habits, [174](#)

Mountains, how the trees climb them, [13](#);  
why you always hear a rattle of stones in the mountains at sunrise, [43](#);  
how the winds help trees to climb the western slopes, [55](#);  
how the mountains help the rain to come down and why so many rivers rise in mountains, [56](#);  
why the bones of the monsters are found in the mountains, [31](#);  
how the mountains helped kill off the monsters, [32](#);  
farm villages of the marmots in the mountains, [124](#);  
team-work between mountains and pebbles, [240](#)

Nature Study, its great value, [231](#);  
how it is taking the place of cruel sport, [232](#)

New England, why its soil is so versatile and dependable, and how it helps grow farm boys into famous men, [239](#)

New Zealand, home of a bird that is a very late riser but nevertheless gets the worm, [167](#)

Oven-Birds, of South America, how they differ from the American oven-birds, [172](#);

their remarkable adobe homes and their friendliness toward man, [172](#)

Pebbles, how they help feed the Wisconsin cows, [239](#), [240](#);

teamwork between mountains and pebbles, [240](#)

Philippines, one of the regions where mound-birds live, [174](#), [176](#)

Ploughing, Nature's system: work of the squirrels, [14](#);

of the elephants and their ancestors among prehistoric monsters, [27](#);

of the Moas, [28](#);

of the Dinosaurs, [29](#);

storm ploughs of the winds, [46](#);

use of the plough to prevent soil waste, [70](#);

the great ploughs of the Ice Ages, [237](#)

Pocket Gopher, Thompson-Seton's "master ploughman," [128](#);

why he has that queer expression on his face, [128](#);

how he spends the winter, [218](#)

Pocket-Mouse, [130](#), [131](#)

Pot Holes, soil-grinding mills of the rivers, [61](#)

Prairie-Dog, his watch tower and how it protects him from his enemies, [126](#);

his great sociability, [127](#)

Rains, their work in making and transporting soil, [44](#), [55](#)

Rivers, work of the river mills in soil making, [60](#)

Roots, how lichens get along without them, [4](#);

how and why they work at different levels, [11](#);

how they make their way about (you won't wonder that Darwin said their actions suggested intelligence!), [186](#)

Sand, how it helps the soil to breathe, [59](#)

Seeds, how they determine the order of march of the trees, [12](#);  
use of screw-propellers and other devices, [42](#), [49](#), [51](#);  
how and why baby plants back into the world, [190](#);  
how they tried to change a sprouting seedling's mind but couldn't, [195](#);  
how "rag babies" tell the fortune of corn, [199](#)

Shrews, their work as ploughmen, [115](#);  
where they spend the winter, [218](#)

Siberia, there you will find the voles and their root cellars, [212](#)

South America, home of the four-footed farmers that wear armor, [120](#);  
and of the viscacha, [127](#);  
a good place to look for flamingoes, [166](#);  
and for oven-birds, [171](#)

South Sea Islands, one of the regions in which you find birds that hatch their babies with an incubator, [174](#)

Squirrels, how they help the trees to march, [14](#);  
the winding streets of Ground-Squirrel Town, [123](#);  
marmots, the largest of the squirrel family, [124](#);  
how the tree-squirrels spend the winter, [218](#)

Swallows, their habits and their service as soil makers, [177](#)

Termites, insects improperly called "white ants";  
their habits in relation to the history of the soil, [100](#)

Terracing, how employed to prevent waste of soil, [71](#)

Texas, you can still find armadillos there, [120](#)

Trees, their settled order of march into new lands, [8](#);  
how the winds and the rains help trees to climb the western slopes of mountains, [55](#);

how waste of trees causes waste of soil, [69](#)

Turtles, how turtles differ from tortoises;  
habits of both these water farmers, [137](#);  
how turtles differ from crabs in their notions about laying eggs, [142](#)

Viscachas, South American relatives of the prairie-dogs;  
their villages and their athletic fields, [127](#);  
how they rescue their buried comrades, [128](#)

Volcanoes, their contribution to soil making, [39](#);  
how they help the plant world to get its breath, [40](#);  
team-work between volcanoes and lichens, [235](#)

Voles, four-footed farmers who fill root cellars for the winter, [212](#)

Wasps, their habits in relation to the history of the soil, [102](#)

Weather and the groundhog's shadow, [219](#)

Weeds, as soil makers, [9](#)

Winds, how they helped Mr. Lichen to discover the world, [1](#);  
how they help the trees to march, [12](#);  
their work in making, mixing, and transporting soil, [37](#)

Winter in the animal world, under the ground, [204](#)

Woodchuck (Groundhog), picturesque home of a Connecticut woodchuck, [134](#);

Mr. Woodchuck's winter quarters and his shadow, [219](#)

Wyoming, one of the homes of the prehistoric monsters, [27](#)

## FOOTNOTES:



- [1] "The Strange Adventures of a Pebble."
- [2] "The Strange Adventures of a Pebble."
- [3] All these things put together are called "weathering."
- [4] Muir. "[The Mountains of California](#)."
- [5] "The Strange Adventures of a Pebble."
- [6] That is to say, no descendants worthy of them. It is now thought some of the modern reptiles may be degenerate descendants of the big reptiles of old.
- [7] Muir: "[The Mountains of California](#)."
- [8] "[Outlines of Earth's History](#)."
- [9] "The Two Majesties." This painting, by a great French realist, shows a lion getting home rather late, after his night out, stopping for a look at the rising sun; a thing with which, owing to his habits, he is not very familiar.
- [10] When you study French, if you want to read this book—like most French works on science it is very interesting—ask for Perrier's "Organization des Lumbricus Terrestris."
- [11] Just listen to this: "Worms," says Mr. Darwin, in that remarkable book of his, "are indifferent to very sharp objects, even rose thorns and small splinters of glass."
- [12] In the world of science, the ant goes by her Latin name, *Formica*, and the whole family is known as the *Formicidae*. To a Roman boy *Formica* simply meant "ant." *Fusca* is also Latin, and means "dark"; so you can see this part of the story is about a species of dark ant. As a matter of fact he is dark brown.
- [13] The scientific name for this particular kind of ant is *Lasius niger*.
- [14] A "bad" conductor is often a *good* thing, as you'll see by looking it up in the dictionary.
- [15] The whole story is told in the famous book, "[The Mason Bee](#)," by Henri Fabre. He was the teacher.
- [16] The boys were a great help. You ought to see what Fabre himself says about them in that famous book of his.
- [17] You've often noticed them, haven't you? Now read Fabre's wonderful book and see how much you *didn't* notice.
- [18] "And that's once too many," as the old farmer said; and we must agree with him when we think only of the damage they do.
- [19] One of my friends in the faculty of the University of Chicago tells me there are still a good many armadillos in Texas.
- [20] Isn't that the way a toad swallows an angleworm? Or how *does* he do it?

[21] Observers find that flamingoes can be successfully approached by putting on the skin of a cow or a horse.

[22] [CHAPTER VI.](#)

[23] "Admire," in those days, meant "to wonder at."

[24] By the way, the funny thing is that, while the bacteria that live on roots of the legumes are plants and not animals, most of them *do* move about.

[25] Rev. H. S. McCook: "The Agricultural Ant of Texas."

[26] Strictly speaking, I presume this was the same Apollo who carried the sun about in his chariot, and "Destroyer of Mice" was one of his many nicknames.

[27] Here imagine Brer Bear putting on his specs and reading from the book.

[28] That's the name of the Englishman I've just been quoting. He's a famous artist, but, like most cultivated Englishmen, can also write a good book when he feels like it.

[29] Ingersoll: "Wild Neighbors."

[30] Adapted from Langdon Smith.

[31] Whittier's "Barefoot Boy."

[32] "Pictured Knowledge."

[33] "The Strange Adventures of a Pebble."

[34] "As You Like It."

[35] Martin: "Physiography of Wisconsin."

#### Transcriber's note:

In the scanned version of this book, there is apparently a printer error in the acknowledgments for sources of illustrations (page x) where the author refers to an illustration on page 125. There is no illustration on page 125 in the original text, so the hyperlink in this ebook has been connected to the closest illustration, (caption: This Must Be a Pleasant Day) which is on page 126 in the original text.

Another possible printer error occurred on page 52, where the phrase "branches and holes" appears in the original text. In an effort to relate the context of the phrase, this has been changed to "branches and boles" in this text.

Full-page illustrations have been moved to the nearest paragraph so as not to interrupt the flow of the text. Some page numbers are missing as a result.

Where appropriate, internal hyperlinks within the index link directly to the pertinent chapter heading, section, or illustration referred to on the referenced page. In cases where no appropriate heading was available on the page, the top of the referenced page is linked. In some cases illustrations have been moved from the original location in order to avoid breaks in paragraphs, and to place them more closely to the related paragraph. For example, the index reference for "Hornbills" (page 169) links directly to the illustration now located on page 170.

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